

SPACETIME EVOLUTION OF JETS WITH THE STAR EXPERIMENT

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ABSTRACT OF THE DISSERTATION

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Most of all stable matter is made of protons and neutrons in the atomic nuclei. Protons and neutrons, or hadrons are also composite objects made of quarks and gluons, or partons which are as far as we know, fundamental particles of the strong nuclear force or just the strong force. The field theory Quantum Chromodynamics (QCD) formalizes properties of the strong force that makes it impossible for partons to escape hadrons under “normal” circumstances. However, when matter is heated up to extremely high temperatures or compressed to extremely high pressure, the same property of QCD will lead to the hadronic matter undergoing a phase transition into a partonic state consisting of free quarks and gluons outside the hadronic volume, called Quark-Gluon Plasma (QGP).

QGP is created in the laboratory by colliding nuclei (atoms stripped of all electrons) at near light speeds. In the very early stages of these collisions, before QGP formation, partons undergo hard scattering and produce highly energetic outgoing partons. In vacuum, these hard scattered partons would undergo successive fragmentations and eventually evolve into spatially dense ensembles of hadrons called jets. However in heavy ion collisions, these partons go through the QGP, undergoing extra radiative and collisional energy losses. These

extra interactions change the spatial compositions of hadrons inside jets, compared to the vacuum case. This phenomenon is called jet quenching, a commonly used probe of QGP, with the vacuum case approximated by equivalent proton-proton ($p + p$) collisions.

Two jet analyses are presented in this thesis. The first, jet-hadron correlations in Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV, explores the dependence of jet quenching on length of the path travelled in QGP, by calculating changes in average azimuthal distribution of final state hadrons relative to axes of differently oriented jets.

The second analysis characterizes angular spread of the jet's energy through observables called generalized angularities. The analysis begins with $p + p$ collisions at $\sqrt{s} = 200$ GeV showing distributions of various generalized angularities, which will eventually become baselines for heavy-ion measurements. Next, we groom away the soft, wide-angle radiation from these jets to separate the effects of early and late evolution. We close with initial studies for ongoing measurement of heavy-ion measurements of generalized angularities.

Along the way, various computational techniques studied for these analyses are also discussed.

Acknowledgments

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Dedication

A dedication here.

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Chapter 1

Introduction

Since ancient times, humans have pondered the nature of matter, as evidenced by the fact that between 2000-3000 years ago, the Greeks and Indians independently developed the philosophical school of *Atomism*. The words “atomism” and “atoms” are etymologically linked to the Greek word *atomon* (indivisible), first used by Democritus in 500 B.C., to denote the most fundamental, indivisible unit of matter. The atomon of Democritus differed from the matter they came from quantitatively but not qualitatively (an atom of wood would still be an almost infinitesimal piece of wood). Later around 340-330 B.C., Aristotle’s *hylomorphic* (from Greek words *hyle* for matter and *morphe* for form) worldview challenged Democritus’s assumption of the atomon’s indivisibility, as it was simply not verifiable in ancient Greece. The ingenuity of Aristotle’s hylomorphism lied in his abstraction of the unknown, he theorized the *elachista* (Greek word for minima, later translated into Latin as *minima naturalia*) not as an indivisible unit of matter, but simply a *size scale* beyond which matter could no longer be structured into recognizable forms like flesh, wood, bronze, etc.

Meanwhile, sometime between 600-200 B.C., the Indian philosopher Kanada developed a non-theistic, atomistic approach to the Vedas in the Sanskrit text *Vaishesika Sutra* (Aphorisms on particularity). Kanada postulated that even the smallest division of any material that still resembles the matter, *Anu* (Sanskrit word for atom) would further comprise of *Paramanu* (Sanskrit for ultimate-atom/sub-atom) Interestingly, Kanada’s paramanu were qualitatively different than ordinary matter and were of four kinds, two with mass and two without, and their indivisibility came from their immeasurability, making his theories resemble a middle ground between Aristotle and Democritus. Kanada also went into concepts

like rotational invariance (the paramanu must be spherical to be the same from all directions). However, even though the words Anu and Paramanu are still used to mean “atom” and “nuclear/sub-atom” respectively in almost every language spoken in today’s India, the ultimate objective of Kanada’s sutras had little to do with formalizing natural phenomena and were mostly a metaphysical reading of Vedic teachings like *Dharma* and *Moksha* (Vedic version of the Buddhist *Nirvana*).

Around 2000 years later, Aristotle’s minima naturalia became the foundation of *corpuscular theories* from European natural philosophers like Issac Newton, Rene Descartes and Robert Boyle. Particularly, in 1704 Issac Newton’s *Opticks* [24] described the *Corpuscular theory of light*, conceptualizing light as a rectilinear stream of discrete “corpuscles” with a finite velocity, an early version of the modern *photons*. With the perfected versions of Newton’s optics explaining phenomena like reflection, refraction, polarization, etc. the corpuscular school of thought was dominant in scientific circles of 18th century Europe. Although Thomas Young and his double-slit experiment in 1801 [25] ousted the corpuscular theory of light from annals of serious scientific discourse for rest of 19th century, the resurrection of atomism as a serious scientific philosophy was already being scripted in John Dalton’s notebooks. Between 1803 and 1808, Dalton proposed that every chemical element consists of identical, indivisible atoms that combine in fixed, whole-number ratios, explaining the laws of chemical combination and turning atomism into a quantitative, testable theory [26].

The indivisibility of Dalton’s atoms lasted less than a century. In 1897, J.J. Thomson showed that cathode rays are made of negatively charged particles far lighter than any atom, the *electron*, the first fundamental particle to be discovered and the first sign that atoms have an internal structure [27]. Soon after, Planck (1900) and Einstein (1905) revived Newton’s corpuscles in a new form: light is emitted and absorbed in discrete quanta, later called *photons* [28, 29]. In 1911, Rutherford’s analysis of alpha particles scattered off a thin gold foil revealed that almost all the mass of an atom is concentrated in a tiny, positively charged *nucleus* [30], which by 1932 was known to consist of *protons* (Rutherford, 1919) [31]

and *neutrons* (Chadwick, 1932) [32]. In 1928, Dirac’s relativistic equation for the electron predicted antimatter [33], and its first example, the *positron*, was found by Anderson in cosmic rays in 1932 [34]. The following decades brought a flood of unexpected particles, the *muon* [35], the *pion* [36], the *strange* particles [37] and the *neutrino* [38], and, with the new accelerators of the 1950s, dozens of short-lived hadrons, a “particle zoo” too large to be fundamental.

Order was restored in 1964, when Gell-Mann and Zweig proposed that hadrons are built from a few *quarks* [39, 40], which soon had to be given a new quantum number, *color* [41, 42]. The deep-inelastic scattering experiments at SLAC in 1968–1969, in a sense Rutherford’s experiment repeated at a much finer scale, found that the proton indeed contains point-like constituents, the *partons*, later identified with quarks [43, 44]. In 1973, Gross, Wilczek and Politzer showed that a non-Abelian gauge theory of color, *Quantum Chromodynamics* (QCD), is asymptotically free, i.e. its coupling becomes weak at short distances, making QCD the theory of the strong interaction [45, 46]. Between 1974 and 1977, the discoveries of the *charm* quark (through the J/ψ), the *tau* lepton and the *bottom* quark hinted at a third generation of matter [47–50]. Quarks and gluons themselves are never observed in isolation, but their traces are: two collimated sprays of hadrons, *jets*, were seen in electron-positron annihilation at SLAC in 1975 [51], and the three-jet events observed at the PETRA collider at DESY in 1979 were the first direct evidence of the *gluon* [52]. The W and Z bosons of the unified electroweak theory were discovered at CERN in 1983 [53–55], the *top* quark at Fermilab in 1995 [56, 57] and the *tau neutrino* in 2000 [58], completing three generations of quarks and leptons. Finally, the discovery of the *Higgs boson* at CERN in 2012 [59, 60] completed the particle content of the Standard Model, described in the following section.

1.1 The Standard Model

According to the Standard Model (SM), all matter results from interactions between fundamental particles: lepton and quarks (fermions), and mediators (bosons). Table 1.1 shows

		Leptons					Quarks		
		l	Q	(L_e, L_μ, L_τ)	q	Q	$(F_d, F_u, F_s, F_c, F_b, F_t)$		
First generation	{	e^-	-1	(1, 0, 0)	d	-1/3	(-1, 0, 0, 0, 0, 0)		
		ν_e	0	(1, 0, 0)	u	2/3	(0, 1, 0, 0, 0, 0)		
Second generation	{	μ^-	-1	(0, 1, 0)	s	-1/3	(0, 0, -1, 0, 0, 0)		
		ν_μ	0	(0, 1, 0)	c	2/3	(0, 0, 0, 1, 0, 0)		
Third generation	{	τ^-	-1	(0, 0, 1)	b	-1/3	(0, 0, 0, 0, -1, 0)		
		ν_τ	0	(0, 0, 1)	t	2/3	(0, 0, 0, 0, 0, 1)		

Table 1.1: Three generations of leptons (left) and quarks (right).

fermions of the standard model, their anti-particle equivalent would have all the signs reversed for the quantum numbers. For example, an anti-electron (e^+) and anti-electron-neutrino ($\bar{\nu}_e$) lepton number would flip to $(-1, 0, 0)$ and charges (Q s) would be +1 and 0 respectively.

The mediators/bosons exchanged between the other fundamental particles give rise to the fundamental interactions: electromagnetism from the photon (γ), weak force from $W^\pm$ and Z bosons, the gluon (g) for the strong force, and the Higgs boson (h) for imparting mass to the fundamental particles. Various interaction vertices possible in SM, that become the building blocks of all fundamental interactions are shown in 1.2.

1.2 Quantum Chromodynamics

Quantum Chromodynamics (QCD) is the component of the SM which describes the strong interactions of quarks and gluons. The QCD Lagrangian can be expected as:

$$\begin{aligned}
\mathcal{L}_{\text{QCD}} = & \underbrace{\bar{q}^\alpha (i\gamma^\mu \partial_\mu \delta_{\alpha\beta} - g\gamma^\mu t_{\alpha\beta}^a A_\mu^a - m\delta_{\alpha\beta}) q^\beta - \frac{1}{4} F_{\mu\nu}^a F_a^{\mu\nu}}_{\mathcal{L}_{\text{classical}}} \\
& - \underbrace{\frac{1}{2\lambda} (\partial^\mu A_\mu^a)^2}_{\mathcal{L}_{\text{gauge-fixing}}} + \underbrace{\bar{c}^\alpha \partial^\mu (\partial_\mu \delta_{\alpha\beta} + igT_{\alpha\beta}^a A_\mu^a) c^\beta}_{\mathcal{L}_{\text{ghost}}}
\end{aligned} \tag{1.1}$$

$\mathcal{L}_{\text{classical}}$ is the classical Lagrangian density shows interaction of a quark field q^α with 3 color degrees of freedom interacting with gluon fields A_μ^a with 8 color degrees of freedom. g and m are dimensionless and correspond to QCD coupling constant, and quark-mass

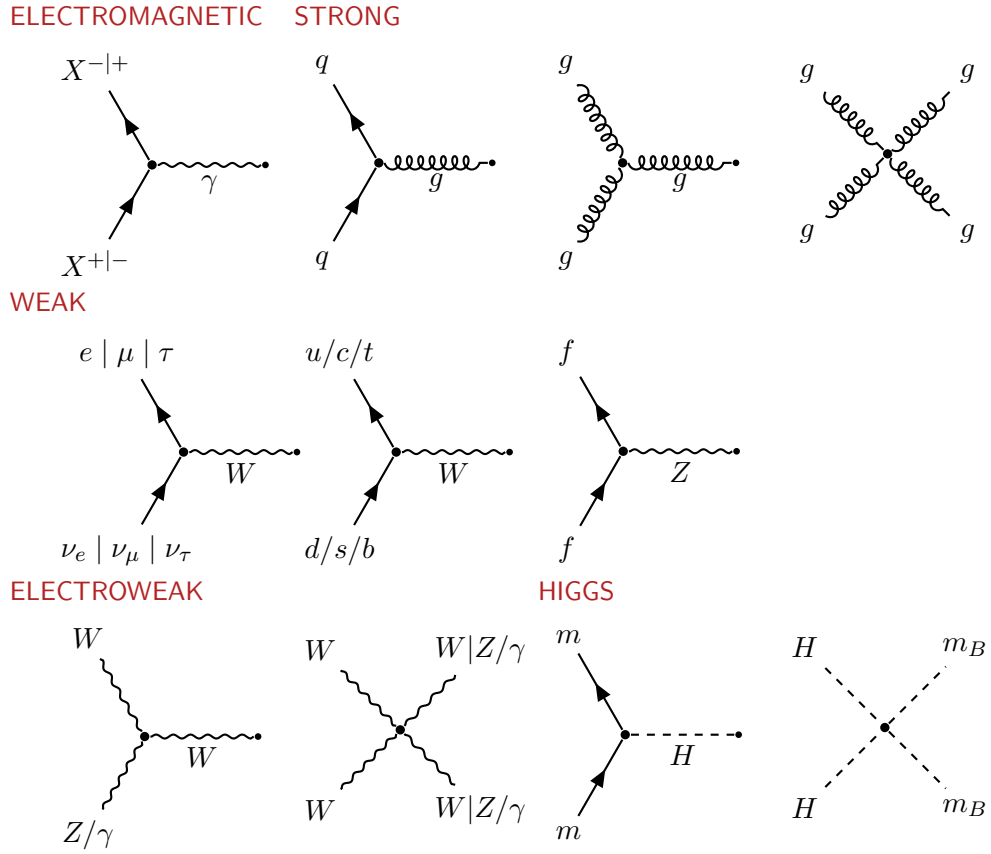


Table 1.2: Feynman diagrams of various interaction vertices possible in the Standard Model

respectively. $F_{\mu\nu}^a$ are the gluon tensors defined as,

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a - gf_{abc}A_\mu^b A_\nu^c. \quad (1.2)$$

To properly define gluon propagators, we need to make a choice of gauge. The $\mathcal{L}_{\text{gauge-fixing}}$ parametrizes the class of covariant gauges with the gauge parameter λ . In non-Abelian theories, this gauge fixing term leads to creation of unphysical degrees of freedom, which are canceled out by $\mathcal{L}_{\text{ghost}}$ which contains complex, scalar fermion fields c^α .

The matrices t^a and T^a are the fundamental and adjoint representations of SU(3) in form of traceless 3×3 and 8×8 hermitian matrices satisfying:

$$\begin{aligned} [t^a, t^b] &= if_{abc}t^c, \quad \text{tr}(t^a t^b) = \frac{1}{2}\delta^{ab} \\ [T^a, T^b] &= if_{abc}T^c, \quad (T^a)_{bc} = -if^{abc}. \end{aligned} \quad (1.3)$$

Comparing with quantum electrodynamics (QED), where the photon field tensor lacks the last term of Eqn. 1.2 as it is an Abelian, $U(1)$ gauge theory, we can explain the presence of three/four-gluon vertices and absence of such vertices for the photon in Table 1.2. The lagrangian can be used to formulate *Feynman rules* for calculating amplitude contributions from various propagators and vertices, which are collected in Tab. A.1.

1.2.1 The Running Coupling

$$\alpha_s(Q^2) = \frac{\alpha_s(Q_0^2)}{1 + \beta_0(\alpha_s(Q_0^2)) \ln(Q^2/Q_0^2) + \mathcal{O}(\alpha_s^2)} \quad (1.4)$$

Eq. 1.4 show a monotonically decreasing relationship between α_s and Q^2 . At high momentum transfers ($Q \gg \Lambda$), the coupling strength vanishes ($\alpha_s \rightarrow 0$). In this regime, quarks and gluons behave as quasi-free particles, making perturbative QCD (pQCD) calculations highly reliable (*Asymptotic Freedom*). At low energy scales ($Q \ll \Lambda$), the coupling diverges. In this non-perturbative QCD (npQCD) limit, isolated color charges cannot exist

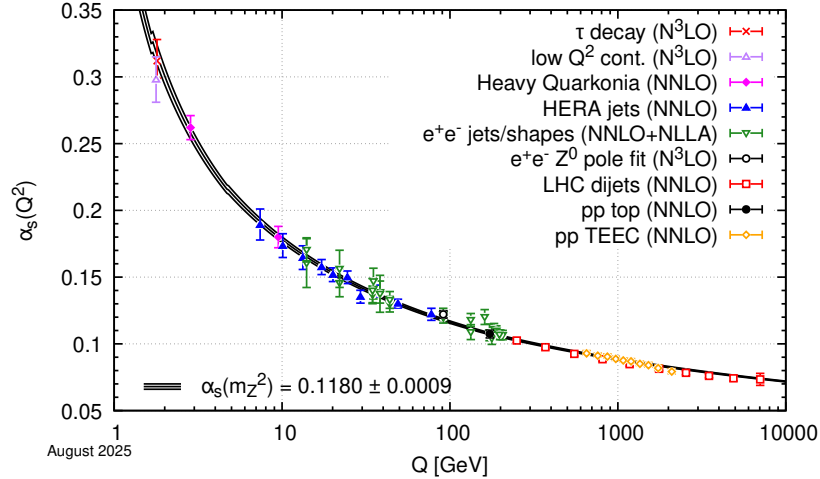


Figure 1.1: Summary of determinations of α_s as a function of the energy scale Q compared to the running of the coupling computed at five loops taking as an input the current Particle Data Group (PDG) world average, $\alpha_s(m_Z^2) = 0.1180 \pm 0.0009$ [1]

and partons are permanently bound into color-neutral hadrons (*Color Confinement*).

1.3 Hadronization and event generators

Coherent branching does not only suppress soft emission, it also organizes the color flow of the shower. Eq. A.20 shows that a pair of partons close in angle radiates as its parent, carrying the parent's charge, so the color of a branching is handed down the tree along with its momentum. Applying this at every branching down to the cut-off t_0 , neighbouring partons are the ones that are color connected, and the final state has sorted itself into color singlet groups whose invariant masses are set by t_0 and Λ , and not by the scale of the hard process that started it. This property is called *preconfinement* [61]. The preconfinement singlets can either decay directly into hadrons, giving the *cluster* models [62], or stretch a confining color flux tube between color connected partons and break it up, giving the *string* models [63].

QCD event generators chain these processes, matrix elements for the hard processes from eq. A.1, parton showers, additional semi-hard scatterings among the remaining constituents of the incoming hadrons making up the *underlying event*, hadronization, and finally the

decays of unstable hadrons. Everything below the hard process carries parameters that are not fixed by theory, and a set of values for them obtained by fitting to data is called a *tune*. Two generators starting from the same matrix element can therefore still disagree, and the spread between them is the usual way of gauging how much of a prediction is genuinely perturbative.

This thesis uses PYTHIA [64] and HERWIG [65], which differ in how they deal with the coherence of Sec. A.2.4. HERWIG orders its shower in angle and imposes Eq. A.15 directly, then hadronizes the preconfined singlets as clusters. PYTHIA instead orders in transverse momentum and radiates off color dipoles rather than off individual partons, which suppresses soft wide-angle emission without ordering in angle at all, and hadronizes with strings.

Both generators are used with tunes made for RHIC energies [66–68], as the tunes in widest use are fitted mostly to LHC data and overshoot the underlying event at $\sqrt{s} = 200$ GeV. PYTHIA-6 and PYTHIA-8 share the string hadronization and the transverse-momentum ordered shower, and differ mainly in their treatment of multiple parton interactions, color reconnection and beam remnants, all of which control the multiplicity and the soft, wide-angle part of an event.

Ch. 3 also uses JEWEL [69], which adds scattering of the shower partons off a thermal medium to this picture. The response of the detector to the generated particles is simulated separately using GEANT [70], and Ch. ?? describes how its effects are taken back out of a measurement.

1.3.1 Quark Gluon Plasma (QGP)

Asymptotic freedom of QCD means that matter at extremely high temperatures and densities (high Q^2) undergo a *deconfinement* phase transition, with partonic degrees of freedom replacing hadronic degrees of freedom. This state of deconfined quarks and gluons is called QCD/quark-gluon plasma due to analogous phenomena observed in atomic physics [71].

Thermodynamically, when temperature (T) or baryon number density (n_B) approaches

the confinement scale Λ introduced in Sec. 1.2.1 ($T \sim \Lambda \sim \mathcal{O}(10^{12})K$, and $n_B \sim \Lambda^3 \sim 1 \text{ fm}^{-3}$) the boundaries between hadrons become ill defined resulting in hadrons “melt” into QGP [72]. For reference, the core of our sun “only” has $T \sim \mathcal{O}(10^7)K$ and $n_B \sim \mathcal{O}(10^{-14})\text{fm}^{-3}$.

These orders of T and n_B are the theorized conditions prevalent in the universe a few μs after Big-Bang. The only natural phenomena in the present-day, cold universe that can achieve these conditions are the cores of neutron stars [73] or neutron star mergers [74]. Artificially, the required T and n_B can be achieved at facilities like the Large Hadron Collider (LHC) and Relativistic Heavy Ion Collider (RHIC) through collisions of heavy-ions (like Au, Pb, etc.) at near-light speeds, the aptly named *little bangs*. More on this will be discussed in Sec. 1.6.

1.4 Relativistic collisions of nuclei

Outgoing partons from hard processes undergo many successive splitting and fragmentation processes, becoming a collimated “spray” of hadrons called *jets* which are detected as dense clustering of energy in calorimeters. In principle, a *jet finding algorithm* should be able to faithfully get the kinematics of the initiating parton given the theory discussed in sec.A.2, which show partons cleanly branching into partons and eventually fragmenting into final-state hadrons that can be traced back to the shower partons. However reality is rarely this straightforward. First, the fragmentation functions that connect the partons and final state hadrons cannot be calculated in experiments. Second, it is not trivial to establish connection between final state particles and initial state partons even at the parton level, before we have the complications from hadronization and detector level reconstruction. For example, in a clean reaction like $e^+e^- \rightarrow q\bar{q}$. If one of the quark emits a gluon, then depending on the emission angle of the gluon, we might cluster 2 or 3 jets in the final state. We will further define jet finding in sec. 1.5

As discussed in sec. 1.3.1, collisions of heavy-ions at near-light speeds can create the

conditions for forming QGP. Sec. 2.1 discussed RHIC and sec. 2.2 describe the STAR detector system. Before moving on to analyses of heavy-ion collisions, we need to establish a few more concepts. Nuclei like $^{197}_{79}\text{Au}$, $^{96}_{40}\text{Zr}$, $^{96}_{44}\text{Ru}$, etc. are complex arrangements of protons and neutrons that have a multitude of non-trivial initial configurations leading up to their collisions which have significant effects on the final state particle distributions. Therefore, any measurement done using heavy-ion events need to report these initial configurations of the colliding nuclei. We discuss these in sec. 1.6.

1.5 Finding Jets

To address the prevalent ambiguity of jet definitions, the Snowmass Accord [75] stated that, experimental and theoretical measurements had to use the same jet finding algorithm on final state hadrons in order to be comparable. Thus clustered jets are not-quite the proxies for partons in the same way as parton showers, but regions in the final state phase space that are most likely to have contributions from parton showers. Even after we define a jet, another ambiguity comes from the decision of what constituents in the event must be used as inputs into the jet finding algorithm. Especially, in heavy-ion collisions jet-medium interactions make that choice non-trivial. For example, is a soft particle inside the jet cone coming from soft gluon radiation from the jet or the medium? Keeping these considerations in mind, a “good” jet finding algorithm is:

- *Collinear safe*, meaning it would find the same jet with the same $p_{T,\text{jet}}$ regardless of fragmentation details
- *Infrared safe*, meaning it would find the same jets, even in the presence of a large number of soft partons

The most commonly used family of infrared and collinear safe jet algorithms, sequentially recombines constituents in an event according to the following prescription:

1. For every pair of particles $\{i, j\}$ with momenta $(\mathbf{p}_i, \mathbf{p}_j) \equiv \{(p_{T,i}, \eta_i, \varphi_i), (p_{T,j}, \eta_j, \varphi_j)\}$

$$d_{ij} = \min(p_{T,i}^{2p}, p_{T,j}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \text{ and } d_i = p_{T,i}^{2p} \quad (1.5)$$

where, $\Delta R_{ij}^2 = (\eta_i - \eta_j)^2 + (\phi_i - \phi_j)^2$ is the (η, ϕ) distance between particle i and j . R is the resolution parameter of the algorithm, which decides the angular extent of clustered jets.

2. $i_{\min} = \min(\{d_{ij}\} \cup \{d_i\})$ is found. $d_{\min} \in \{d_{ij}\}$, the particles are combined into one jet candidate, adding their energies and momenta according to some *recombination scheme*, and the first step is repeated.
3. $d_{\min} \in \{d_i\}$, this is recorded as a final state jet candidate. It is removed from the list and the first step is repeated.

These steps are iterated until no particles remain. This algorithm becomes the k_T *algorithm* [76] for $p = 1$, *Cambridge-Aachen* [77] for $p = 0$ and the most widely used *anti- k_T algorithm* [78] for $p = -1$. The difference in these algorithms are shown in fig. 1.2. The *FastJet* library [79] efficiently implements these algorithms and a few more, and is the de-facto standard in the High Energy Physics (HEP) community. It also provides various recombination schemes.

- *E-scheme*, just adds the four momenta of the jet candidates. it is the default in FastJet and the most commonly used.
- *Boost-invariant $p_t/p_t^2/E_t/E_t^2$ schemes* involve a preprocessing stage rescaling the energy to be equal to the 3-momentum for the p_t and p_t^2 schemes or the 3-momentum to

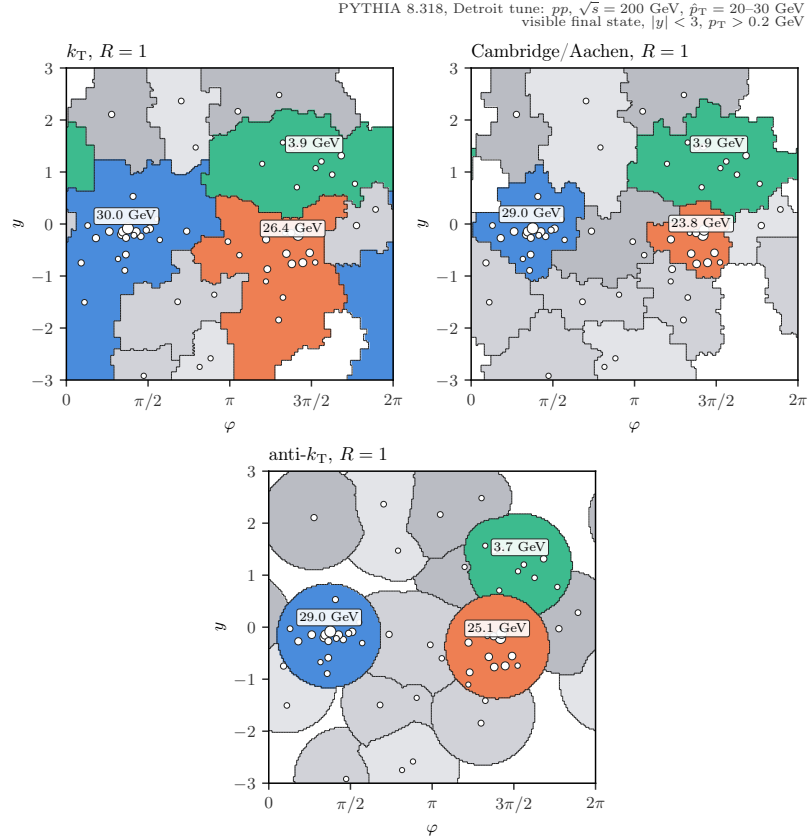


Figure 1.2: Jet areas of one PYTHIA 8 pp event at $\sqrt{s} = 200$ GeV (Detroit tune) with the k_T (top left), Cambridge/Aachen (top right) and anti- k_T (bottom) algorithms, $R = 1$. Coloured regions are the three hardest jets, the same colour marking the same jet in every panel; other jets are grey. Circles are particles, their area growing with p_T .

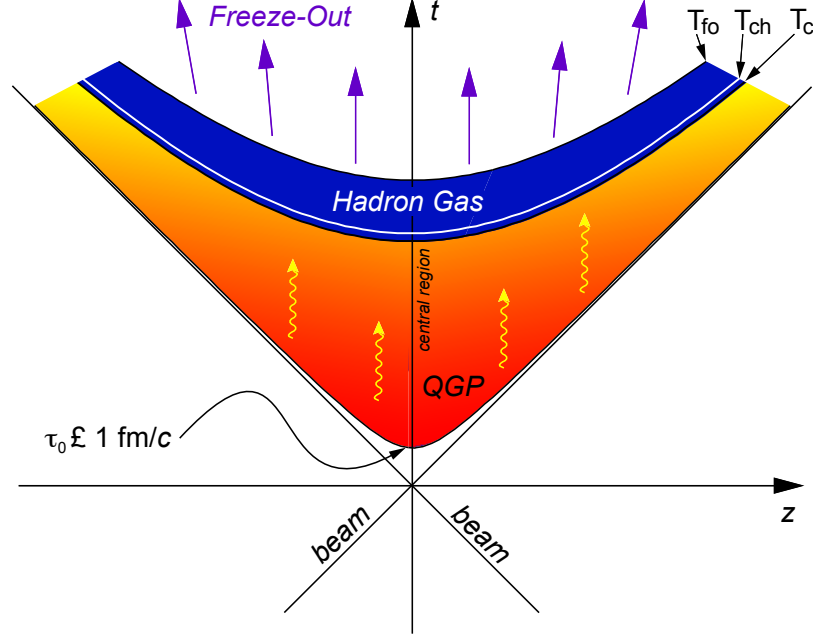


Figure 1.3: A light cone diagram showing the stages of a heavy ion collision. The abbreviation T_{fo} is for the thermal freeze-out temperature, T_{ch} is for the chemical freeze-out temperature, and T_c is for the critical temperature where the phase transition between a hadron gas and a QGP occurs. τ_0 is the formation time of the QGP. Taken from [2]

be equal to the energy in the E_t and E_t^2 schemes. Then for all schemes the recombination p_r of p_i and p_j is a massless 4-vector satisfying

$$p_{t,r} = p_{t,i} + p_{t,j}, \quad (1.6a)$$

$$\varphi_r = \frac{w_i \varphi_i + w_j \varphi_j}{w_i + w_j}, \quad (1.6b)$$

$$y_r = \frac{w_i y_i + w_j y_j}{w_i + w_j}, \quad (1.6c)$$

where w_i is $p_{t,i}$ for the p_t and E_t schemes, and is $p_{t,i}^2$ for the p_t^2 and E_t^2 schemes.

- *Winner-takes-all (WTA) scheme* uses the four momentum of the higher momentum jet candidate

1.6 Heavy-Ion Collisions

Estimates of the energy density indicate that heavy ion collisions at RHIC and LHC reach energy densities ranging between 1-12 GeV/fm³ [80, 81], well in excess of the 0.2-1 GeV/fm³ required for QGP formation [82, 83].

The heavy ion collision and the evolution of the fireball, as depicted in Figure 1.3, has several stages, and the measurement of the final state particles can be affected by one or all of these stages depending on their production mechanism and interaction time within the medium. Immediately after the collision, the system is in a non-thermalized, pre-equilibrium state where, partons undergo hard scattering, dominated by $2 \rightarrow 2$ QCD processes with the outgoing partons back-to-back in azimuth ($\Delta\varphi \approx \pi$). At $\tau \approx 0.5 - 1$ fm/c, the system thermalizes into the QGP state which expands following hydrodynamic laws. During the expansion, the system cools down until it reaches the temperature ($T_c \approx 156$ MeV from lattice QCD studies) at which a crossover phase transition to a hadronic gas state occurs. By $\tau \approx 10$ fm/c after the collision, confinement is restored and colored partons hadronize into colourless hadrons. When the chemical freeze-out temperature ($T_{ch} \approx 165$ MeV) is reached, the particle abundances of the system are frozen. Elastic interactions can still occur till the kinetic freeze-out ($T_{fo} \approx 100$ MeV), where the particle momentum spectra are established.

1.6.1 Initial State of Colliding Nuclei

The initial state of the incoming nuclei is not precisely known, but its properties impact the production of final state particles. The incoming nuclei are often modeled as either an independent collection of nucleons called a Glauber initial state [3], or a wall of coherent gluons called a Color Glass Condensate [84]. The widely used model for determining these quantities is the Glauber Monte Carlo (Glauber MC) model [85], which assumes Wood-Saxon density distribution of nucleons. The nuclei are then assigned a random impact parameter

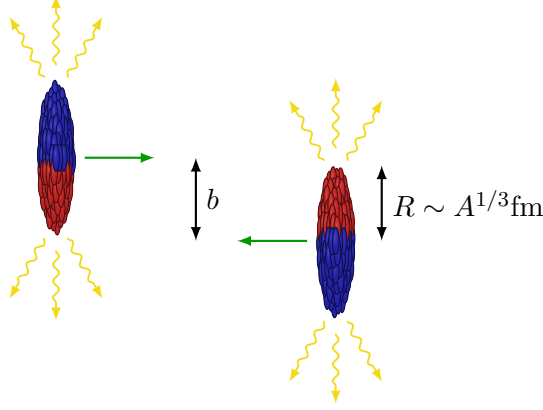


Figure 1.4: Typical heavy-ion collision with an impact parameter of b . The participating nuclei are colored red and the spectators are colored blue.

and translated to collide. If the distance between centers of nucleons is less than $\sqrt{\sigma_{\text{inel}}^{\text{NN}}/\pi}$, where $\sigma_{\text{inel}}^{\text{NN}}$ is the inelastic nucleon–nucleon cross section, the nucleons have undergone a binary collision. Nucleons that have undergone at least one binary collision are said to have participated in the reaction. This procedure is followed to obtain the number of participant nucleons (N_{part}) and the number of binary collisions (N_{coll}) event-by-event.

A schematic diagram of the geometry of heavy-ion collisions is shown in Fig. 1.4.

1.6.2 Centrality Determination

As the impact parameter (b) of collisions, the distance between the geometrical centers of the colliding nuclei, is not measurable in experiments, a proxy called *centrality* is defined. Centrality is characterized by the charged particle multiplicity of events, connected to the impact parameter through a simulation defining it as a function of N_{part} , N_{coll} and impact parameter b from Glauber model is mapped with the experimental data through a particle production model called the Two-Component Model [86]. Assuming that particle production is governed by contribution from hard component ($\propto N_{\text{coll}}$) and soft component (proportional to $\propto N_{\text{part}}$), it simulates the multiplicity distribution according to

$$\frac{dN_{\text{ch}}}{d\eta} = n_{pp} \left[x N_{\text{coll}} + (1 - x) \frac{N_{\text{part}}}{2} \right] \quad (1.7)$$

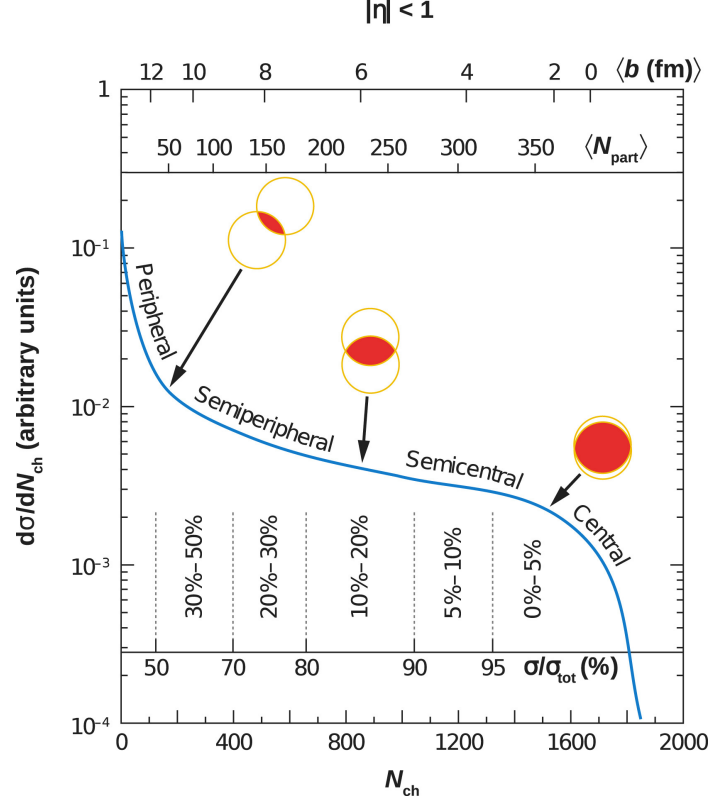


Figure 1.5: An illustration of the total inclusive charged particles multiplicity N_{ch} and number of participant (N_{part}) and impact parameter (b) in heavy-ion collisions. Taken from [3]

where the quantity n_{pp} is the multiplicity from a $p + p$ collision of the same center-of-mass energy. x is the contribution of hard component. n_{pp} is drawn from a negative binomial distribution with parameters NBD ($n_{pp}; \langle n_{pp} \rangle, k$), where $\langle n_{pp} \rangle$ is the average multiplicity in the $p + p$ collision and k is the width of the distribution. The simulated charged multiplicity distribution is then fitted with the corresponding multiplicity distribution from the experiment. $\langle n_{pp} \rangle$, k , and x are kept as free parameters in the fitting to experimentally measured charged particle multiplicity distribution. Initial guess values for x could be set close to those determined in different experiments for faster convergence of fitting [87, 88].

An event belongs to a centrality class, say 0%-5% if its charged particle multiplicity falls in the top 5th percentile. Fig. 1.6.2 illustrates collision centrality definition in heavy-ion collisions by comparing the charge particle multiplicities with Glauber MC + Two-component model simulation. Centrality classes are classified for the simulated distribution.

$\langle N_{\text{part}} \rangle$, $\langle N_{\text{coll}} \rangle$ in those centrality classes are calculated.

1.6.3 Anisotropic flow

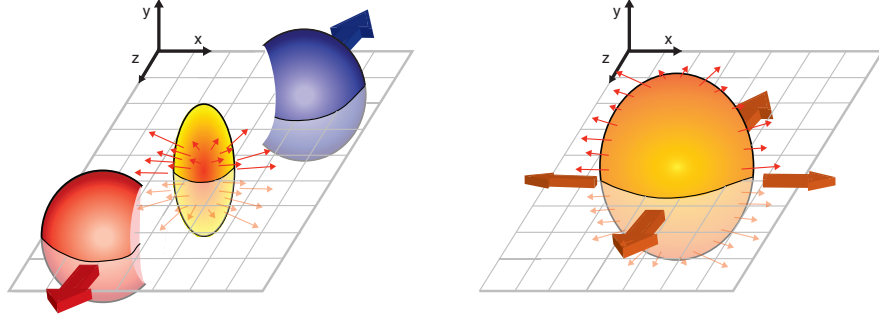


Figure 1.6: Schematic diagrams showing the initial overlap region (left) and the spatial anisotropy generated by this anisotropic overlap region. This anisotropy can be quantified using the Fourier coefficients of the momentum anisotropy. Taken from [2]

Contrary to initial naive expectations of a gas-like QGP, the QGP formed in these collisions was shown to behave like a liquid of quarks and gluons [89]. Thus, mean free path of particles in QGP fireball is much smaller than the fireball's system size. Hence, in non-central heavy ion collisions the initial state coordinate space asymmetry generates a pressure gradient in the direction of the impact parameter b , schematically shown in fig. 1.6.3. This spatial asymmetry translates into an azimuthal momentum anisotropy as the system expands, meaning that particles are more likely to be emitted along the plane where the pressure gradients are the highest. This plane is defined by the beam axis and the impact parameter of the collision, and called the *reaction plane* (Ψ_{RP}). The anisotropy can be characterized by the Fourier decomposition of the azimuthal particle distribution with respect to Ψ_{RP} [90, 91]:

$$\frac{dN}{d(\phi - \Psi_{\text{RP}})} = \frac{N_0}{2\pi} \left(1 + 2 \sum_{n=1}^{\infty} v_n \cos[n(\phi - \Psi_{\text{RP}})] \right), \quad (1.8)$$

The *anisotropic flow* coefficients v_n , the n^{th} order term in the Fourier decomposition of the

final state particle azimuthal distribution is given by,

$$v_n = \langle \cos([n(\phi - \Psi_{RP})]) \rangle \quad (1.9)$$

Due to the symmetry of the collision geometry, the most prominent contribution is from the second order coefficient v_2 , called *elliptic flow*.

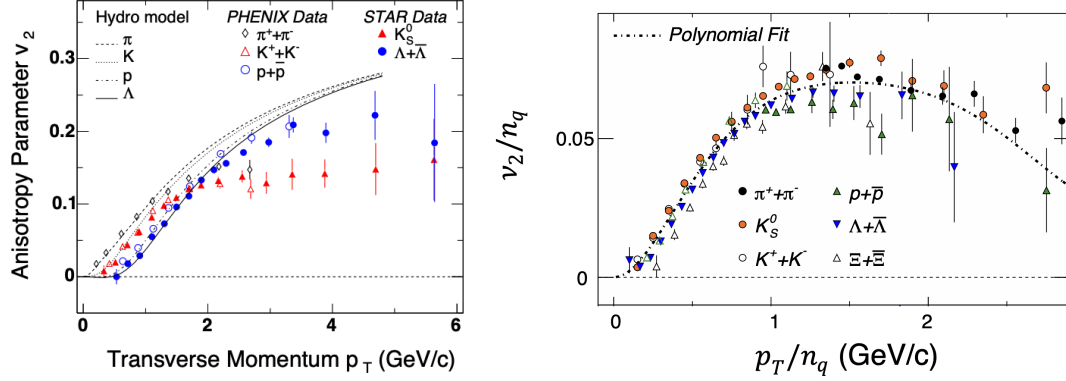


Figure 1.7: **Left panel:** Elliptic flow v_2 for several particle species in Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV from STAR and PHENIX experiments. Predictions from hydrodynamic models are shown as lines. Taken from [4]. **Right panel:** Number of constituent quarks (n_q) scaled v_2 as a function of p_T/n_q . Taken from [5].

v_2 was measured by the STAR and PHENIX for several particle species and is shown in Fig. 1.7. Large positive v_2 values indicate collective phenomena associated with a strongly coupled fluid, or a *signature of QGP*. A scaling of the v_2 values with the number of constituent quarks (2 for mesons, 3 for baryons) yielded a universal curve as a function of p_T/n_q suggesting the flow dynamics were developed in a medium with quark and gluon degrees of freedom, i.e. QGP.

1.7 Jets in QGP

As discussed in sec. 1.6, hard scatterings of partons that produce jets happens before the system thermalizes into QGP (parton producing a 10 GeV jet scatters at $\tau \sim 1/E \sim 0.2\text{fm}/c$). Thus the ensuing parton shower traverses the QGP, and resulting jet is *modified/quenched* compared to an appropriate *baseline*. The two main mechanisms for jet quenching are

collisional and *radiative* energy losses. Collisional energy loss happens due to the elastic scattering between the parton and the medium constituents and usually dominates for lower energy partons. Radiative energy loss happens due to induced gluon radiation (*bremssstrahlung*) through inelastic scatterings with the medium particles [92–94]. For partons with $p_{T,\text{parton}} \geq 5 \text{ GeV}/c$, this is usually the dominant source of energy loss [95].

The baseline measurement is done in a collision system where either QGP is not formed or has negligible lifetime compared to central heavy-ion collisions. Two possible choices arise for the baseline, first is $p + p$ collisions from which we can calculate the *Nuclear Modification Factor* (R_{AA}) for distributions of any observable $\mathbf{O}$ calculated in $p + p$ and $A + A$ collisions, given by

$$R_{AA}(\mathbf{O}, b) = \frac{dN_{AA}/d\mathbf{O}}{\langle T_{AA}(b) \rangle \cdot d^2\sigma_{pp}/d\mathbf{O}} \quad (1.10)$$

where $T_{AA}(b) = N_{\text{coll}}(b)/\sigma_{NN}^{\text{inel}}$ is the nuclear overlap function which determines the number of participating nucleons and number of binary collisions N_{bin} for a heavy ion collision as a function of its impact parameter (b) [96]. The second common baseline used in absence of a $p + p$ baseline is peripheral collisions of heavy-ion collisions. The *Relative Nuclear Modification Factor* R_{CP} is given by,

$$R_{CP}(\mathbf{O}, b) = \left. \frac{dN/d\mathbf{O}}{\langle N_{\text{bin}}(b) \rangle} \right|_{\text{cent}} \times \left. \frac{\langle N_{\text{bin}}(b) \rangle}{dN/d\mathbf{O}} \right|_{\text{peri}} \quad (1.11)$$

Before infrastructure became available to reliably reconstruct a sufficient number of jets in heavy-ion collisions, the effects of the medium on high energy partons were studying the *Di-hadron correlation*. Leading hadron (highest p_T in event) is used as a jet proxy and the azimuthal angular correlation ($\Delta\varphi$) of all charged hadrons in the event is studied. Fig. 1.8 shows one of the first measurements performed by STAR [6], di-hadron correlations compared between $p + p$, $\text{Au} + \text{Au}$, and $\text{d} + \text{Au}$ collisions. The $p + p$ and $\text{d} + \text{Au}$ both show peaks around $\Delta\varphi = 0$ and $\Delta\varphi = \pi$ as expected, given that hard scatterings are dominated by $2 \rightarrow 2$ Feynman diagrams. However, for $\text{Au} + \text{Au}$ collisions, the peak around

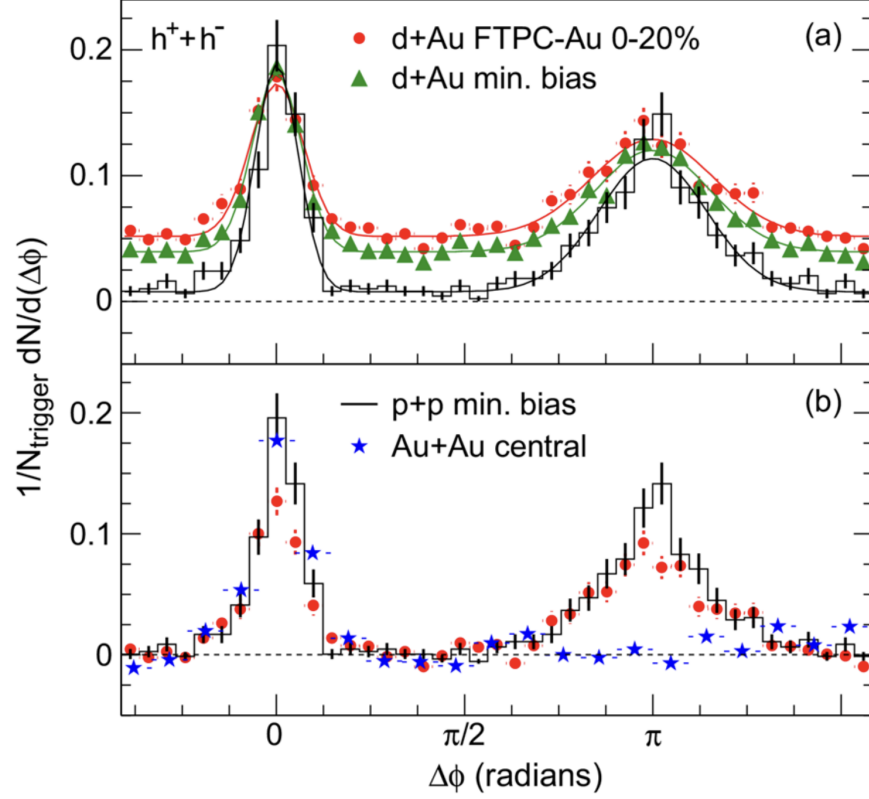


Figure 1.8: Measurement of the angular correlation between all charged hadrons and the leading hadron in the event for $p + p$ (black lines), $d + Au$ (red and green markers), and $Au + Au$ events (blue markers), performed by the STAR experiment. The vanishing of the away-side yield around $\Delta\phi \approx \pi$ in $Au + Au$ events is evidence of particles getting quenched in the QGP medium. Taken from [6].

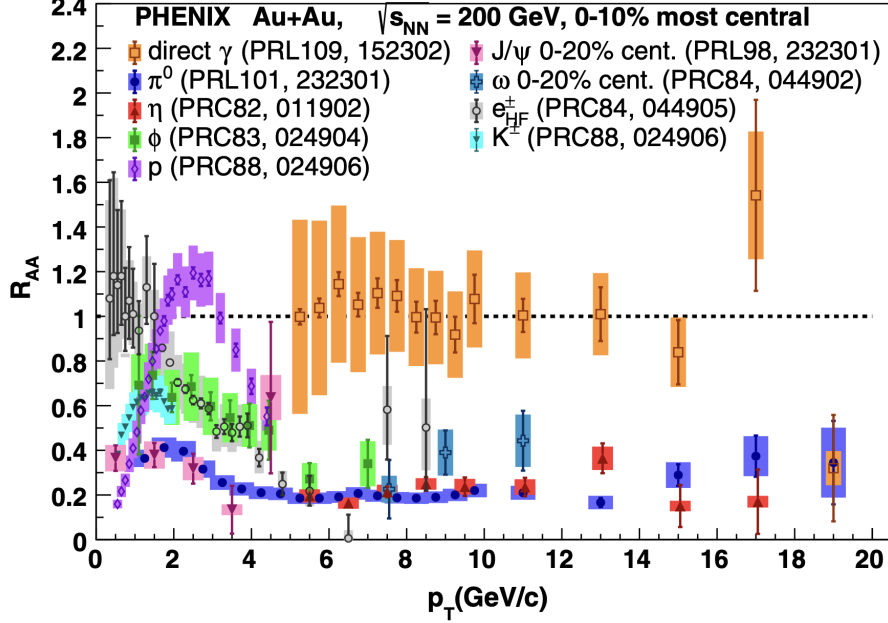


Figure 1.9: Nuclear modification factor R_{AA} as a function of p_T for identified particle species in Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the PHENIX experiment. Taken from [7].

$\Delta\varphi = \pi$ completely vanishes. This was the expectation because depending on the distance from the center of the collision where the initial back-to-back partons were produced, one of the partons would travel a larger distance in the QGP medium and therefore lose more energy than the other one.

A closer look at jet quenching was given by the PHENIX experiment through a collection of R_{AA} measurements discussed in eq. 1.7 for identified particles spectra, shown in Fig. 1.9. Whereas, direct photons (γ) do not interact strongly, and hence have no yield modification in the medium. All strongly interacting particles show varying levels of suppression from 0.2 - 0.6, again confirming their energy loss in the presence of QGP.

Although measurements with high p_T hadrons help us understand jet quenching mechanisms in the medium, they suffer from survival bias, as these hadrons arise from a subsample of jets which usually undergo the least amount of quenching [97]. Fully reconstructed jets, however, are mostly unbiased probes of energy flow at high p_T , and offer a wealth of substructure observables which allow us to look at energy flow within the jet more differentially.

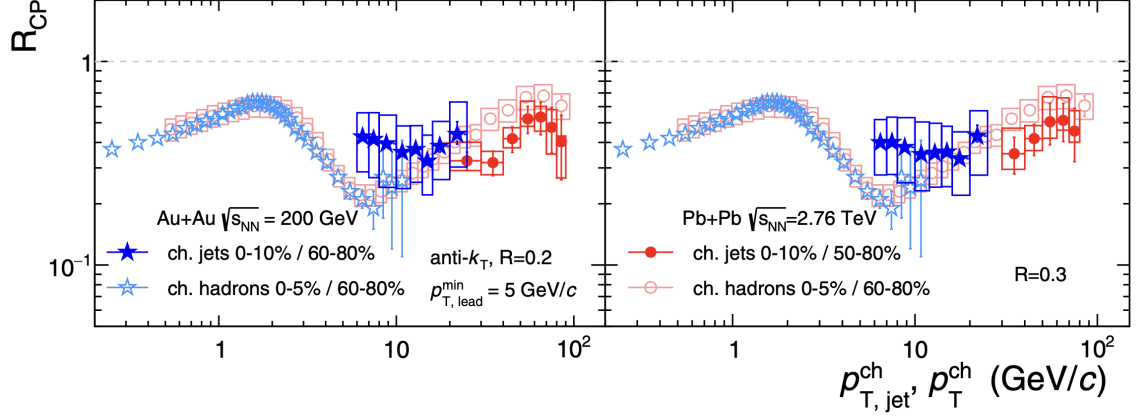


Figure 1.10: R_{CP} distributions for charged hadrons (open markers) and jets (solid markers) from RHIC (blue star markers) and LHC (red circle markers) experiments. Taken from [8] and references within [9–11].

This motivates the studies presented in chapters 3 and 4.

The STAR experiment measured the R_{CP} of the p_T spectra of charged particle jets (clustered charged tracks only and contain no calorimeter hits (towers) as input) from central and peripheral Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV shown in fig. 1.10 [8]. Measured for charged jets was compared to R_{CP} for charged hadrons from STAR at the same energy, and to the measured R_{CP} for charged hadrons and jets from the ALICE and ATLAS experiments at $\sqrt{s_{NN}} = 2.76$ TeV. First, the charged hadrons R_{CP} is found to be the same over the same range of p_T at RHIC and LHC energies (which are an order of magnitude apart). The charged jet R_{CP} is also found to have similar values between RHIC and LHC energies with roughly 60% of the energy lost outside the jet cone in central collisions with respect to peripheral collisions, with no significant p_T dependence in contrast to the p_T dependence observed for the charged hadrons R_{CP} . This may be because the inclusive hadron distribution usually arises from the leading hadron in the corresponding jet, and is therefore is sensitive to the fragmentation function. This fragmentation process can therefore introduce the p_T dependence of R_{CP} in hadrons.

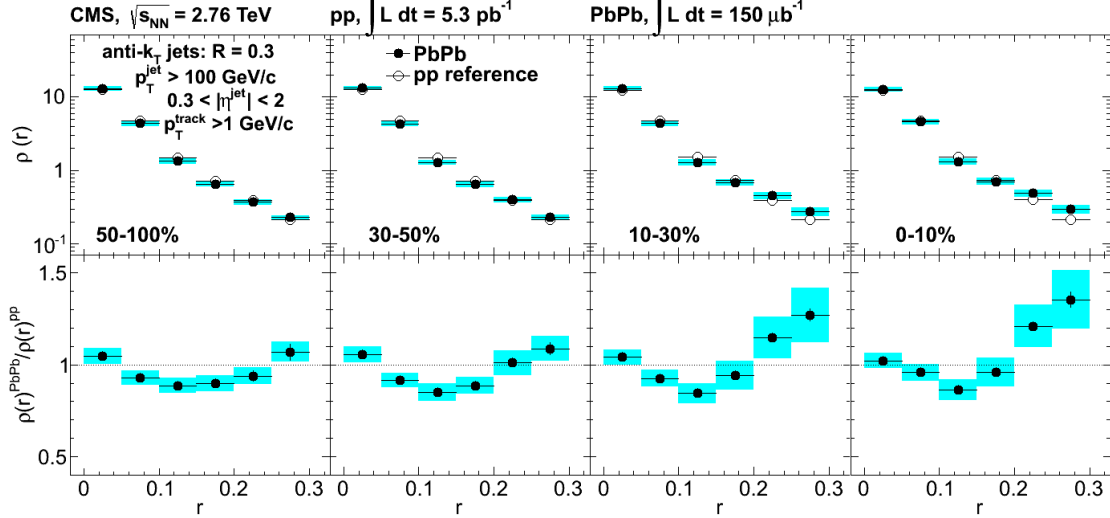


Figure 1.11: Differential jet shapes in Pb+Pb and $p+p$ collisions for four Pb+Pb centralities. Each spectrum is normalized so that its integral is unity. This shows that there are more particles in jets in central collisions and these modifications are primarily at large angles relative to the jet axis, as expected from partonic energy loss. Taken from [2]

Going a bit into the jet substructure, CMS measurement of jet's average radial momentum profile around the jet-axis, called *differential jet shape* defined as,

$$\rho(r) = \frac{1}{\delta r} \frac{1}{N_{jet}} \sum_{jets} \frac{\sum_{track \in [r_a, r_b]} p_T^{track}}{p_T^{jet}} \quad (1.12)$$

where the jet cone is divided rings of width δr which have an inner radius r_a and an outer radius r_b . From the results shown in fig. 1.11, the differential jet shapes in the most central Pb+Pb collisions are broadened in comparison to measurements done in $p + p$ collisions at the same center of mass energy [98]. These measurements demonstrate that there is an enhancement in the modification with increasing angle from the jet axis, indicating a broadening of the jet profile and a depletion near $r \approx 0.2$.

1.8 Outline

Angular ordering makes the angular distribution of energy inside a jet a record of the shower that produced it, with the wide-angle radiation coming from the early, hard splittings and

the core of the jet being filled in later. This thesis measures that distribution. It is measured first in $p + p$ collisions, where it is a test of the vacuum shower and a baseline, and then in Au+Au collisions, where a deviation from that baseline is the imprint of the medium the shower has gone through.

Ch. 2 describes RHIC and the STAR detector, in particular the TPC and the BEMC which measure the charged and neutral parts of a jet. Ch. 1.4 defines jets as the output of a jet finding algorithm rather than as partons, introduces SoftDrop grooming, and sets up the quantities used to describe a heavy-ion collision: the Glauber picture of the initial state, centrality, the event plane, and the nuclear modification factors used to quantify jet quenching. Ch. ?? collects the analysis techniques, mainly the classifiers used to tag jets and the unfolding procedure used to remove detector effects from a measured distribution. Ch. 3 presents jet-hadron correlations relative to the event plane in Au+Au collisions, which probe the dependence of energy loss on the path length traversed in the medium. Ch. 4 presents generalized angularities, inclusive and groomed, in $p + p$ collisions and then in Au+Au collisions. Ch. 5 summarizes both measurements and what they suggest about where to go next.

Chapter 2

Solenoidal Tracker At Relativistic Heavy Ion Collider

2.1 Relativistic Heavy-Ion Collider (RHIC)

The Relativistic Heavy Ion Collider (RHIC), at Brookhaven National Laboratory (BNL), is the only collider with dedicated running for heavy ion research (the CERN Large Hadron Collider runs heavy ions for only about one month per year), and the only polarized proton collider ever built. Two concentric accelerator rings 2.4 miles in circumference containing a total of 1,740 superconducting magnets afford RHIC the capability to independently accelerate and collide ion beam species from ${}^1_1\text{H}(p)$ to ${}^{238}_{92}\text{U}$ at energies between 3.85 GeV and 254.9 GeV per nucleon and different spin polarizations for protons (p), as shown in Fig. 2.1. These capabilities make RHIC the most flexible and capable collider in the world for transformative studies of extreme states of nuclear matter and the origin of the proton spin.

On February 6th 2026 RHIC ended operations with collisions of ${}^{16}_8\text{O}$ nuclei at $\sqrt{s_{\text{NN}}} = 200$ GeV recorded by the *Solenoidal Tracker At RHIC (STAR)* and *sPHENIX* detectors. Over its 25 years of active existence, it has performed ~ 300 trillion collisions of 49 unique pairs of nuclei, collected ~ 450 petabytes of data, whose analysis has led to over 650 publications. Eventually, one of RHIC's ion storage rings will be removed and replaced with a new ring for storing accelerated electrons inside the existing accelerator tunnel, and the other major components of the RHIC accelerator complex will live on as parts of the upcoming Electron-Ion Collider (EIC).

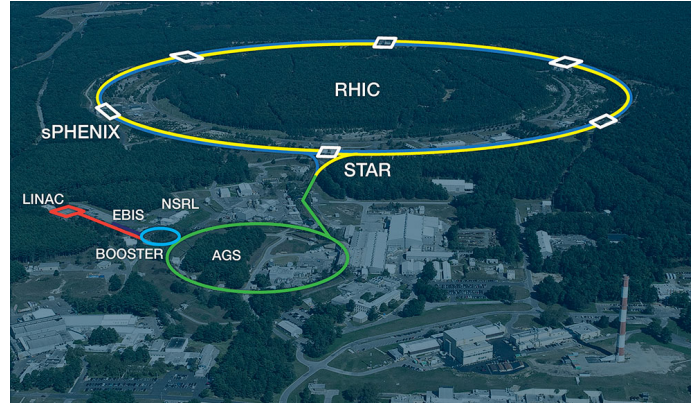


Figure 2.1: An aerial view of the Relativistic Heavy Ion Collider (RHIC), showing the locations of the facilities used for accelerating different ions, and the positions of the STAR and sPHENIX detectors. [12]

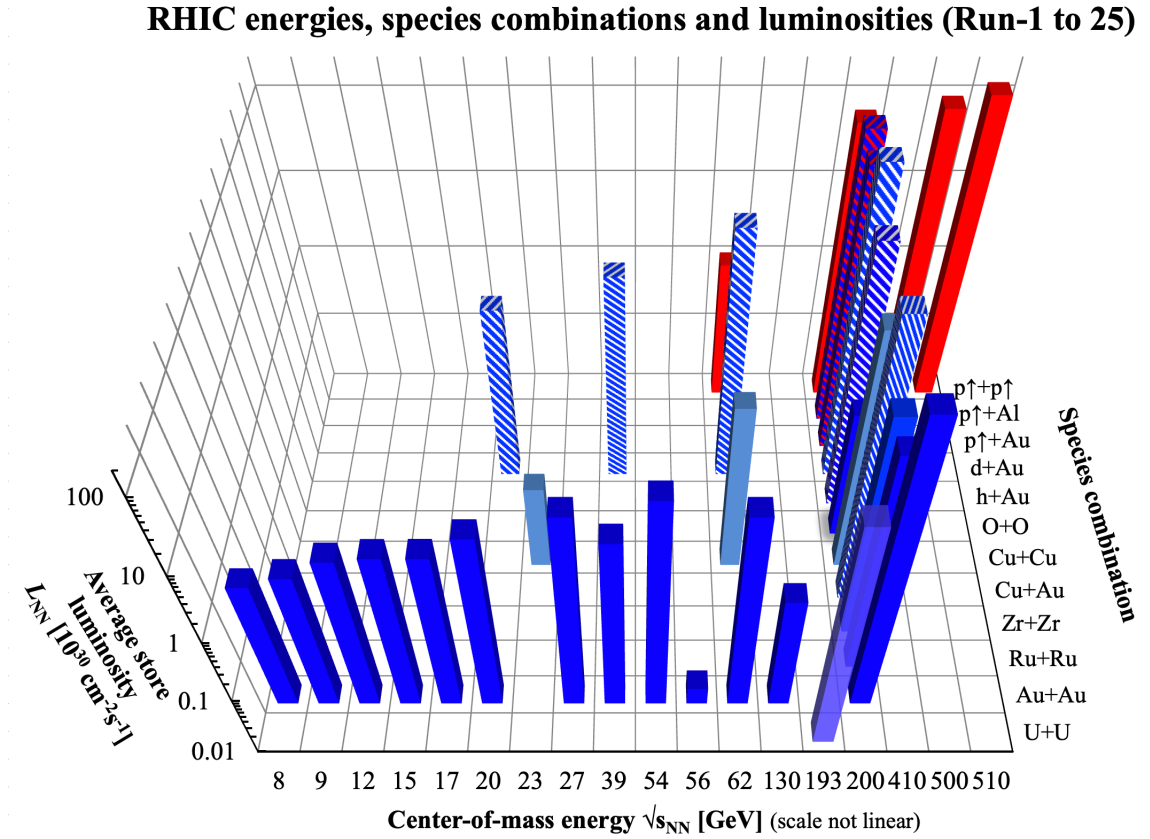


Figure 2.2: Summary of RHIC energies, species combinations, and luminosities for runs between 2000 and 2026. [13]

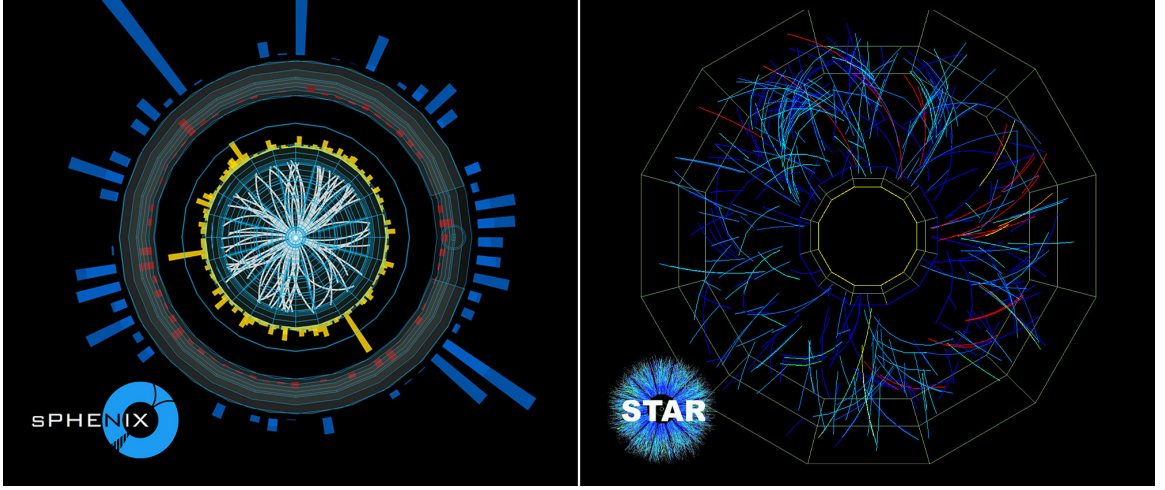


Figure 2.3: Some of the final O+O collisions at $\sqrt{s_{NN}} = 200$ GeV captured by the sPHENIX and STAR detectors at the RHIC. [14]

2.1.1 From stable matter to little bangs

Before collisions, the nuclei have to be created first by taking the corresponding atoms and stripping off all their electrons. That process looks different for different nuclei.

- **For protons** it is *Optically Pumped Polarized H^- Ion Source (OPPIS)* fed into the *Linear Accelerator (LINAC)*. The LINAC accelerates them to ~ 200 MeV and injects into the *Booster Synchrotron* at which point, the two electrons are stripped away by a carbon foil. They are accelerated to upto 2.5 GeV and fed into the *Alternating Gradient Synchrotron (AGS)* which accelerates the proton beam to about 25 GeV.
- **For heavy-ions** like $^{197}_{79}\text{Au}$, a *Laser Ion Source (LION)* extracts Au^+ ions using laser ablation, which go into *Electron Beam Ion Source (EBIS)* which ionizes them to Au^{32+} and accelerates them to 2 MeV per nucleon using electron beams. Then they enter the booster synchrotron which accelerates them to about 95 MeV per nucleon. While exiting from the synchrotron, the atoms are stripped of more electrons by passing them through a stripping foil, and atoms in the Au^{77+} state are fed to the AGS. The beam's kinetic energy up to 8.86 GeV per nucleon for these Au atoms and the final two K shell electrons are stripped.

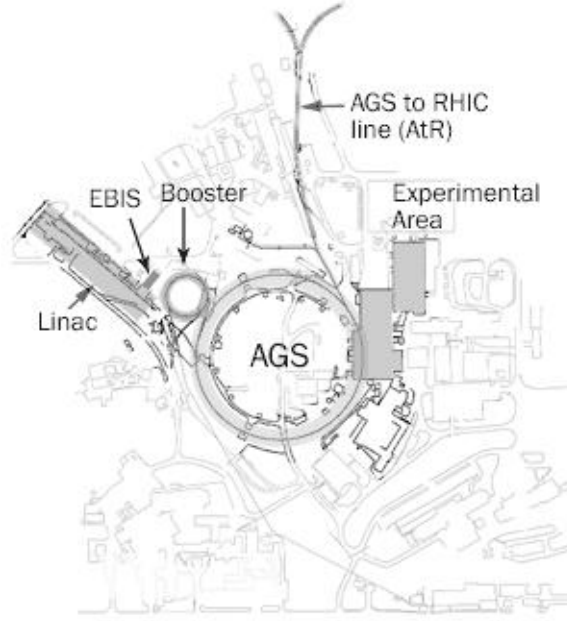


Figure 2.4: Schematic view of the AGS complex, showing various stages the ions go through before going into RHIC

when the ion beam reached top speed in the AGS, it was taken down another beamline called the AGS-to-RHIC (AtR) transfer line. At the end of this line, a switching magnet sent the ion bunches down each of two beamlines to fill RHIC's two ion storage rings. Bunches directed left filled the clockwise RHIC ring (blue) while those directed right traveled counterclockwise in the second RHIC ring (yellow). In the rings, superconducting radio-frequency (RF) cavities accelerate the ion bunches to energies of 100 GeV per nucleon for heavy ions (255 GeV for protons), with four cavities operating at 28.15 MHz to generate standing waves of electric fields for the acceleration and ten 197 MHz cavities maintaining beam stability during storage. Superconducting dipole magnets with 3.5 T fields keep the beams on a circular path, while quadrupole magnets focus the ion beams. A total of 1,740 superconducting magnets are used in the two rings at RHIC, cooled by liquid Helium at < 4.6 K. An aerial view of the RHIC complex with the different subsystems is shown in Fig 2.1.

There are 6 interaction regions (IR) around the RHIC complex, where the two beams cross and allow for detectors to be placed. 4 such detectors have existed during RHIC's

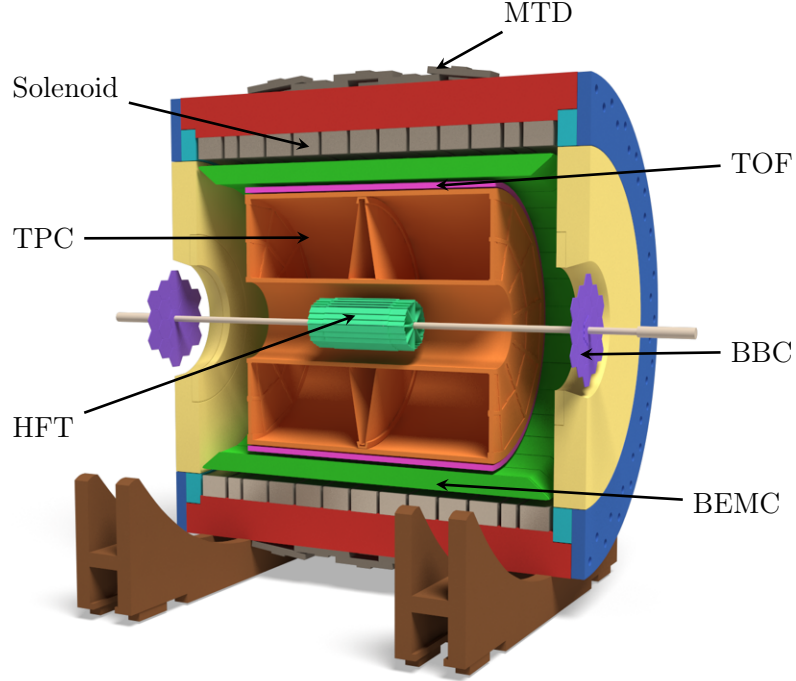


Figure 2.5: Schematic of the STAR detector, depicting the various detector subsystems.

operation: At 10 o'clock and 2 o'clock, the PHOBOS and BRAHMS experiments operated only for the first few years of RHIC (ending in 2006). At 8 o'clock, the PHENIX experiment collected data from 2000 to 2016 and was then transformed into the sPHENIX detector which began operations in 2023. STAR sits at the 6 o'clock position, and has been the only experiment running for all 25 years of RHIC's existence.

2.2 The STAR detector

STAR [99] is designed to provide hermetic coverage around the interaction point, emphasizing precise tracking capabilities in high multiplicity environments, for its main physics mission of obtaining a fundamental understanding of the microscopic structure of QCD interactions at high energy densities.

It is a highly versatile, multipurpose detector with several subsystems used for triggering, tracking and identifying the particle species for charged tracks, and calorimetry. These subsystems are surrounded by a room-temperature solenoid magnet 6.2 m long and 5.26 m

in diameter which generates a magnetic field of 0.5 T in the detector volume. The detector is cylindrical with full coverage in the azimuthal direction in the mid-rapidity region along with wide coverage in pseudorapidity. A schematic diagram of the STAR detector with its various detector subsystems is shown in Fig. 2.5. As the only experiment running continuously since the inception of RHIC, STAR has undergone major upgrades that have augmented its tracking capabilities and increased its acceptance in the forward rapidity regions. The main detectors used for this analysis are the *Time Projection Chamber* (TPC) and *Barrel Electromagnetic Calorimeters* (BEMC).

2.2.1 Detector Coordinate System

Before we get to the components of the STAR detector system, we will introduce the coordinate system used for the calculations involved. STAR is cylindrical around the beamline, so a convenient choice for z -axis is the beamline going east-west. $z = 0$ is set at the *Interaction Point* (IP), center of the detector volume, as shown in Fig. 2.6

Kinematics of ultra-relativistic particles are described by its four-momentum, (E, p_x, p_y, p_z) . However, these coordinates are not additive under Lorentz boosts in the z -direction, making calculations involving relativistic entities somewhat unwieldy. Since p_x, p_y are already invariant under Lorentz boosts along the z -axis, we can just convert them to the convenient cylindrical coordinates

$$p_T = \sqrt{p_x^2 + p_y^2}, \quad \varphi = \arctan\left(\frac{p_y}{p_x}\right) \quad (2.1)$$

Energy (E) can be replaced with the particle's mass (m), or transverse mass $m_T = \sqrt{p_T^2 + m^2}$. We can make the z -coordinate additive under Lorentz Boosts along z -axis by representing a particle's direction along the z -axis by *rapidity*

$$y = \frac{1}{2} \ln \frac{E + p_z}{E - p_z} \quad (2.2)$$

However since y is mass-dependent and most detectors don't provide robust particle

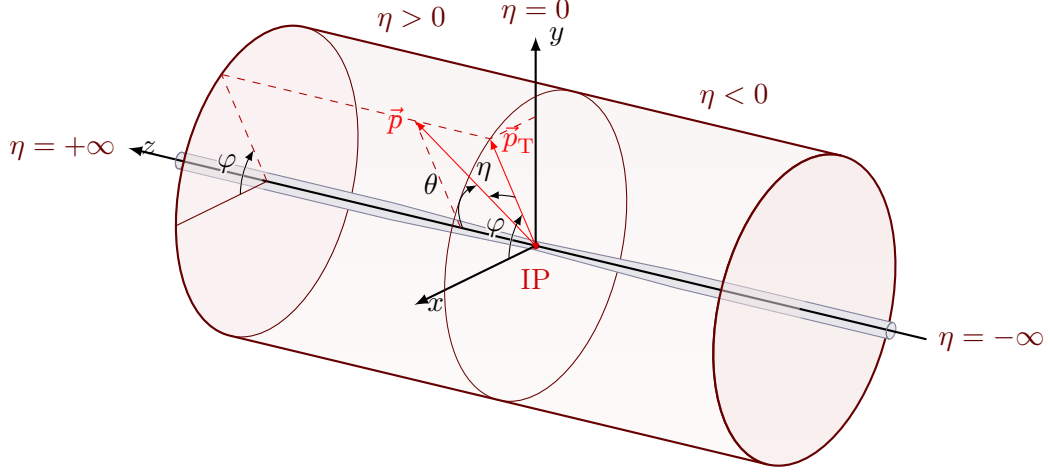


Figure 2.6: The (p_T, η, φ) coordinate system, drawn for a cylindrical collider detector. The beam axis defines z and the interaction point (IP) sits at the origin. A particle of momentum $\vec{p}$ is described by its transverse momentum $\vec{p}_T$, its azimuthal angle φ in the transverse plane, and its pseudo-rapidity η , fixed by the polar angle θ with respect to the beam.

identification at all kinematic ranges. We usually define the z -direction in a detector using *Pseudorapidity*, which is the massless limit of rapidity

$$\eta = \lim_{m \rightarrow 0} y = \frac{1}{2} \ln \frac{p + p_z}{p - p_z} = -\ln \left(\tan \frac{\theta}{2} \right) \quad (2.3)$$

Where θ is the polar angle between the particle's direction and the z -axis as shown in Fig. 2.6. Thus a particle's four-momentum is given by $(m_T, p_T, \eta, \varphi)$ or (p_T, η, φ, m) . See Appendix B.2 for a more detailed derivation of coordinates.

2.2.2 TPC

Time Projection Chamber (TPC) [16] was the the primary tracking detector for STAR, providing particle identification and momentum information for charged tracks by measuring their ionization energy loss (dE/dx) inside the TPC volume. As shown in Fig. 2.7 the TPC was a cylindrical detector 420 cm long with an outer (inner) radii of 200 (50) cm, concentric to the 0.5 T solenoid. Its acceptance covered a pseudorapidity range of $|\eta| < 1.0$, along with full azimuthal coverage. Particle momenta could be measured accurately over a range of 0.1

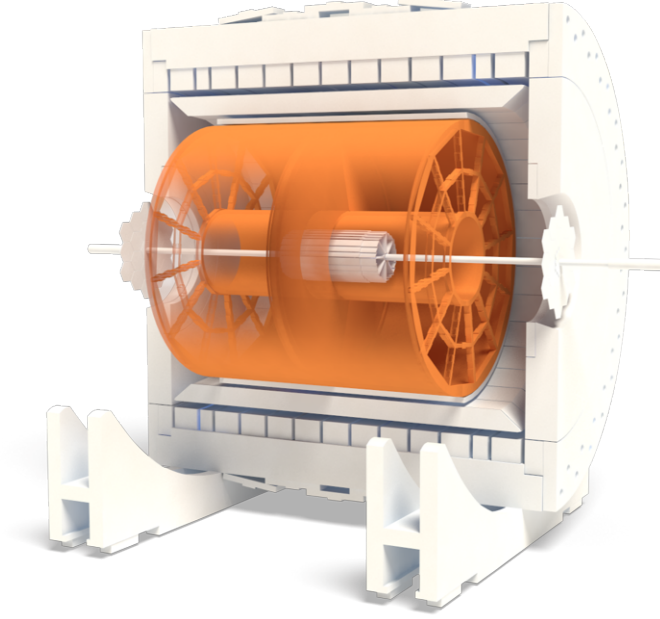


Figure 2.7: The Time Projection Chamber at STAR. [15].

to 30 GeV/ c , and particle identification was limited between 0.1 and 1 GeV/ c . The TPC volume contained an ionizable gas mixture called P10 (10 % methane, 90 % argon), chosen for its high drift velocity of 5.45 cm/ μ s in the well-defined, uniform electric field of ≈ 135 V/cm at STAR. The uniform electric field is defined between the thin conductive central membrane at $z = 0$ maintained at 28 kV acting as a cathode, and the two end caps each 210 cm away from the center at 0 V acting as anodes. The drift velocity can change over time based on atmospheric pressure fluctuations, therefore, calibrations are typically done using lasers every few hours during data-taking. To ensure minimal contamination by impurities like oxygen and water vapor maintained at 2 mbar above atmospheric pressure.

The charged particles leave ionization trails were captured by Multi-Wire Proportional Counters (MWPCs) with a segmented pad plane used in TPC readout. As particles ionize the TPC gas, freed electrons drift toward the MWPCs under the effect of the electric field. Signal contamination was prevented using a gated grid that initially restricted electron flow, opening only if a collision is deemed significant, allowing electrons to enter the MWPCs. Positive ions generated inside the MWPC are prevented from entering the TPC's drift

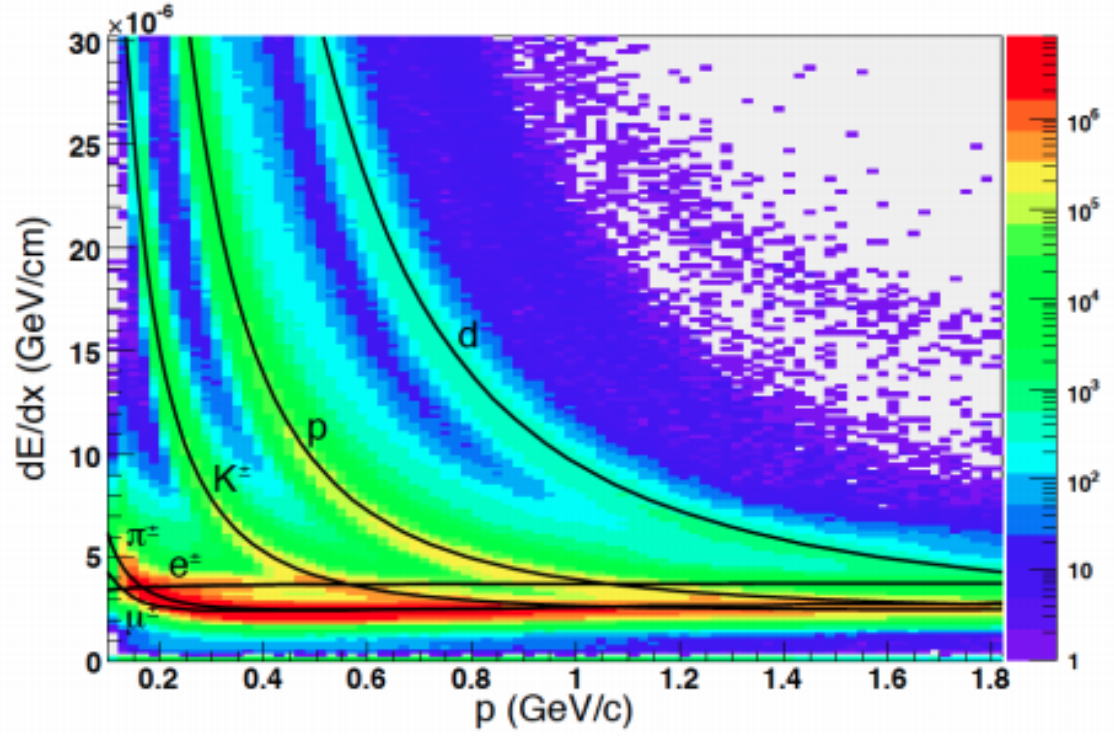


Figure 2.9: The ionisation energy loss for charged particles from TPC as a function of momenta. The letters represent different particles: $\mu^\pm$ for muons, $e^\pm$ for electrons, $\pi^\pm$ for pions, $K^\pm$ for Kaons, p for protons, and d for deuterons. The solid lines are the expected value of dE/dx for the corresponding particle species from the Bichsel function [17].

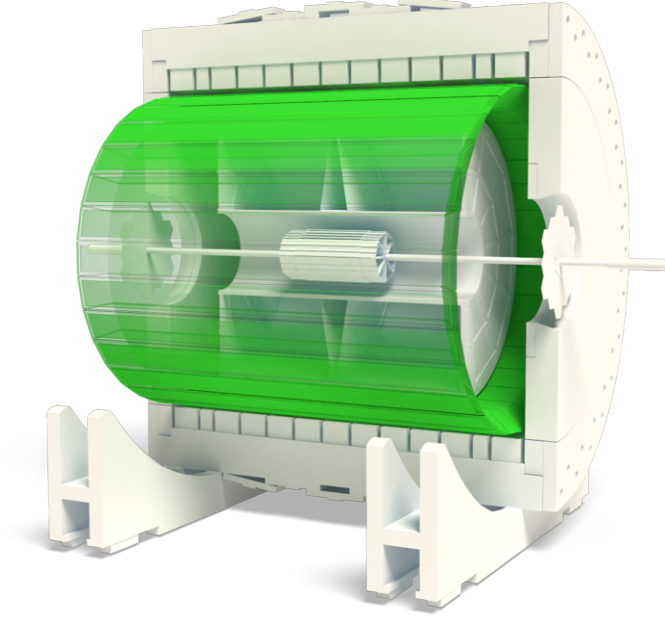


Figure 2.10: The Barrel Electromagnetic Calorimeter at STAR. Taken from [15].

function [17] which are shown as solid lines in Fig. 2.9. For each particle species, the band has non-zero thickness due to the finite momentum resolution, and selection criteria based on simulations are used to assign particle ID. The particle identification from TPC is limited to low momenta tracks (about 1 GeV/c) only beyond which the bands for different species start to overlap.

2.2.3 BEMC

The STAR Barrel Electromagnetic Calorimeter (BEMC) [100] designed to measure particle energies, particularly for electron and neutral particle identification. BEMC was a fast detector, used as a trigger to select events containing a hard process. It also provided the neutral component of the jet energy that couldn't be reconstructed by the TPC in jet-related measurements. As shown in Fig. 2.10, it was cylindrical detector located between the TPC and magnet coils. It covers the pseudorapidity range $|\eta| < 1$ and the full azimuth range ($0 \leq \phi \leq 2\pi$), matching the TPC's coverage.

The BEMC had inner and outer radii of 223.5 cm and 263 cm respectively and consisted

of 120 modules, each containing 40 towers, totaling 4800 towers. Each of the towers covers a range of $\Delta\eta \times \Delta\phi = 0.05 \times 0.05$, approximately $10 \text{ cm} \times 10 \text{ cm}$ near $\eta = 0$, growing slightly with increasing $|\eta|$ due to their projective geometry toward the center of the detector ($z = 0$). Each tower was built with 20 alternating layers of 5-mm thick lead, acting as absorbers to initiate electromagnetic showers, and 21 layers of plastic scintillators, which detect the photons produced by these showers. Together, the layers created a depth of around 20 radiation lengths (X_0), allowing the calorimeter to capture nearly all of the energy from electromagnetic showers. This depth is enough to contain 95 % of the energy from electrons and photons above 1 GeV. Within these layers, the shower's energy was most concentrated around 5-7 radiation lengths in, where the Shower Maximum Detector (SMD), a gas (90 % Argon, 10 % CO_2)-filled proportional counter improved spatial resolution to help differentiate between electromagnetic and hadronic shower profiles. The BEMC has an energy resolution that improves with particle energy, around 17% for 1.5 GeV electrons and 10% for 3 GeV electrons in head-on Au + Au collisions. This follows a typical calorimetric trend where energy resolution decreases as energy increases ($\sigma_E/E \approx 1.5\% + 15\%/\sqrt{E}$), providing complementarity with TPC where the resolution increases with increasing momentum ($\sigma_{p_T}/p_T \propto p_T$).

Chapter 3

Jet-hadron correlations relative to the event plane

In this analysis, the coordinate system used to depict the distribution of associated particles is defined relative to a reconstructed jet, also called a *trigger jet*. The distribution is thus given by:

$$\frac{1}{N_{\text{trig}}} \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta}, \quad (3.1)$$

where N_{trig} is the number of trigger jets, $N_{\text{assoc,jet}}$ is the number of associated particles, $\Delta\phi$ ($= |\phi_{\text{jet}} - \phi_{\text{assoc}}|$) is the azimuthal angle of the associated particles relative to the trigger jets, and $\Delta\eta$ ($= |\eta_{\text{jet}} - \eta_{\text{assoc}}|$) is the difference in the pseudorapidities of the trigger jet and associated particle. When studying the $\Delta\phi$ distributions, a peak is observed at $\Delta\phi = 0$ which is the triggered jet, referred to as the *near-side*. Another peak which is seen near $\Delta\phi = \pi$ is referred to as the *away-side*. The away-side consists of the hadrons associated with the trigger jet and make up the second jet which results from the early hard-scatter.

The goal of this analysis is to study the conditional yield of associated particles, the width of the near- and away-side peaks (quantified using the gaussian width) as a function of the angle between the jet axis and the event plane. The yield is estimated by:

$$\text{Yield} = \frac{1}{N_{\text{trig}}} \int_c^d \int_a^b \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta} d\Delta\phi d\Delta\eta. \quad (3.2)$$

The integration limits are chosen based on practical considerations, including the detector acceptance and binning of histograms.

3.1 Analysis Method

This measurement utilizes data collected during the 2014 run from Au+Au collisions at nucleon-nucleon center-of-mass energy of $\sqrt{s_{\text{NN}}} = 200$ GeV by the STAR experiment [101] at RHIC. Events referred to as signal (same) events are required to contain a HT trigger in the BEMC [102]. Minimum-bias (MB) triggered events based on coincidence of Zero Degree Calorimeters (ZDC coincidence), Vertex Position Detectors and Beam-Beam Counters signals are used to estimate the pair-acceptance effects via a mixed-event (ME) technique [103]. For this analysis, 9.4M HT-triggered and 4.0M MB collision events in the 20–50% centrality range are used.

3.1.1 Data set and event selection

The data consist of the mid- and low-luminosity subsets of the 2014 Au+Au production (830 and 548 runs, respectively). Runs are rejected when run-averaged quantities of the MB events, namely the mean primary-vertex position along the beam $\langle v_z \rangle$ and in the transverse plane $\langle V_r \rangle$, the ZDC coincidence rate $\langle \text{ZDCx} \rangle$, the reference multiplicity $\langle \text{refMult} \rangle$, and the mean track transverse momentum, lie more than three standard deviations from their average over all runs; such outliers indicate detector or beam conditions that are not representative of the data set. The HT2 trigger requires a single BEMC tower with a transverse energy above approximately 5.4 GeV.

Events are required to have a reconstructed primary vertex within $|v_z| < 24$ cm of the center of the detector. The pseudorapidity acceptance of the TPC changes with the vertex position, and the MB sample used for the event mixing (Sect. 3.1.5) contains very few events at large $|v_z|$, which would limit the precision of the acceptance correction; the value of 24 cm was chosen by balancing the available statistics against the size of the resulting systematic uncertainties. Events containing a track with $p_T > 30$ GeV/ c are rejected, since such tracks are dominated by cosmic rays and mis-reconstructed tracks for which the TPC momentum resolution is poor.

The centrality of each event, defined in Sect. 1.6.2, is determined from the charged-particle multiplicity measured in the TPC within $|\eta| < 0.5$ (refMult) using the standard STAR procedure: refMult is corrected event-by-event for its dependence on the vertex position and on the luminosity, and the multiplicity distribution of the most peripheral events is corrected for the trigger inefficiency of the VPD using a Monte Carlo Glauber simulation. The 20–50% centrality range used here corresponds approximately to corrected refMult values of 75–250. To increase the number of events available for the event mixing by more than 60%, events from two MB triggers with different online VPD vertex requirements (VPDMB5 and VPDMB30) are combined. The VPDMB5 events are reweighted in bins of refMult and v_z so that their multiplicity distributions match those of the VPDMB30 events, which compensates for the different VPD resolution and the offset of the online VPD vertex cut, and they are only used for $|v_z| < 16$ cm; events in which both triggers fired are used only once.

3.1.2 Track and tower selection

Charged tracks are reconstructed in the TPC and are required to be primary tracks, i.e. to be consistent with originating from the primary vertex. The selection criteria, listed in Tab. 3.1, are designed to reject background tracks and to ensure a good momentum measurement. The distance of closest approach (DCA) of the track to the primary vertex is required to be below 3 cm, which suppresses secondary particles from weak decays and interactions with the detector material, as well as tracks from pile-up collisions. A minimum of 15 fit points in the TPC ensures a reliable curvature, and hence momentum, measurement, while requiring that at least 52% of the possible fit points be used removes *split tracks*, a single particle reconstructed as two track segments that would otherwise be counted twice. Tracks below 0.2 GeV/ c curl up in the magnetic field before crossing enough pad rows to be reconstructed reliably, and tracks above 30 GeV/ c have a poor momentum resolution. The acceptance $|\eta| < 1.0$ matches that of the BEMC. Only tracks with $p_T > 2.0$ GeV/ c are used in the jet reconstruction (Sect. 3.1.4), while all selected tracks with $p_T > 1.0$ GeV/ c serve as associated

particles.

Table 3.1: Track selection criteria.

Criterion	Value
Primary track	yes
DCA to the primary vertex	< 3 cm
Number of fit points, $N_{\text{hits,fit}}$	≥ 15
$N_{\text{hits,fit}}/N_{\text{hits,poss}}$	≥ 0.52
Pseudorapidity	$ \eta < 1.0$
Transverse momentum	$0.2 < p_T < 30$ GeV/ c

The single-track reconstruction efficiency, $\epsilon(p_{T,\text{assoc}}, \eta_{\text{assoc}})$ in Eq. 3.5, is the probability that a charged particle within the acceptance is reconstructed and passes the track selection. It is determined from simulated pions, kaons and protons embedded into real Au+Au events, passed through the GEANT simulation of the STAR detector and the full reconstruction chain, as the fraction of embedded particles that are matched to a reconstructed track. Since the efficiency decreases with the detector occupancy, it is evaluated as a function of p_T and η in bins of centrality and of luminosity. It is approximately constant above 5 GeV/ c , so the value at 4.5 GeV/ c is used for all tracks above 5 GeV/ c . The correction is applied track-by-track to the associated hadrons in both the same and the mixed events. A relative uncertainty of 5% is assigned to the efficiency; it accounts for the imperfect description of the TPC response in the embedding and is the dominant contribution to the global scale uncertainty of the results (Sect. 3.1.8).

The BEMC towers are required to have $0.2 < E_T < 30$ GeV and $|\eta| < 1.0$, with the tower transverse energy calculated with respect to the primary vertex. Towers that are hot (firing too often or with too large energies), cold, or dead during the run period are excluded using a list of 452 bad towers, determined from their firing rates and energy distributions. In addition, events in which the HT2 trigger was fired by one of these bad towers are rejected, so that a bad tower can neither contribute to a jet nor trigger an event.

Charged particles that leave a track in the TPC also deposit energy in the BEMC: charged hadrons as minimum-ionizing particles or partial hadronic showers, and electrons

with nearly all of their energy. Without a correction this energy would be counted twice in the jet reconstruction. Following Ref. [19], the momenta of all tracks that extrapolate to a given tower are subtracted from its energy,

$$E_{\text{tower}}^{\text{corr}} = E_{\text{tower}} - f_{\text{had}} \sum_{\text{matched}} p c, \quad (3.3)$$

with $f_{\text{had}} = 1$, and the tower is removed if its corrected energy falls below the threshold required for the jet constituents. Although subtracting the full track momentum over-corrects the charged hadrons, which deposit on average only about a third of their energy in the BEMC, it removes the large tower-by-tower fluctuations of the deposited energy and therefore improves the jet energy resolution.

3.1.3 Event plane reconstruction

The event plane ($\Psi_{2,EP}$), the experimental estimate of the reaction plane described in App. C, is calculated event-by-event using the Modified Reaction Plane (MRP) method [18, 104, 105] using charged tracks with $0.2 < p_{T,\text{track}} < 1.0$ GeV/ c measured within the TPC. The impact of highly energetic jets on the calculation of the event-plane orientation is reduced by removing the particles within the pseudorapidity strip ($|\Delta\eta| < 0.4$) across $\Delta\phi$ surrounding the leading jet. This procedure also removes a significant portion of the away-side jet, located opposite in azimuth. The event plane is obtained from the p_T -weighted second-order flow vector $\mathbf{Q}_2$ of these tracks (Eqs. C.1 and C.2); the p_T weights are used because the flow coefficients grow approximately linearly with p_T at low p_T . Restricting the event-plane tracks to $p_T < 1.0$ GeV/ c also avoids auto-correlations with the associated hadrons, which are measured above 1.0 GeV/ c . The non-uniform acceptance and efficiency of the TPC make the distribution of the reconstructed $\Psi_{2,EP}$ non-flat; this is corrected by re-centering $\mathbf{Q}_2$ and by the shift method up to the 20th order (App. C), with both corrections determined separately for 10% centrality intervals and 4-cm v_z intervals.

The distributions of the associated tracks relative to the trigger jet are measured in three bins in the angle between the trigger jet and the event plane,

$$\begin{aligned}
 \text{in-plane:} \quad & |\Psi_{2,EP} - \phi_{\text{jet}}| < \pi/6, \\
 \text{mid-plane:} \quad & \pi/6 < |\Psi_{2,EP} - \phi_{\text{jet}}| < \pi/3, \\
 \text{out-of-plane:} \quad & |\Psi_{2,EP} - \phi_{\text{jet}}| > \pi/3,
 \end{aligned} \tag{3.4}$$

where the angle difference is folded into $[0, \pi/2]$. The analysis is restricted to 20–50% central Au+Au collisions to achieve the highest event-plane resolution and therefore the analysis will be most sensitive to any path-length dependencies.

The second- (fourth-) order event-plane resolutions relative to the second-order event plane ($R_2\{\Psi_{2,EP}\}$ and $R_4\{\Psi_{2,EP}\}$ respectively) as a function of collision centrality are shown in Fig. 3.1. The resolution is peaked around the 20-30% and 30-40% centrality. Because of the finite number of particles, the reconstructed event plane only approximates the true symmetry plane; the resolution $R_n\{\Psi_{2,EP}\}$ (Eq. C.4) quantifies how well it does so. It is determined from the data with the two-sub-event method (Eqs. C.5 and C.6), in narrow centrality intervals that are combined over 20–50% weighted by the number of events. For 20–50% central events, the event-plane resolutions $R_2\{\Psi_{2,EP}\}$ and $R_4\{\Psi_{2,EP}\}$ are 0.56 and 0.28, respectively. The statistical uncertainties of the resolution are below 1%; varying the resolution by $\pm 1\%$ has a negligible effect on the final $\Delta\phi$ correlations. Measured values for the event-plane resolution are in good agreement with prior STAR studies [18]. These resolutions are evaluated to correct the observed flow coefficients which arise in the fits of the combinatorial background discussed in Sect. 3.1.6.

The finite resolution also mixes jets between the three orientation bins, which reduces any event-plane dependence of the signal itself. This effect is not corrected for, since an unfolding of the orientation bins, or a Fourier decomposition of the signal, would require a precision the measurement does not have and would add further uncertainties. Its size can instead be estimated by writing the event-plane dependence of the signal as a Fourier series

in $\phi_{\text{jet}} - \Psi_{2,EP}$ with coefficients v_n^{quench} , which the finite resolution reduces to $R_n v_n^{\text{quench}}$. Integrating over the orientation bins of Eq. 3.4 and keeping the dominant $n = 2$ term gives $Y_{\text{out}}/Y_{\text{in}} \approx 1 - (6\sqrt{3}/\pi)R_2 v_2^{\text{quench}}$. For an anisotropy of the jet constituents of $v_2^{\text{quench}} = 0.1$, comparable to the measured anisotropy in the number of jets, the out-of-plane to in-plane (mid-plane to in-plane) yield ratio would be 0.67 (0.83) with a perfect event plane, but only 0.81 (0.91) with $R_2 = 0.56$. The finite resolution therefore pulls any event-plane dependence of the yields toward unity, which must be kept in mind when interpreting the yield ratios in Sect. 3.2.

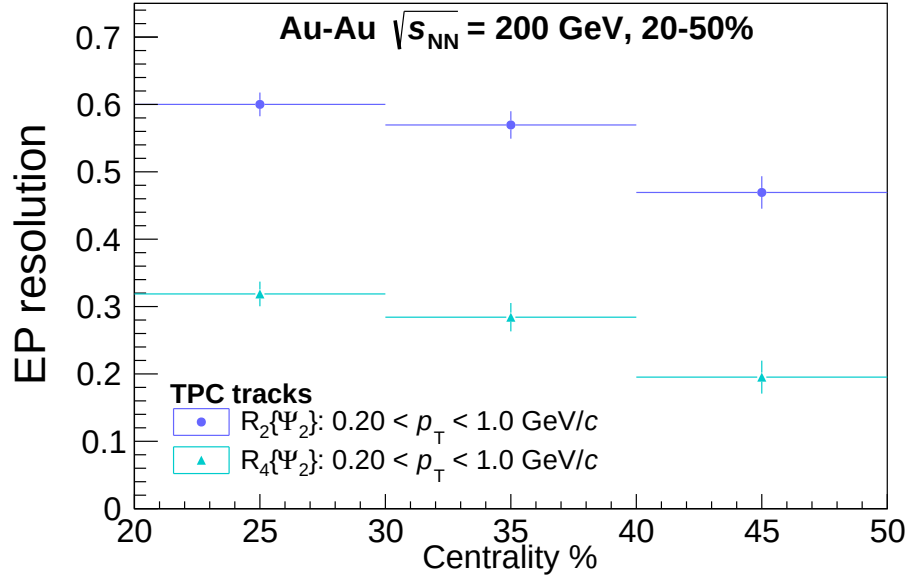


Figure 3.1: Event-plane resolution: Second-order (fourth-order) harmonic relative to the event plane, $R_2(\Psi_2)$ ($R_4(\Psi_2)$), respectively. The approach follows the Modified Reaction Plane (MRP) method [18] utilizing the charged tracks of the TPC for event-plane reconstruction and resolution calculation for tracks ranging from 0.2-1.0 GeV/c. Combined over 20–50% centrality, $R_2 = 0.56$ and $R_4 = 0.28$.

3.1.4 Jet reconstruction and selection

Full jets are reconstructed by measuring charged-tracks in the TPC and collecting neutral-particle information from the BEMC. The anti- k_T algorithm clusters these particles into jets by reconstructing the jet momenta as the quadratic sum of their constituent momenta using

a boost-invariant p_T^2 recombination scheme. Tracks used for reconstructing jets are assumed to be pions while the towers to have arisen from massless particles. All jets measured in this work are clustered with resolution parameter of $r = 0.4$. The area of jets, A_{jet} , is found with FastJet using active ghost particles [106]. Partially reconstructed jets at the edge of the acceptance are rejected by applying a fiducial cut, $|\eta_{\text{jet}}| < 1.0 - r$, to assure all jets fall within the acceptance of the detectors.

Jets produced in heavy-ion collisions sit on top of a large amount of underlying event. The jet signal can be found beneath tens to hundreds of particles resulting from various other processes. To reduce the influence of these background particles, this analysis requires tracks (towers) with $p_T(E_T) > 2.0 \text{ GeV}/c$ for jet reconstruction. At RHIC energies this selection reduces the median background transverse-momentum density per unit A_{jet} , $\langle \rho \rangle$, down to $\approx 0.6 \text{ GeV}/c$. This high-constituent selection, referred to as a “hard-core” jet selection reduces fluctuations, fake jets, and background jet particles [107]. To further reduce contributions from the background and to match the trigger condition, the jets are required to contain a constituent tower that fired the HT trigger ($E_{T,\text{tower}} > 5.4 \text{ GeV}$). Jets containing a track with $p_T > 30 \text{ GeV}/c$ are rejected, as such tracks are likely to be misreconstructed. The remaining jets are studied in classes of jets with $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$ and $20 < p_{T,\text{jet}} < 40 \text{ GeV}/c$. The numbers of trigger jets in the two classes are shown in Fig. 3.2 for each orientation relative to the event plane. There are about 31 000 trigger jets with $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$ in 20–50% central collisions, and more jets are found in-plane than out-of-plane. This ordering is expected both from the path-length dependence of jet quenching and from the larger in-plane background, which adds more momentum to in-plane jets and moves more of them above the selection threshold; the latter effect is addressed by the jet-energy-shift correction (Sect. 3.1.8).

No event-by-event subtraction of the underlying event, ρA_{jet} , is applied to the jet momenta: with $2 \text{ GeV}/c$ constituents, too few background k_T jets are found in an event to

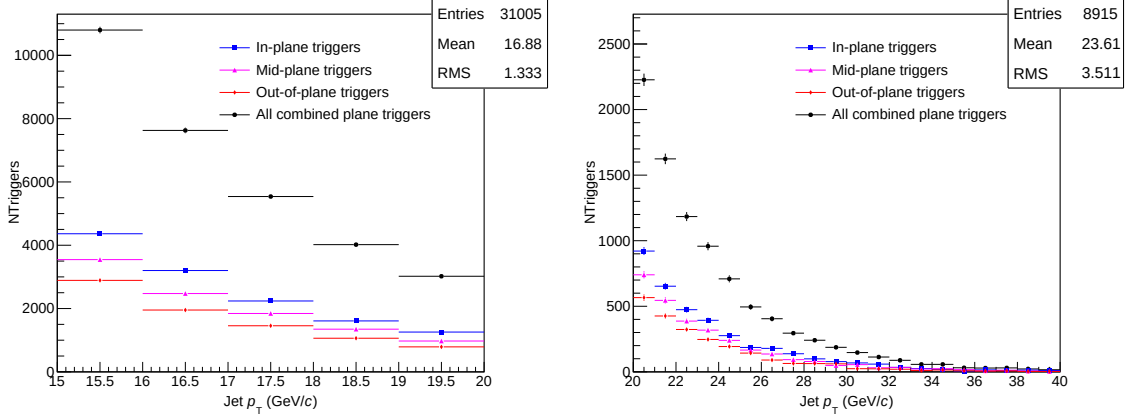


Figure 3.2: Number of trigger jets as a function of the reconstructed $p_{T,\text{jet}}$ for $15 < p_{T,\text{jet}} < 20$ GeV/ c (left) and $20 < p_{T,\text{jet}} < 40$ GeV/ c (right) full jets in 20–50% central Au+Au collisions, for in-plane (blue), mid-plane (magenta) and out-of-plane (red) jets and all orientations combined (black).

estimate ρ reliably, and the remaining background contribution is small. The residual background and its dependence on the jet orientation relative to the event plane are instead accounted for by the jet-energy-shift correction described in Sect. 3.1.8. The hard-core and trigger-tower requirements select jets that fragmented hard and lost little energy, which favors jets produced close to the surface of the medium and heading outward. This *surface bias* of the trigger jet increases, on average, the path length of the recoiling away-side jet through the medium.

3.1.5 Jet-hadron correlation

The measurement of the correlation function (distribution of charged hadrons relative to reconstructed jets) described in Eqn. 3.1 requires several corrections. The correlation function is measured in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$) as:

$$\frac{1}{N_{\text{trig}}} \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta} = \frac{1}{N_{\text{trig}}} \frac{1}{\epsilon(p_{T,\text{assoc}}, \eta_{\text{assoc}})} \frac{1}{a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)} \left(\frac{d^2 N_{\text{assoc,jet}}^{\text{meas}}}{d\Delta\phi d\Delta\eta} - \frac{d^2 N_{\text{assoc,jet}}^{\text{bkgd}}}{d\Delta\phi d\Delta\eta} \right). \quad (3.5)$$

$N_{\text{assoc,jet}}$ gives the number of pairs of trigger jets and the associated hadrons, $N_{\text{assoc,jet}}^{\text{meas}}$

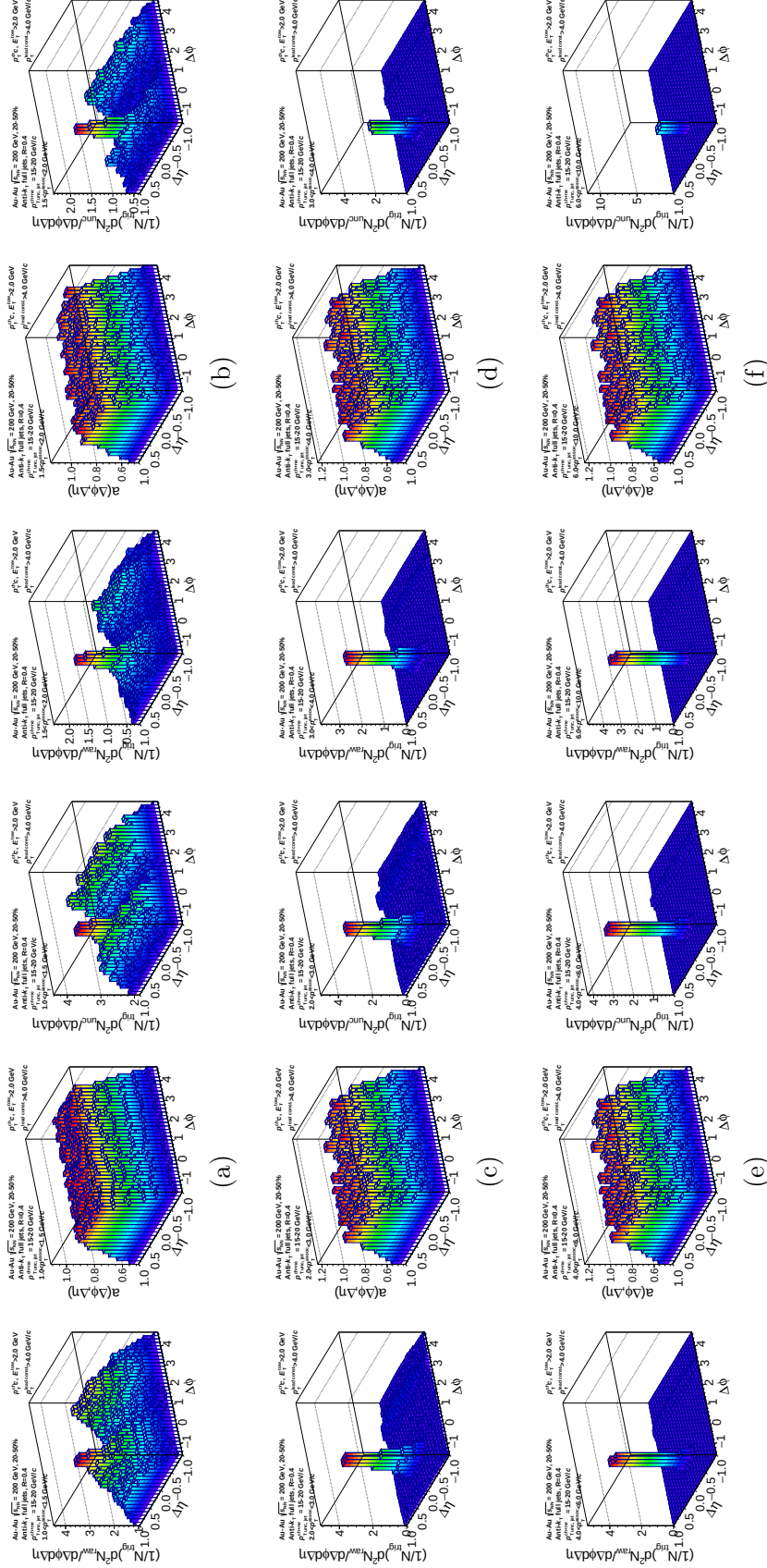


Figure 3.3: Two-dimensional jet-hadron correlations in $\Delta\phi$ and $\Delta\eta$ for in-plane $15 < p_{T,jet} < 20$ GeV/c jets in 20–50% central Au+Au collisions, for the associated-hadron momenta (a) $1.0 < p_{T,assoc} < 1.5$ GeV/c, (b) $1.5 < p_{T,assoc} < 2.0$ GeV/c, (c) $2.0 < p_{T,assoc} < 3.0$ GeV/c, (d) $3.0 < p_{T,assoc} < 4.0$ GeV/c, (e) $4.0 < p_{T,assoc} < 6.0$ GeV/c and (f) $6.0 < p_{T,assoc} < 10.0$ GeV/c. Each group shows, from left to right, the raw same-event distribution per trigger jet, the normalized mixed-event pair acceptance $a(\Delta\phi, \Delta\eta)$, whose trapezoidal shape in $\Delta\eta = \eta_{jet} - \eta_{assoc}$ follows from $|\eta_{assoc}| < 1.0$ and $|\eta_{jet}| < 0.6$, and the acceptance-corrected correlation function, which still contains the combinatorial background.

and $N_{\text{assoc,jet}}^{\text{bkgd}}$ are the number of pairs measured and the pair characterized as background respectively. $\epsilon(p_{\text{T,assoc}}, \eta_{\text{assoc}})$ is the single-track reconstruction efficiency¹. The pair acceptance, $a(p_{\text{T,assoc}}, \Delta\phi, \Delta\eta)$, is calculated from the raw pairs that we measure from a trigger jet associated with charged hadrons from mixed events.

The correlations are determined in bins of centrality, reconstructed trigger-jet transverse momentum ($p_{\text{T,jet}}$), associated-hadron transverse momentum ($p_{\text{T,assoc}}$), and bins of the trigger jet relative to the event plane (in, mid, out, all combined angles) defined in Eq. 3.4. Figure 3.3 shows, for in-plane $15 < p_{\text{T,jet}} < 20$ GeV/ c jets, how the raw same-event distributions are turned into acceptance-corrected correlation functions by the mixed-event correction described below. The corrected correlation functions contain a large combinatorial background ($N_{\text{assoc,jet}}^{\text{bkgd}}$), which must be subtracted. This subtraction procedure is described in Sect. 3.1.6 .

Event mixing and pair acceptance

The pair-acceptance correction, $1/a(p_{\text{T,assoc}}, \Delta\phi, \Delta\eta)$, accounts for the detector and tracking inefficiencies, and is found by correlating jets from HT-triggered events with associated hadrons from MB events of the same event class. In addition to providing the acceptance correction, the mixed-event procedure will also help remove the trivial correlation due to an η dependence in the single-particle track distributions [108]. Taking the ratio of the single-event pairs by the mixed-event pairs removes the detector effects of acceptance and efficiency. The pair acceptance will serve as the dominant effect, given that there is little η dependence in both the tracks and jets across the acceptance range of this analysis. Since the trigger jets are restricted to $|\eta_{\text{jet}}| < 0.6$ and the associated hadrons to $|\eta_{\text{assoc}}| < 1.0$, the number of possible pairs is constant for $|\Delta\eta| < 0.4$ and falls linearly to zero at $|\Delta\eta| = 1.6$, which gives the pair acceptance its trapezoidal shape in $\Delta\eta$, illustrated with a toy model in Fig. 3.4 and visible in the measured mixed-event distributions of Fig. 3.3.

¹Acceptance effects such as gaps between sector boundaries are included in this correction, however, we refer to $\epsilon(p_{\text{T,assoc}}, \eta_{\text{assoc}})$ as an efficiency correction in order to distinguish it from $a(p_{\text{T,assoc}}, \Delta\phi, \Delta\eta)$.

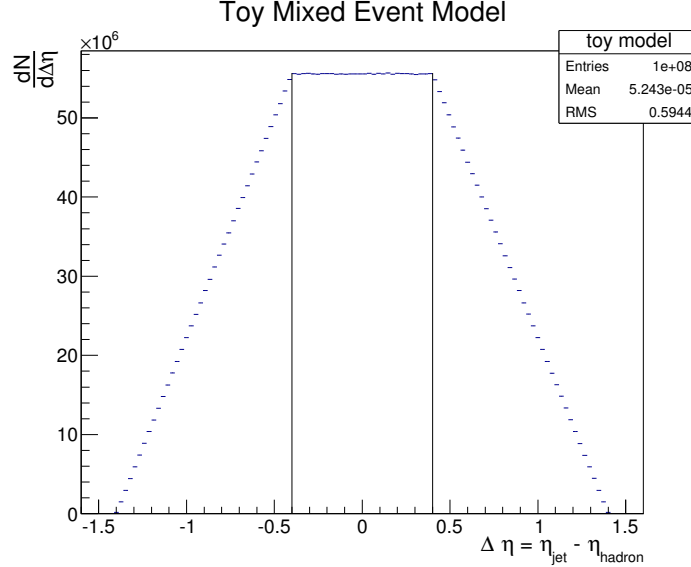


Figure 3.4: Toy model of the pair acceptance in $\Delta\eta = \eta_{\text{jet}} - \eta_{\text{hadron}}$ for uncorrelated trigger jets and hadrons, each uniformly distributed in pseudorapidity (here $|\eta_{\text{jet}}| < 0.5$ and $|\eta_{\text{hadron}}| < 0.9$). The number of pairs is constant where the smaller acceptance is fully contained in the larger one and falls linearly to zero at the largest possible separation. The analysis uses the plateau region $|\Delta\eta| < 0.4$ to normalize the measured mixed-event distributions.

Because the trigger jet and the associated hadrons of a mixed pair come from different events, they are correlated only through this acceptance.

The mixing is done with a pool of MB events, binned in 10% centrality and 4-cm v_z intervals, that holds up to 25 000 tracks; each trigger jet is correlated with the tracks of all events in the pool of its event class, and the mixed-event pairs are weighted by the inverse of the number of mixed events.

The event-mixing procedure used in this analysis is well described in Ref. [103]. The mixed events used for calculating $a(p_{\text{T,assoc}}, \Delta\phi, \Delta\eta)$ in this work are required to be within the same 10% centrality class and to have a vertex position within 4 cm along the direction of the beam (z). They are constructed separately for 20-30%, 30-40%, and 40-50% centrality classes and combined accordingly. High-momentum tracks are nearly straight, so the detector acceptance does not change significantly at high- $p_{\text{T,assoc}}$, and thus all associated

momentum bins greater than 2.0 GeV/ c are combined to increase statistics. There is no difference in efficiency and acceptance within uncertainties for different orientations of the jet relative to the event plane, and therefore the same correction $a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)$ is applied for all angles relative to the event plane. The acceptance $a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)$ is normalized to 1 at its maximum, determined using the region of approximately constant acceptance ($|\Delta\eta| < 0.4$). For each $p_{T,\text{assoc}}$ bin, the projection of the flat plateau region (integrated over the region $|\Delta\eta| < 0.4$) was fit with a constant over the whole $\Delta\phi$ range. The associated uncertainties of the fits were used for the systematic uncertainty on the mixed-event normalization, which is added in quadrature and reported as the scale uncertainty. This systematic uncertainty is under 1% (1.25%) in all $p_{T,\text{assoc}}$ bins used for the reported results of jets with $15 < p_{T,\text{jet}} < 20$ GeV/ c and $20 < p_{T,\text{jet}} < 40$ GeV/ c .

A shape uncertainty in $\Delta\phi$ due to the z_{vtx} binning, similar to that in $\Delta\eta$, could lead to an additional shape uncertainty in the correlation functions. To test for this, the ratio of the 1D $\Delta\phi$ projection with the nominal z_{vtx} binning and the binning described above was calculated for each $p_{T,\text{jet}}$, $p_{T,\text{assoc}}$, and centrality bin. The variations about 1.0 are smaller than the statistical errors associated with the points. This uncertainty was therefore considered negligible.

There is an additional shape uncertainty associated with the application of the acceptance correction due to slight changes in the correlation function at large $\Delta\eta$ in the acceptance with v_z position. The background level is determined from the level of the correlation function at large $\Delta\eta$, leading to a scale uncertainty in the background subtraction dependent on $p_{T,\text{assoc}}$. This uncertainty is from the differences between the nominal (z-integrated) and the z-vertex method (z-binned) for correcting the mixed events on the level of the background in the $0.6 < \Delta\eta < 1.2$ range, and signal plus background, in the $|\Delta\eta| < 0.6$ range. The large $\Delta\eta$ region is used to determine the background so any uncertainties in the level of the correlation function in this region lead to an uncertainty in the level of the background in the signal region. This is expressed as an additional scale band on the final results.

In the nominal (z-integrated) method the same- and mixed-event distributions are summed over all v_z bins before the division, because dividing the sample into 4-cm v_z bins, as would be ideal given the v_z dependence of the acceptance, enlarges the statistical fluctuations of the mixed-event normalization in each bin. In the z-binned method the acceptance correction is instead applied separately in each bin i ,

$$\frac{1}{N_{\text{trig}}} \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta} = \frac{1}{N_{\text{trig}}} \sum_i \frac{1}{\epsilon a_i(\Delta\phi, \Delta\eta)} \frac{d^2 N_i}{d\Delta\phi d\Delta\eta}, \quad (3.6)$$

where N_i and a_i are the same-event pairs and the pair acceptance in v_z bin i . For each method, the ratio of the integrals of the correlation function Y over the signal+background ($|\Delta\eta| < 0.6$) and background-dominated ($0.6 < |\Delta\eta| < 1.2$) regions is computed, and their ratio defines the scale factor

$$\alpha = \frac{\int_{\text{sig}} Y_{\text{nom}} d\Delta\eta / \int_{\text{bkgd}} Y_{\text{nom}} d\Delta\eta}{\int_{\text{sig}} Y_{\text{z vtx}} d\Delta\eta / \int_{\text{bkgd}} Y_{\text{z vtx}} d\Delta\eta} \quad (3.7)$$

for each $p_{\text{T,assoc}}$ bin. The scale band is obtained by multiplying the correlation function by α and by $2 - \alpha$. This uncertainty is determined by varying the binning of the mixed events in v_z and is correlated for different angles relative to the event plane and for different bins in $p_{\text{T,assoc}}$. Above $p_{\text{T,assoc}} > 3 \text{ GeV}/c$, this uncertainty is negligible because the background is small.

3.1.6 Flow modulation of combinatorial background

The combinatorial background ($N_{\text{assoc,jet}}^{\text{bkgd}}$) from Eq. 3.5 can be parametrized for trigger jets restricted to the orientation $\mathfrak{R} = \text{in}/\text{mid}/\text{out-of-plane}$ relative to the event plane as [90, 109]:

$$\left(\frac{1}{\pi} \frac{dN_{\text{assoc,jet}}^{\text{bkgd}}}{d\Delta\phi} \right)_{\mathfrak{R}} = B^{\mathfrak{R}} \left(1 + \sum_{n=2}^{\infty} 2v_{n,\text{assoc}} v_{n,\text{jet}}^{\mathfrak{R}} \cos(n\Delta\phi) \right), \quad (3.8)$$

where $v_{n,\text{assoc}}$ and $v_{n,\text{jet}}^{\mathfrak{R}}$ are the Fourier coefficients of the azimuthal angle distribution (flow

coefficients) of background associated particles and trigger jets restricted to the orientation $\mathfrak{R}$ respectively. $B^{\mathfrak{R}}$ gives the background level amplitude for orientation $\mathfrak{R}$. The trigger jets being restricted to orientation $\mathfrak{R}$ modifies their nominal flow coefficients $v_{n,\text{jet}}$'s into the $v_{n,\text{jet}}^{\mathfrak{R}}$ from Eq. 3.8 which are related to each other as [90]:

$$v_{n,\text{jet}}^{\mathfrak{R}} = \begin{cases} \frac{v_{n,\text{jet}}}{\beta^{\mathfrak{R}}} + \cos(n\phi_s^{\mathfrak{R}}) \frac{\sin(nc)}{nc} \frac{R_n}{\beta^{\mathfrak{R}}} + \\ \frac{1}{\beta^{\mathfrak{R}}} \sum_{k=2,4,6,..} (v_{(k+n),\text{jet}} + v_{|k-n|,\text{jet}}) \cos(k\phi_s^{\mathfrak{R}}) \frac{\sin(kc)}{kc} R_k & \text{if } n \text{ is even,} \\ v_{n,\text{jet}} & \text{if } n \text{ is odd.} \end{cases} \quad (3.9)$$

Where, $\beta^{\mathfrak{R}}$ is the $\mathfrak{R}$ dependence of $B^{\mathfrak{R}}$ given by,

$$B^{\mathfrak{R}} \propto \beta^{\mathfrak{R}} = 1 + \sum_{k=2,4,6,..} 2v_{k,\text{jet}} \cos(k\phi_s^{\mathfrak{R}}) \frac{\sin(kc)}{kc} R_k \quad (3.10)$$

Here $\phi_s^{\mathfrak{R}}$ and c are the center and half-width of the $|\Psi_{2,EP} - \phi_{\text{jet}}|$ range for jets restricted to orientation $\mathfrak{R}$. Odd $v_{n,\text{jet}}$'s mainly arise from initial state fluctuations and are therefore uncorrelated with the second-order event-plane and remain constant when the trigger jet is moved relative to the event plane, while even $v_{n,\text{jet}}$'s will change [109, 110]. The $\mathfrak{R}$ dependent background shape is dependent upon the event-plane resolutions (R_n), which are fixed at the measured values (Sect. 3.1). Extended details into the derivations of relevant equations can be found in Refs. [90, 111, 112].

Collective particle flow plays a major role in understanding the underlying-event background. This background consists of particles created from mechanisms unrelated to the hard process that led to a jet. Some of the jet signal is correlated with our event plane due to the path-length dependence of partonic energy loss, while soft hadrons are predominantly correlated with the event plane due to hydrodynamical flow that also contribute to the bulk-particle production.

Reaction plane fit

To remove the combinatorial background comprised by contributions from the underlying event, the reaction plane fit (RPF) developed in Ref. [109] is applied in this work. The measured jet-hadron correlation signal is decomposed into a near-side and an away-side, with the former being narrow in $\Delta\phi$ and $\Delta\eta$ and the latter narrow in $\Delta\phi$, but broad in $\Delta\eta$. The narrowness of the near-side implies the signal is negligible at large $\Delta\eta$, where the background dominates. Applying RPF, we defined our ‘signal + background’ region for $|\Delta\eta| < 0.6$ and further $0.6 \leq |\Delta\eta| < 1.2$ as a background dominated region where the signal was assumed to be negligible; the outermost $\Delta\eta$ bins, up to $|\Delta\eta| = 1.6$, are not used because the small pair acceptance there makes the corrected correlation function fluctuate strongly. The $\Delta\phi$ projection of the background-dominated region is fit with the parametrization of Eq. 3.8 over $|\Delta\phi| < \pi/2$, where the away-side jet does not contribute, simultaneously for in-plane, mid-plane, and out-of-plane trigger jets. The three orientations share the jet and associated-hadron flow coefficients, and their different background shapes (Eqs. 3.9 and 3.10) constrain the fit much more strongly than a fit to a single orientation would. The correlation function for all orientations combined is not included in the fit, as it would double count the same pairs. The fitted background is then extrapolated over the full $\Delta\phi$ range and subtracted from the signal+background region.

RPF improves upon prior background subtraction techniques [108, 113] by avoiding problems due to contamination from jets on both the near- and away-side by using the near-side at large $\Delta\eta$ and also using the dependence of the flow-modulated background on the angle of the trigger jet relative to the event plane to constrain the background shape and level.

The fits include the terms up to $n = 4$. At high $p_{T,\text{assoc}}$ the combinatorial background is small and few tracks are found at large $\Delta\eta$ from the trigger jet, so the higher-order terms can no longer be constrained and the fit is restricted to $n = 3$; this is the case for $p_{T,\text{assoc}} > 4$ GeV/ c with $15 < p_{T,\text{jet}} < 20$ GeV/ c jets and $p_{T,\text{assoc}} > 3$ GeV/ c with

$20 < p_{T,\text{jet}} < 40$ GeV/ c jets. Therefore, the RPF fits consist of six parameters (B , $v_{2,\text{jet}}$, $v_{2,\text{assoc}}$, $(v_{3,\text{jet}} \times v_{3,\text{assoc}})$, $v_{4,\text{jet}}$, and $v_{4,\text{assoc}}$) for low $p_{T,\text{assoc}}$, and four (B , $v_{2,\text{jet}}$, $v_{2,\text{assoc}}$ and $(v_{3,\text{jet}} \times v_{3,\text{assoc}})$) for high $p_{T,\text{assoc}}$, from which the orientation-dependent $B^{\mathfrak{R}}$ and $v_{n,\text{jet}}^{\mathfrak{R}}$ follow. The fits, with their quality shown as $(\text{data} - \text{fit})/\text{fit}$, are shown for $15 < p_{T,\text{jet}} < 20$ GeV/ c jets in Fig. 3.6, and the extracted background level and $v_{2,\text{assoc}}$ are shown in Fig. 3.5. The background level falls by more than three orders of magnitude between the lowest and the highest $p_{T,\text{assoc}}$ bin, and the extracted $v_{2,\text{jet}}$ and $v_{2,\text{assoc}}$ are consistent with earlier STAR measurements of the jet and charged-hadron v_2 , despite the differences in method and event selection.

The uncertainty of the background is obtained from the covariance matrix σ_{ij} of the fit parameters p_i : for any quantity f derived from the fitted background, such as the background in a $\Delta\phi$ bin or its integral,

$$\sigma_f^2 = \sum_{i=0}^N \sum_{j=0}^N \frac{\partial f}{\partial p_i} \frac{\partial f}{\partial p_j} \sigma_{ij}. \quad (3.11)$$

Since the three orientations are fit simultaneously, this uncertainty is correlated point-to-point in $\Delta\phi$ and between orientations.

Comparison of in-, mid-, and out-of-plane jet-hadron correlations is performed after background subtraction to explore the effects related to event-plane orientation. The background-subtracted correlation functions are shown in Figs. 3.7 and 3.8 for $15 < p_{T,\text{jet}} < 20$ GeV/ c and $20 < p_{T,\text{jet}} < 40$ GeV/ c jets, respectively. Each panel corresponds to one $p_{T,\text{assoc}}$ bin and shows the in-plane, mid-plane, and out-of-plane jets and all orientations combined. The uncertainties from the RPF background subtraction are propagated using Eq. 3.11 and are non-trivially correlated point-to-point and between different bins relative to the event plane. These are shown in gray. The uncertainty from the acceptance correction, described in Sect. 3.1.5 is displayed by the red uncertainty band. The uncertainties on the event-plane resolution are negligible relative to that of the background subtraction and statistical uncertainties of the final results. Additional uncertainties uncorrelated with each other, but

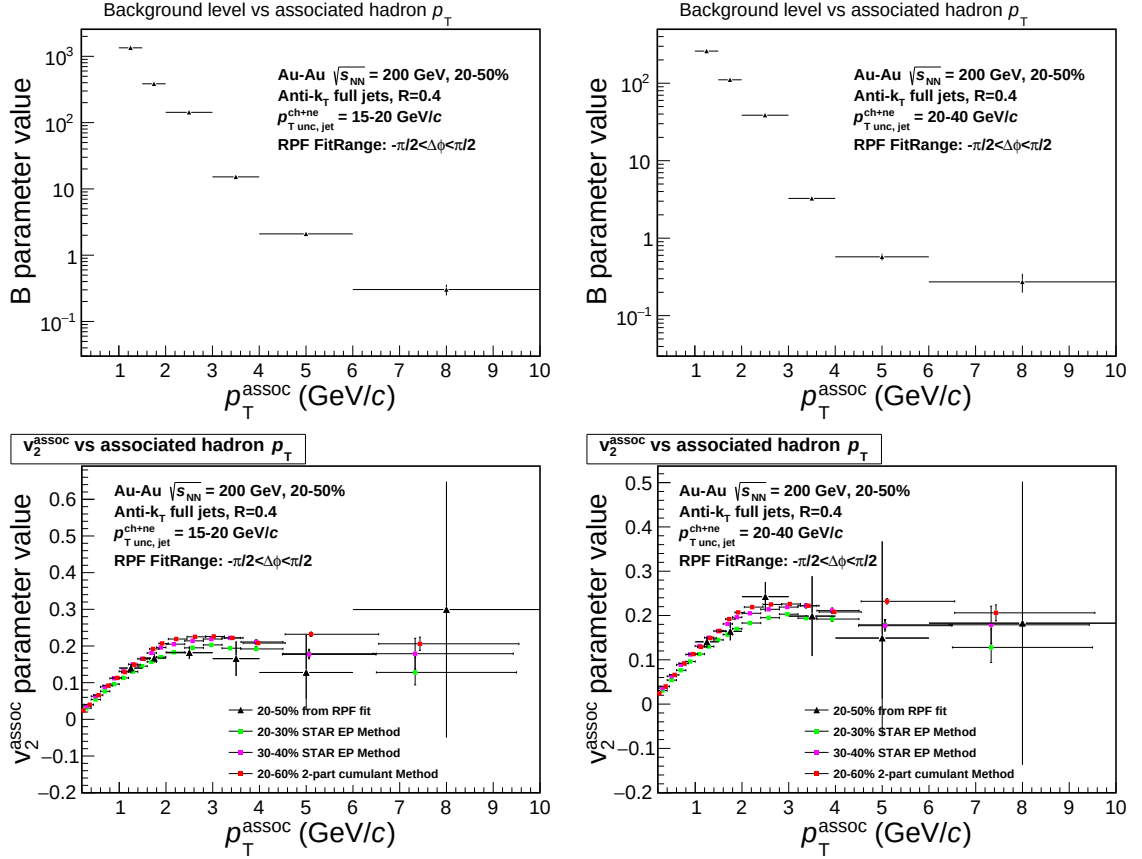


Figure 3.5: Parameters of the RPF fits as a function of $p_{T, \text{assoc}}$ for $15 < p_{T, \text{jet}} < 20$ GeV/ c (left) and $20 < p_{T, \text{jet}} < 40$ GeV/ c (right) full jets in 20–50% central Au+Au collisions: the background level B (top) and the second-order flow coefficient of the associated hadrons, $v_{2, \text{assoc}}$ (bottom). The fits are performed over $|\Delta\phi| < \pi/2$ in the background-dominated region $0.6 < |\Delta\eta| < 1.2$.

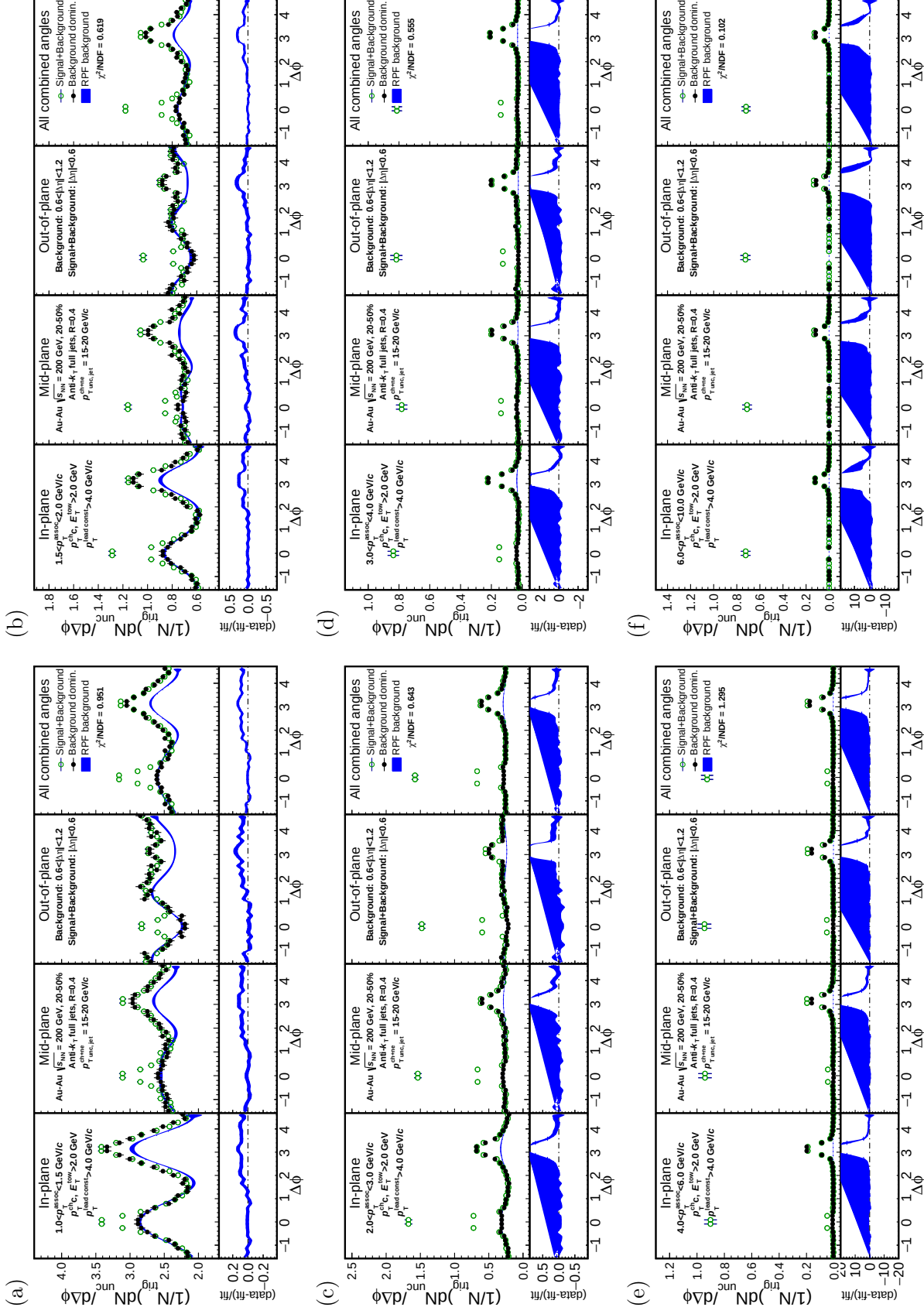


Figure 3.6: RPF fits for $15 < p_{T, \text{jet}} < 20$ GeV/c full jets in 20–50% central Au+Au collisions, for (a) $1.0 < p_{T, \text{assoc}} < 1.5$ GeV/c, (b) $1.5 < p_{T, \text{assoc}} < 2.0$ GeV/c, (c) $2.0 < p_{T, \text{assoc}} < 3.0$ GeV/c, (d) $3.0 < p_{T, \text{assoc}} < 4.0$ GeV/c, (e) $4.0 < p_{T, \text{assoc}} < 6.0$ GeV/c and (f) $6.0 < p_{T, \text{assoc}} < 10.0$ GeV/c. Each panel shows, from left to right, in-plane, mid-plane and out-of-plane jets and all orientations combined. Upper panels: $\Delta\phi$ projections of the signal+background region ($|\Delta\eta| < 0.6$, green) and of the background-dominated region ($0.6 < |\Delta\eta| < 1.2$, black), with the RPF fit to the background-dominated region (blue band, fit uncertainty). Lower panels: quality of the fit to the background-dominated region, $(\text{data} - \text{fit})/\text{fit}$.

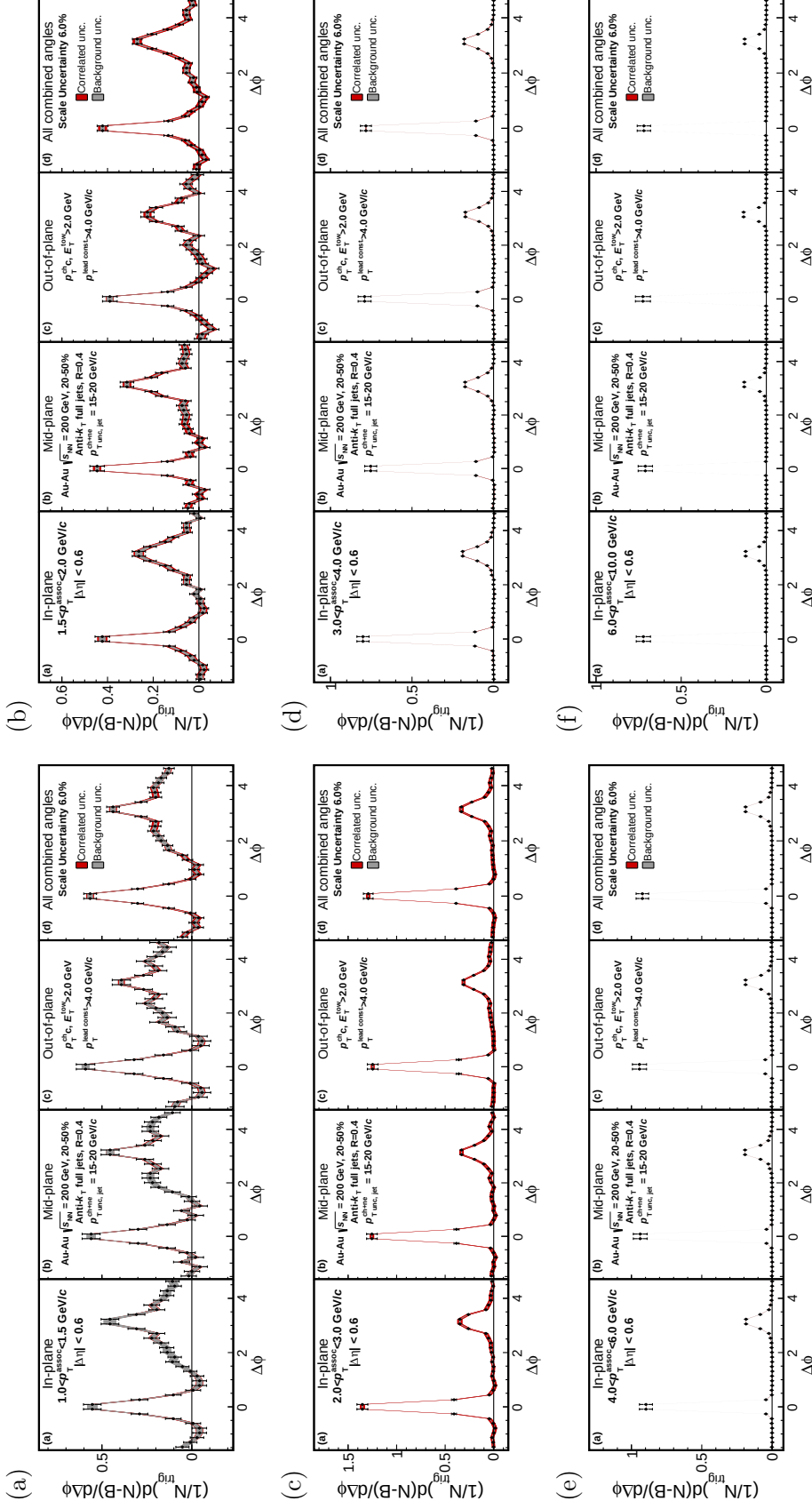
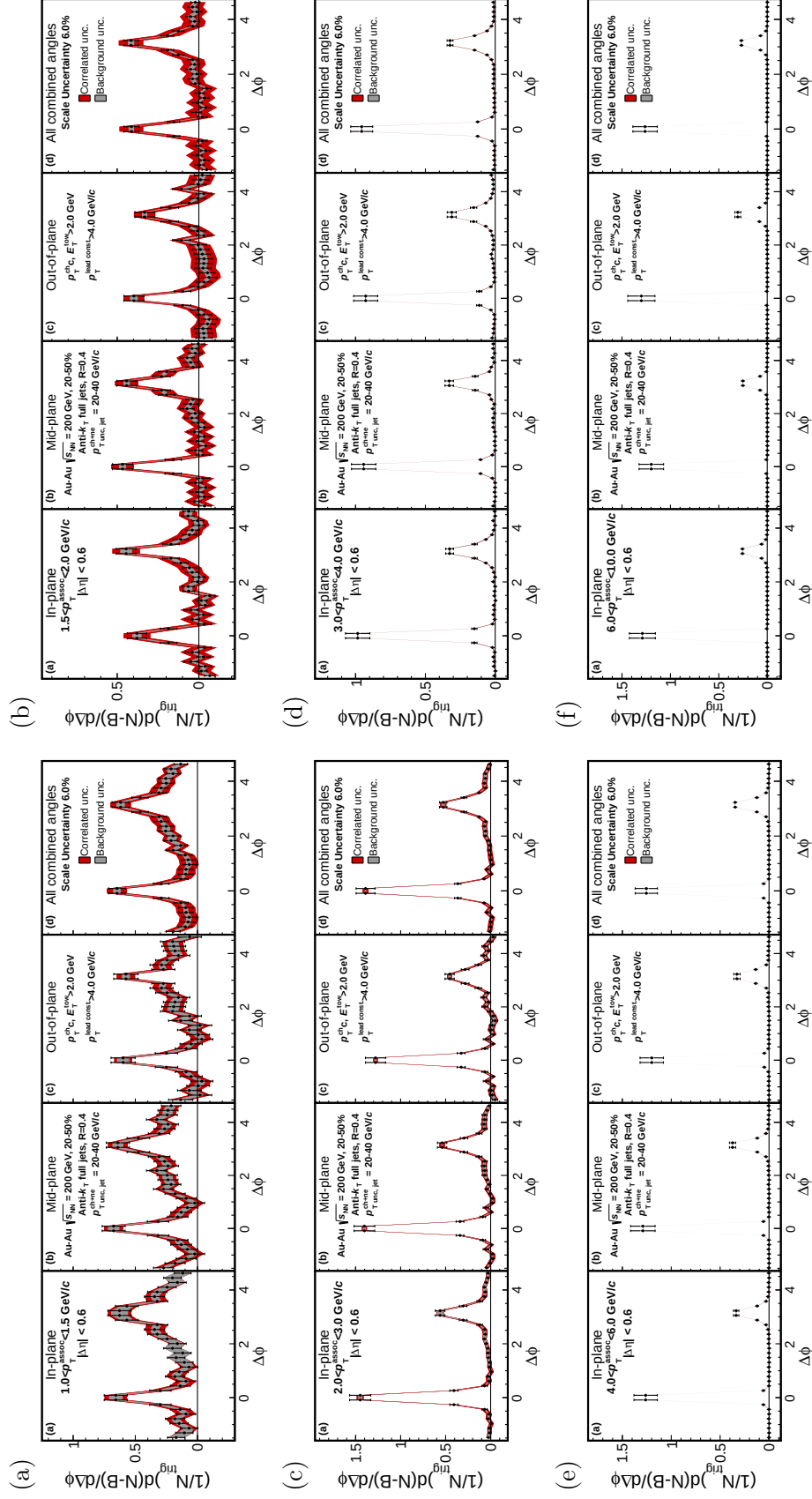


Figure 3.7: Background-subtracted jet-hadron correlation functions in $\Delta\phi$ ($|\Delta\eta| < 0.6$) for $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$ full jets in 20–50% central Au+Au collisions, for (a) $1.0 < p_{T,\text{assoc}} < 1.5 \text{ GeV}/c$, (b) $1.5 < p_{T,\text{assoc}} < 2.0 \text{ GeV}/c$, (c) $2.0 < p_{T,\text{assoc}} < 3.0 \text{ GeV}/c$, (d) $3.0 < p_{T,\text{assoc}} < 4.0 \text{ GeV}/c$, (e) $4.0 < p_{T,\text{assoc}} < 6.0 \text{ GeV}/c$ and (f) $6.0 < p_{T,\text{assoc}} < 10.0 \text{ GeV}/c$. Each panel shows in-plane, mid-plane and out-of-plane jets and all orientations combined. The red band is the correlated scale uncertainty from the mixed-event acceptance correction and the gray band is the uncertainty of the RPF background fit, which is correlated point-to-point; the global scale uncertainty of 6% is not shown.

Figure 3.8: Same as Fig. 3.7, for $20 < p_{T, \text{jet}} < 40 \text{ GeV}/c$ full jets.

correlated for all points in $\Delta\phi$ are given in Tab. 3.2. They are combined in quadrature and listed as the scale uncertainty on the results.

3.1.7 Detector response and embedding closure

The PYTHIA6 Perugia 2012 tune is used to create particle level dijet events embedded in MB Au+Au 200 GeV events at the detector level. The events are generated in bins of the partonic hard-scattering transverse momentum, $\hat{p}_T$, and the bins are combined with weights given by their cross sections divided by the number of generated events. This allows for further comparisons between the jets from the input PYTHIA6 tracks (generator-level jets or GEN-jets) and the jets from the embedded tracks reconstructed by GEANT (reconstructed jets or RECO-jets). RECO level events are analyzed with the same selections and parameters as used by the data analysis. All particles of GEN-level events are required to be in their final state (particles with no further daughters). GEN-level jets have $p_{T,\text{jet}}^{\text{GEN}} > 10 \text{ GeV}/c$ and the RECO-level jets have $p_{T,\text{jet}}^{\text{RECO}} > 5 \text{ GeV}/c$. Further, we require only the RECO-level jets contain a neutral component with $E_T \geq 5.4 \text{ GeV}$ to match the trigger condition applied in data. Our goal is to study the effects of the detector response on jet reconstruction and our analysis. In order to compare the same jet pre- and post-reconstruction, we apply a ‘nearness’ criteria in the $\eta - \phi$ that matches a GEN-jet to one RECO-jet satisfying:

(a) $r_{\text{GEN,RECO}} = \sqrt{(\eta_{\text{jet}}^{\text{GEN}} - \eta_{\text{jet}}^{\text{RECO}})^2 + (\phi_{\text{jet}}^{\text{GEN}} - \phi_{\text{jet}}^{\text{RECO}})^2} \leq 0.4$, $r_{\text{GEN,RECO}}$ being the separation between the gen-jet and the reco-jet in $\eta - \phi$ space, (b) the RECO-jet is the closest to the gen-jet among all reco-jets (minimize $r_{\text{GEN,RECO}}$).

We compare the resulting GEN- and RECO-level jet spectra to calculate the momentum resolution which is given by:

$$p_{T,\text{jet}} \text{ resolution}(\%) = \frac{p_{T,\text{jet}}^{\text{GEN}} - p_{T,\text{jet}}^{\text{RECO}}}{p_{T,\text{jet}}^{\text{GEN}}} \times 100\%. \quad (3.12)$$

Fig. 3.9 shows the $p_{T,\text{jet}}$ resolution for $15 < p_{T,\text{jet}}^{\text{GEN}} < 20 \text{ GeV}/c$ (left) and $20 < p_{T,\text{jet}}^{\text{GEN}} < 40$

GeV/ c (right) $R = 0.4$ full jets. The distributions have been normalized into probability functions and are shown for the 20-50% most central events for all angles of the jet relative to the event plane. Fig. 3.9 further shows an average energy loss of around 15% going from GEN to RECO. This net energy shift is thought to be due to counteracting effects of the tracking inefficiency at the RECO level and there being more tracks in the RECO level from the min-bias pedestal. The $p_{T,\text{jet}}$ resolution for $15 < p_{T,\text{jet}}^{\text{RECO}} < 40$ GeV/ c reco-jets is roughly 10-20%. The event-plane dependence of the $p_{T,\text{jet}}$ resolution was also studied and found to be within 1-2% of each other between different orientations of jets w.r.t event plane. There can be slight differences in the jets reconstructed at lower momenta with $15 < p_{T,\text{jet}} < 20$ GeV/ c for jets at different angles relative to the event plane, due to a low momentum embedded jet overlapping with another jet in the Au+Au data and from statistical fluctuations. As there are more jets from in-plane than out-of-plane orientations in the data, this leads to an apparent difference in the reconstructed jet spectra. Otherwise there are no significant differences between jets at different angles relative to the event plane. A map between $p_{T,\text{jet}}^{\text{GEN}}$ and $p_{T,\text{jet}}^{\text{RECO}}$, called the response matrix, is added to Fig. 3.9 as a smaller inset on the top right. Since the trigger jets are selected at the detector level and the results are not unfolded, these numbers characterize which particle-level jets populate the two reconstructed jet-momentum classes rather than enter as a correction.

The same embedding sample is used for a closure test of the analysis chain. The associated yields and jet-peak widths are extracted from the RECO-level jets with the full analysis procedure, including the mixed-event correction taken from data and the RPF background subtraction, and compared to those of the matched GEN-level jets, whose correlations with the final-state PYTHIA6 particles contain no combinatorial background. The comparison is complicated by the tracking efficiency near a jet, which differs from the average efficiency used in the correction: above 2 GeV/ c it is slightly lower inside $15 < p_{T,\text{jet}} < 20$ GeV/ c jets and slightly higher inside $20 < p_{T,\text{jet}} < 40$ GeV/ c jets than for an average track in the event. After correcting the RECO-level near-side yields for this difference, the RECO/GEN ratio

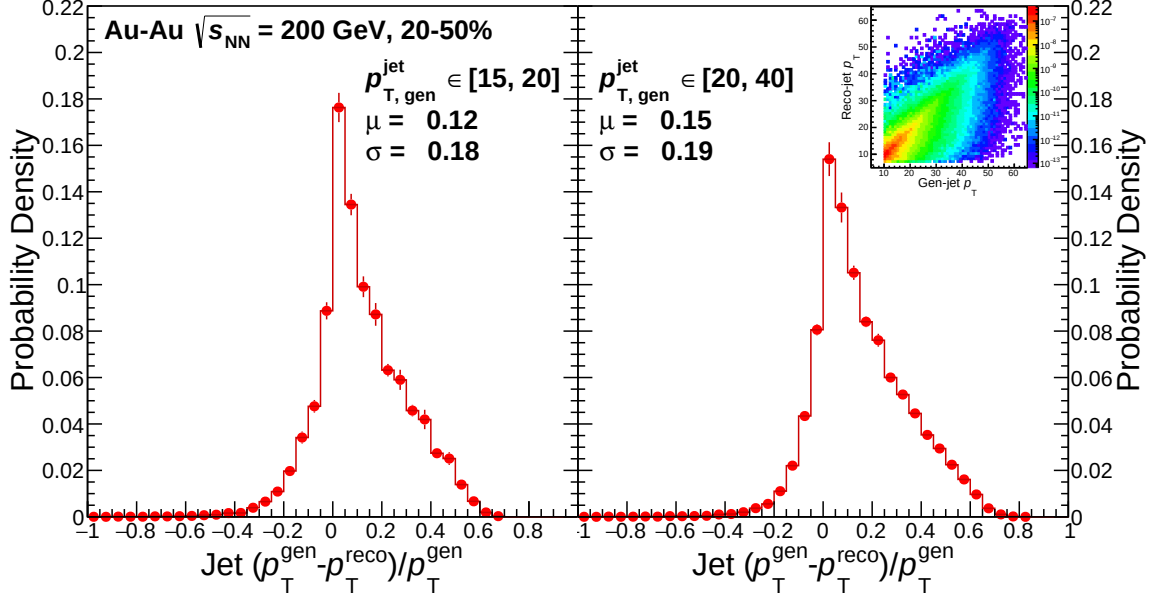


Figure 3.9: $p_{T,jet}$ resolution for $15 < p_{T,jet}^{GEN} < 20$ GeV/ c (left) and $20 < p_{T,jet}^{GEN} < 40$ GeV/ c (right) $R = 0.4$ full jets with the corresponding response matrix as an inset on the top right. Jets are measured from all angles relative to the event plane in the 20-50% most central events. This comparison is for matched GEN-level to RECO-level jets.

of the yields for $20 < p_{T,jet} < 40$ GeV/ c jets, shown in Fig. 3.10, is consistent with unity on the away side. The remaining deviations, on the near side in the highest $p_{T,assoc}$ bin and in the lowest $p_{T,assoc}$ bins for $15 < p_{T,jet} < 20$ GeV/ c jets, were traced to the limited statistics of the low- $\hat{p}_T$ bins of the embedding, which enter with large weights.

3.1.8 Systematic Uncertainties

The systematic uncertainties fall into three classes that differ in how they are correlated, and therefore in how they are shown on the results. (i) Sources that change the overall normalization of the correlation functions, independently of $\Delta\phi$ and of the jet orientation, are combined into a global scale uncertainty: the single-track reconstruction efficiency, the normalization of the mixed events, and the event-plane resolution. (ii) Sources that change the level of the correlation functions in a way that depends on $p_{T,assoc}$ and on the jet orientation are shown as colored scale bands: the shape of the pair acceptance in $\Delta\eta$ and the jet-energy-shift (JES) correction. (iii) The uncertainty of the RPF background fit,

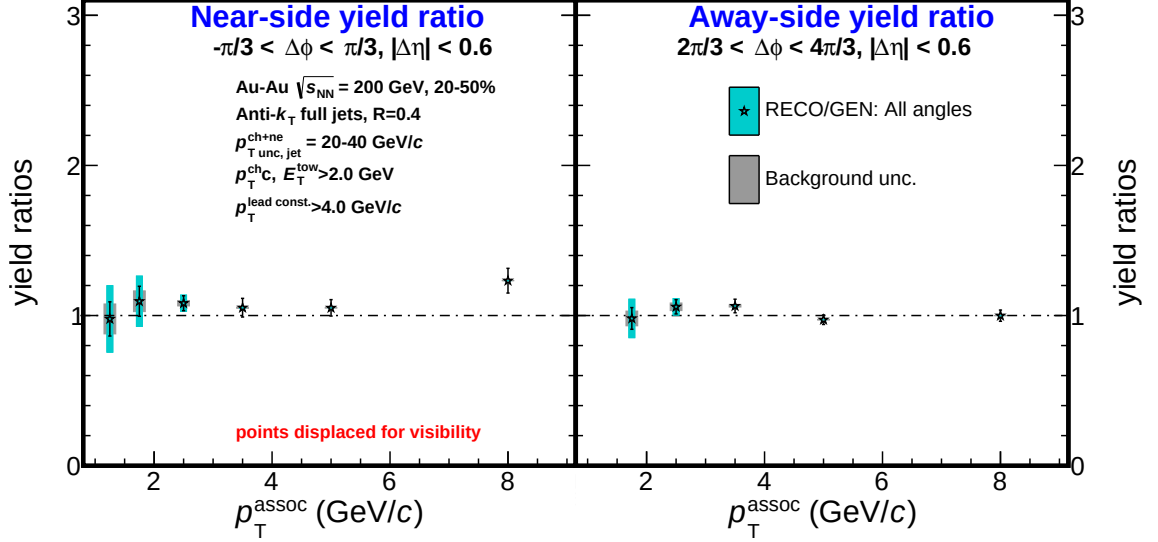


Figure 3.10: Embedding closure test: ratio of the near-side (left) and away-side (right) associated yields extracted from RECO-level jets to those of the matched GEN-level jets vs $p_{T,\text{assoc}}$, for $20 < p_{T,\text{jet}} < 40 \text{ GeV/c}$ full jets embedded in 20–50% central Au+Au events, all orientations relative to the event plane combined. The RECO-level near-side yields are corrected for the tracking efficiency inside jets (see text). The gray bands are the uncertainties of the RPF background fit. Points are displaced for visibility.

obtained from the covariance matrix of the fit, varies from point to point in $\Delta\phi$ and is correlated between points and between orientations; it is shown as a gray band. Each source is described below.

The 5% uncertainty on the single-track reconstruction efficiency (Sect. 3.1.2) directly scales the number of associated hadrons, and hence the yields, while leaving the shape of the correlation functions unchanged. The mixed-event normalization uncertainty is the uncertainty of the constant fitted to the plateau of the pair acceptance (Sect. 3.1.5). The event-plane resolution enters only through the background parametrization; varying it by $\pm 1\%$ changes the results negligibly.

We summarize the systematic uncertainties in Tabs. 3.2, 3.3, and 3.4. Table 3.2 lists the sources of systematic uncertainties which are independent of the angle relative to the event plane. These sources include the single-track reconstruction efficiency (Sect. 3.1.2), uncertainties in the mixed events due to their normalization and shape in $\Delta\phi$ (Sect. 3.1.5), and uncertainties in the event-plane resolution (Sect. 3.1). The uncertainties are added in

quadrature and lead to a 6% uncertainty in the scale of the correlation functions and yields with the single-track reconstruction efficiency being the dominate source. This uncertainty is uncorrelated for different associated-particle momenta.

Table 3.2: Summary of systematic uncertainties which are independent of the angle relative to the event plane and the momentum for both $15 < p_{T,\text{jet}} < 20$ GeV/ c and $20 < p_{T,\text{jet}} < 40$ GeV/ c in 20-50% central Au+Au collisions.

Source	Uncertainty %
Single particle reconstruction efficiency	5
Mixed event (shape $\Delta\phi$)	negligible
Mixed event normalization	< 1.25
Event-plane resolution	< 1

Additional uncertainties, highly dependent on the angle of the jet relative to the event plane and the associated particles' momentum, are summarized in Tabs. 3.3 and 3.4. These include the impact of the scale uncertainty from the acceptance shape (Sect. 3.1.5, Eq. 3.7) and the RPF background fit (Sect. 3.1.6) on the associated yield and jet-peak width results. These uncertainties are compared for two associated particle momentum bins (1.0-1.5 and 3.0-4.0 GeV/ c) to highlight how much of an impact the background has at low momenta. The uncertainty arising from the RPF background subtraction is non-trivially correlated point-to-point in $\Delta\phi$ and for different orientations of the jet relative to the event plane, but is uncorrelated between different $p_{T,\text{assoc}}$ bins. The acceptance-correction uncertainty on the shape in $\Delta\eta$ is also correlated for different orientations of the jet relative to the event plane and uncorrelated across $p_{T,\text{assoc}}$ bins. The acceptance-shape uncertainty is the dominant source at low-momenta, while being more comparable to the background uncertainty at larger momenta.

Since no underlying-event subtraction is applied to the jets (Sect. 3.1.4), background particles that fall into the jet cone shift the reconstructed jet momentum upward, and the shift is larger for in-plane than for out-of-plane jets because the flow-modulated background is larger in-plane. Jets in the different orientations therefore sample slightly different parton momenta for the same reconstructed $p_{T,\text{jet}}$ class. To account for this jet-energy shift (JES)

and to provide an adequate comparison for the jet samples from different orientations relative to event plane, a random-cone study was performed on the minimum bias data of matching centrality selection: cones of radius 0.4 are placed at random positions within the jet acceptance, and the transverse momenta of the charged particles inside them are summed, in bins of the cone orientation relative to the event plane. The mean summed momenta are found to be 0.388 GeV/ c (in-plane), 0.344 GeV/ c (mid-plane), and 0.308 GeV/ c (out-of-plane), with a corresponding RMS of 0.4 GeV/ c . To determine the effect on the results, the correlation functions are also constructed for jet-momentum classes shifted by a fixed amount, and the resulting changes of the yields and widths are rescaled to the shifts found with the random cones. The nominal yields and widths are corrected by these rescaled shifts, and the corresponding rescaled RMS is assigned as the JES uncertainty, which is added in quadrature to the acceptance-shape scale uncertainty.

3.2 Results

Charged particle yields associated with jets are found by integrating the background-subtracted correlation functions according to Eq. 3.2. The integration limits in $\Delta\phi$ for the near-side are $a = -\pi/3$ and $b = \pi/3$, while for the away-side, we have $a = 2\pi/3$ and $b = 4\pi/3$. The integration limits in $\Delta\eta$ are the same for both the near-side and away-side, $c = -0.6$ and $d = 0.6$. The yield is computed as the difference of the integral of the measured signal+background distribution, which carries the statistical uncertainty, and the integral of the fitted background, whose uncertainty follows from Eq. 3.11; the two are reported separately.

Shown in Fig. 3.11 is the near-side (left) and away-side (right) associated yields vs $p_{T,assoc}$ for $15 < p_{T,jet} < 20$ GeV/ c (top) and $20 < p_{T,jet} < 40$ GeV/ c (bottom) full jets in 20-50% centrality collisions. The yields are compared for each orientation of the trigger jet reconstructed relative to the event plane (in/mid/out) for $p_{T,assoc} \in [1.0, 1.5], [1.5, 2.0], [2.0, 3.0], [3.0, 4.0], [4.0, 6.0], [6.0, 10.0]$ GeV/ c .

The main feature is the steeply falling associated yield with increasing $p_{T,assoc}$, which

Table 3.3: Summary of systematic uncertainties on the associated yields and widths calculated from the correlation functions due to the shape uncertainty coming from the shape of the acceptance correction in $\Delta\eta$, the correlated background-fit uncertainty, and the uncertainty associated with the jet energy shift (JES) correction, each varying with event-plane orientation bins. They are displayed for $15 < p_{T,\text{jet}} < 20$ GeV/ c in 20-50% central Au+Au collisions for $1.0 < p_{T,\text{assoc}} < 1.5$ GeV/ c and $3.0 < p_{T,\text{assoc}} < 4.0$ GeV/ c bins. The values are expressed as a percent of the nominal value.

Source	Result	Orientation	Uncertainty %			
			Near-side: $p_{T,\text{assoc}}$ (GeV/ c)		Away-side: $p_{T,\text{assoc}}$ (GeV/ c)	
			1.0-1.5	3.0-4.0	1.0-1.5	3.0-4.0
Acceptance shape	Yield	in-plane	14	1.2	8.1	3.0
		mid-plane	11	1.2	7.6	3.3
		out-of-plane	11	1.1	7.3	3.0
	Width	in-plane	5.2	0.6	4.3	2.4
		mid-plane	5.4	0.5	3.9	2.2
		out-of-plane	4.3	0.4	4.9	2.1
Background fit	Yield	in-plane	11	0.8	6.1	2.1
		mid-plane	7.7	0.9	5.2	2.5
		out-of-plane	7.4	0.7	4.7	2.1
	Width	in-plane	10	0.1	8.2	0.4
		mid-plane	10	0.1	7.5	0.4
		out-of-plane	8.2	0.1	9.3	0.4
JES correction	Yield	in-plane	1.8	3.7	4.9	4.0
		mid-plane	2.8	2.5	1.4	4.2
		out-of-plane	3.7	3.3	4.8	4.2
	Width	in-plane	0.6	0.4	4.0	< 0.1
		mid-plane	0.2	< 0.1	7.9	1.5
		out-of-plane	1.6	0.7	1.9	1.7

Table 3.4: Summary of systematic uncertainties on the associated yields and widths calculated from the correlation functions due to the shape uncertainty coming from the shape of the acceptance correction in $\Delta\eta$, the correlated background-fit uncertainty, and the uncertainty associated with the JES correction, each varying with event-plane orientation bins. They are displayed for $20 < p_{T,\text{jet}} < 40$ GeV/ c in 20-50% central Au+Au collisions for $1.0 < p_{T,\text{assoc}} < 1.5$ GeV/ c and $3.0 < p_{T,\text{assoc}} < 4.0$ GeV/ c bins. The values are expressed as a percent of the nominal value.

Source	Result	Orientation	Uncertainty %			
			Near-side: $p_{T,\text{assoc}}$ (GeV/ c)		Away-side: $p_{T,\text{assoc}}$ (GeV/ c)	
			1.0-1.5	3.0-4.0	1.0-1.5	3.0-4.0
Acceptance shape	Yield	in-plane	35	1.2	22	2.5
		mid-plane	39	1.2	22	2.2
		out-of-plane	59	0.9	37	1.6
	Width	in-plane	30	0.7	8.4	1.9
		mid-plane	22	0.5	15	1.7
		out-of-plane	19	0.4	28	1.2
Background fit	Yield	in-plane	13	1.0	8.2	2.1
		mid-plane	14	0.7	8.0	1.2
		out-of-plane	22	0.9	14	1.5
	Width	in-plane	13	0.1	3.7	0.2
		mid-plane	10	0.1	6.8	0.2
		out-of-plane	8.6	< 0.1	13	0.1
JES correction	Yield	in-plane	0.2	0.4	< 0.1	2.6
		mid-plane	1.1	0.3	2.1	2.9
		out-of-plane	3.1	0.6	1.4	3.5
	Width	in-plane	1.2	0.5	3.3	0.3
		mid-plane	1.2	0.7	1.3	1.5
		out-of-plane	2.2	0.1	0.2	1.9

occurs on both the near- and away-side. Note that associated yields for $p_{T,\text{assoc}} \geq 2.0 \text{ GeV}/c$ include jet constituents, which leads to the discontinuity at $p_{T,\text{assoc}} = 2 \text{ GeV}/c$. Additionally, the use of jets with ‘hard-cores’ and containment of a tower associated to the firing event trigger can lead to a surface bias of the near-side jet. This, however, maximizes the average path length the away-side recoil jets travels, increasing the likelihood of an interaction with the medium. We would expect an in-plane jet and an out-of-plane jet with the same $p_{T,\text{jet}}$ to have different distributions of hard and soft constituents. An in-plane jet with less path travelled in the medium on average would be expected to show higher yields of constituents with higher $p_{T,\text{assoc}}$ than the more quenched out-of-plane jet, which would be expected to show higher yields of constituents with lower $p_{T,\text{assoc}}$. While uncertainties are smaller for jets with $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$, both samples still lack any clear dependence on the event-plane angle. This is an indication that modifications dependent on the average path length are smaller than the experimental uncertainties. On the far right of the near- and away-side panels is the inclusive $p_{T,\text{assoc}}$ bin with $1.0 < p_{T,\text{assoc}} < 10 \text{ GeV}/c$. The associated yields of each event-plane orientation in the inclusive $p_{T,\text{assoc}}$ selection are consistent with each other for the sample where $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$. However, in the jet sample with $20 < p_{T,\text{jet}} < 40 \text{ GeV}/c$, there are indications suggesting potential modifications. This potential modification is apparent on both the near- and away-side with the largest contributions coming from the lowest $p_{T,\text{assoc}}$ bins.

The widths are calculated by fitting a Gaussian, $A e^{-(\Delta\phi - \Delta\phi_0)^2/2\sigma^2}$, to the jet peak centered at $\Delta\phi_0 = 0$ for the near-side and $\Delta\phi_0 = \pi$ for the away-side. The gaussians are fitted separately, with a range of $|\Delta\phi| < \pi/3$ on the near-side and $|\Delta\phi - \pi| < \pi/3$ on the away-side. The Gaussian fit is repeated with different values of the background parameters and the covariance matrix is used to propagate the uncertainties. The scale uncertainties on the widths are given by $\sigma_w^{sc} = \frac{B}{\sigma_B} \times |\alpha - 1|$, where α (Eq. 3.7) is the $p_{T,\text{assoc}}$ -dependent scale factor associated with the acceptance shape when propagating the background determined in the $0.6 \leq |\Delta\eta| < 1.2$ to the region $|\Delta\eta| < 0.6$.

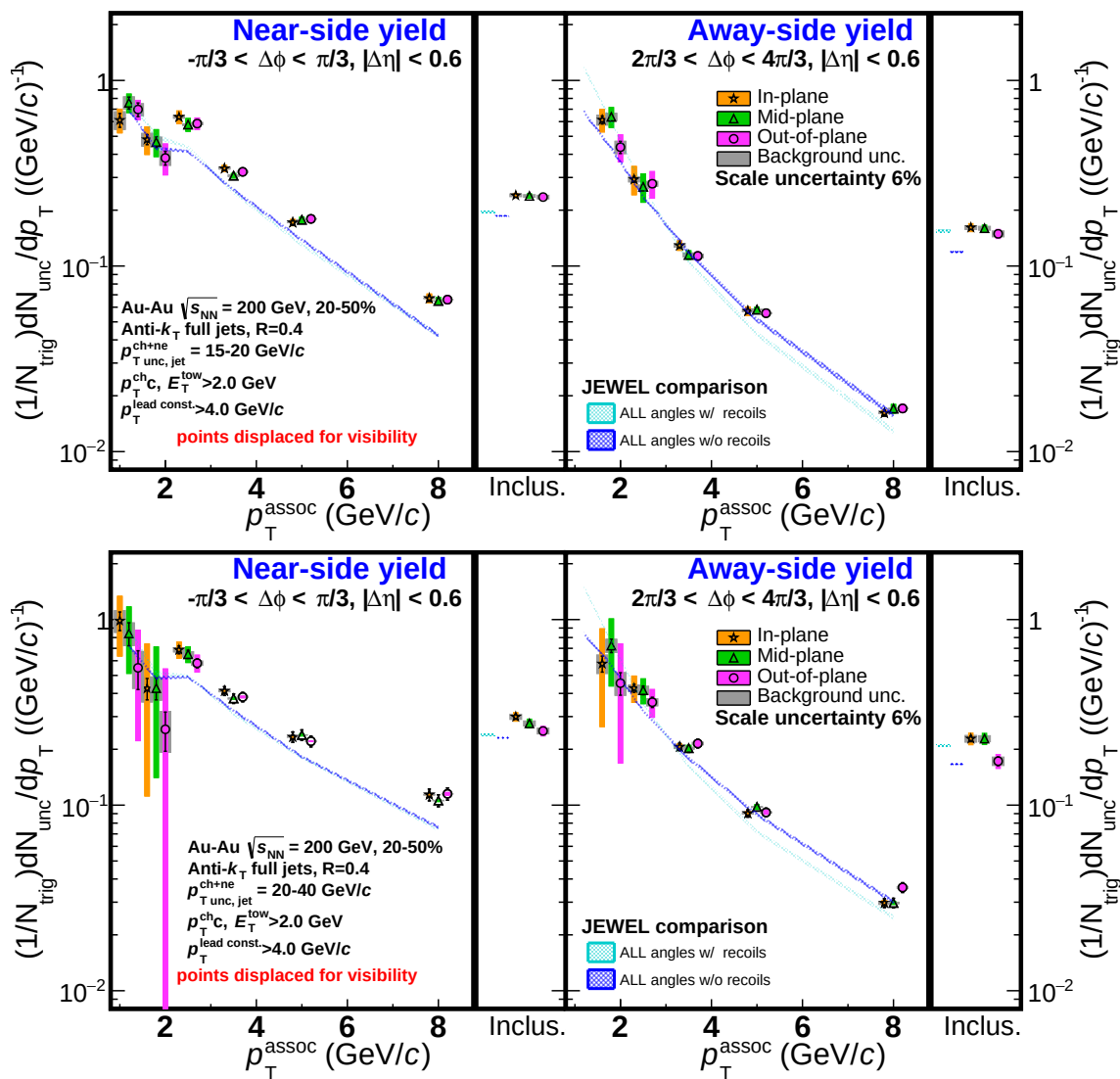


Figure 3.11: Near-side (left) and away-side (right) uncorrected (unc.) associated yield vs $p_{T,\text{assoc}}$ for 15-20 (top) and 20-40 GeV/c (bottom) full jets of 20-50% centrality in Au+Au collisions. The grey bands describe the systematic uncertainties of the background fits and are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the mixed event acceptance shape and JES correction. There is an additional 6% global scale uncertainty. Included on the far right of the near- and away-side panels is the inclusive transverse momentum bin from 1.0-10.0 GeV/c. Points are displaced for visibility.

From Fig. 3.12, it is clear a broadening of the jet peaks is occurring for decreasing $p_{T,assoc}$. This is expected from both collisional energy loss and gluon bremsstrahlung. With out-of-plane jets expected to traverse a longer average path length than in-plane jets, this would lead to additional interactions with the medium and more subsequent re-scatterings resulting in a larger width for jets out-of-plane relative to in-plane. Within the uncertainties there is no clear ordering. This indicates the effect of path-length dependent energy loss is not large enough to be seen by the current precision of the data. On the far right of the near- and away-side panels is the inclusive $p_{T,assoc}$ bin with $1.0 < p_{T,assoc} < 10$ GeV/ c . The widths of each event-plane orientation in the inclusive selection for the the sample with $15 < p_{T,jet} < 20$ GeV/ c are consistent. Conversely, the $20 < p_{T,jet} < 40$ GeV/ c sample reveals indications of potential modifications. This potential modification is apparent on both the near- and away-side and primarily comes from the lowest transverse momentum bins where sample size is the largest.

The measurements presented in Figs. 3.11 and 3.12 are compared to calculations from “Jet Evolution With Energy Loss” known as JEWEL [114], a jet energy loss model based on radiative and collisional energy loss in connection with partons sampled from a longitudinally expanding medium [69]. Model calculations are provided for two regimes, for calculations that a) include recoiled partons, and b) do not include recoiled partons. When no recoils are kept, the lost jet momentum is removed from the entire system. This is useful for modeling energy loss in the hard part of the jet. When recoils are kept, the jets momentum is fully conserved, but this adds both energy and additional background particles to the di-jet. In an experimental analysis, we likely would measure some, but not all of the recoil particles as they are often indistinguishable from background.

Due to the dominant impact of jet-by-jet fluctuations on partonic energy loss over path-length dependence [115, 116], JEWEL only predicts a very slight event-plane dependence which is well below the systematic uncertainty in the measurement. Variations among event-plane orientations were not seen at the 10% level. This is therefore consistent with

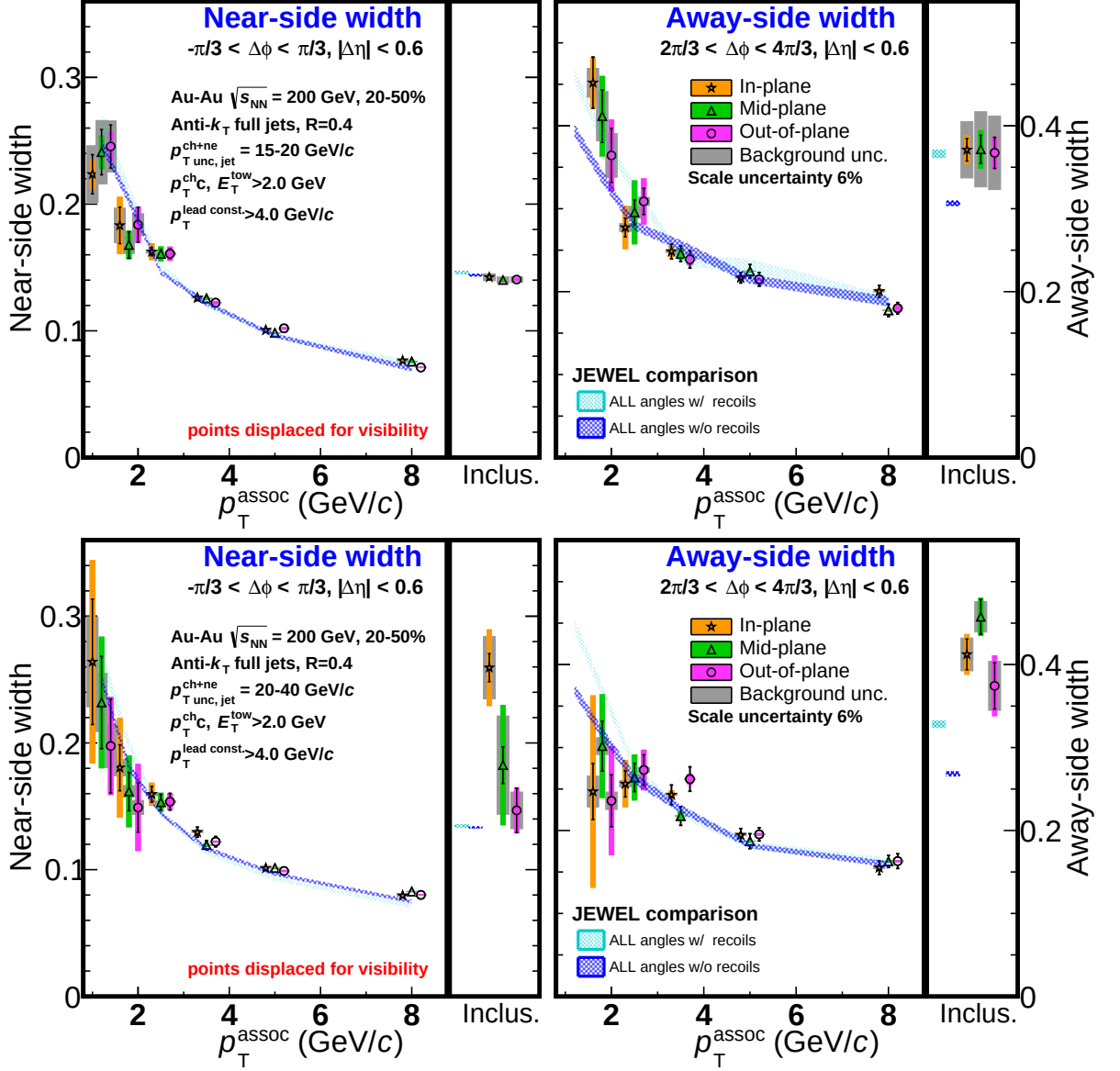


Figure 3.12: Near-side (left) and away-side (right) jet width vs $p_{T,assoc}$ for 15-20 (top) and 20-40 GeV/c (bottom) full jets of 20-50% centrality in Au+Au collisions. The widths are extracted from the gaussian fit to the jet peak. The grey bands describe the systematic uncertainties of the background fits which are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the mixed-event acceptance shape and JES correction. There is an additional 6% global scale uncertainty. Included on the far right of the near- and away-side panels is an inclusive transverse momentum bin from 1.0-10.0 GeV/c. Points are displaced for visibility.

path-length dependence having an insignificant impact compared to jet-by-jet fluctuations in energy loss. Fluctuations in the density of the medium may also suppress observable path-length dependence and are not included in the JEWEL model. However, higher precision JEWEL calculations may be needed to discern any potential event-plane dependent effects. We thus show the JEWEL comparisons corresponding to the sample integrated over all angles relative to the event plane and compare to the results from data. Comparisons show that the away-side is well described in terms of both the associated yields and widths by including recoils at low $p_{T,\text{assoc}}$, while at high $p_{T,\text{assoc}}$, the yields are better described by not including recoils and the widths have similar results to within uncertainties for both cases. When looking at the near-side, the widths are quite similar at high $p_{T,\text{assoc}}$ with slightly larger values when including recoils at low $p_{T,\text{assoc}}$. For the associated yields at high $p_{T,\text{assoc}}$, both JEWEL cases are similar to each other, but underestimate the data and including recoils has larger yields that better match the data at low $p_{T,\text{assoc}}$.

To better quantify and examine the event-plane dependence of the yields, ratios were taken of mid-plane yields relative to in-plane yields and out-of-plane yields relative to in-plane yields. The advantage of taking ratios is a reduction in systematic uncertainties due to cancellation of uncertainties from several sources. The propagation of uncertainties follows Sect. 3.1.6. The yield ratio is expressed as:

$$r = \frac{Y_A}{Y_B} = \frac{Y_A^{\text{meas}} - Y_A^{\text{bkgd}}}{Y_B^{\text{meas}} - Y_B^{\text{bkgd}}}, \quad (3.13)$$

where A and B denote different event-plane orientations of the yield. The statistical errors, coming only from the terms, Y_A^{meas} and Y_B^{meas} , which were completely uncorrelated were calculated as:

$$\sigma_r^{\text{stat}} = \left| \frac{1}{r} \right| \sqrt{\left(\frac{\sigma_A}{Y_A} \right)^2 + \left(\frac{\sigma_B}{Y_B} \right)^2}. \quad (3.14)$$

The scale uncertainties, displayed as colored bands on the yield and width plots, are correlated and completely cancel in the ratio. Uncertainties from the RPF background subtraction are propagated with Eq. 3.11, but now applied to Eq. 3.13, which includes correlated background terms in the numerator and denominator of the ratio, so that part of the background uncertainty cancels. The correlated background uncertainties are given by:

$$\sigma_r^{\text{bkgd}} = \sqrt{\sum_{i=0}^N \sum_{j=0}^N \frac{\partial r}{\partial p_i} \frac{\partial r}{\partial p_j} \sigma_{ij}}. \quad (3.15)$$

Where, p_i 's are the parameters of the RPF fits. Fig. 3.13 shows the near-side (left) and away-side (right) associated-yield ratios of out-of-plane/in-plane and mid-plane/in-plane for 15-20 (top) and 20-40 GeV/ c (bottom) jets.

For $15 < p_{T,\text{jet}} < 20$ GeV/ c , the out-of-plane to in-plane associated-yield ratio shows slight enhancement out-of-plane relative to in-plane at low- $p_{T,\text{assoc}}$, although the effect is small. This can potentially be due to additional induced gluon radiation that out-of-plane and mid-plane jets would experience relative to in-plane jets, possibly from the longer path length traversed by jets that are not in-plane. Deviations of the yield ratios from 1.0 are not statistically significant on the away-side, although a small suppression is seen in-plane relative to both mid-plane and out-of plane on both the near-side and the away-side for $2.0 < p_{T,\text{assoc}} < 5.0$ GeV/ c for out/in. The suppression is expected to occur at a higher momentum fraction (z) of the jet. On the away-side, the effects are favoring a redistribution of energy from high-momentum constituents to lower momentum constituents.

For $20 < p_{T,\text{jet}} < 40$ GeV/ c the near-side ratios are consistent with 1.0 with some movement at the two lowest $p_{T,\text{assoc}}$ bins. On the away-side, mid/in is consistent with 1.0 with a little enhancement and out/in has an enhancement at high- $p_{T,\text{assoc}}$ and suppression at low- $p_{T,\text{assoc}}$. This observation contradicts the expectations. If in-plane jets interact less, we expect ratios to be < 1.0 at high- $p_{T,\text{assoc}}$ and > 1.0 at low- $p_{T,\text{assoc}}$. This is a reminder of the competing effects in the analyzed momentum range, and it is an indication that the

Table 3.5: Results of fits to Fig.3.13 (top panel: 15-20 GeV/ c jets) to a constant c , the χ^2 over the number of degrees of freedom (NDF), the number of standard deviations σ of c from one, and the range of c within a 90% confidence limit (CL).

parameter	Near-side		Away-side	
	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$
c	0.93 ± 0.042	0.949 ± 0.038	0.89 ± 0.065	0.997 ± 0.066
χ^2/NDF	0.68	0.67	0.71	0.18
σ	-1.6	-1.3	-1.7	-0.1
90% CL	$0.86 - 1.00$	$0.89 - 1.01$	$0.78 - 1.00$	$0.94 - 1.05$

expected path length effects due to jet energy loss are dominated by the fluctuations in the medium.

Alternatively, the initial geometry within specific centrality bins may be fluctuating at a larger magnitude than any possible event-plane dependence [117, 118]. The possibility of this occurrence can be studied by looking at the initial configuration and selecting low and high ellipticity events by using the q_n flow vector found within a selected centrality range.

To study the impact of surface bias and event-plane resolution, a check was performed to investigate the systematic change in the ratio of yields (out/in, mid/in) with the angle of the event plane by fitting a constant to Fig. 3.13 for jets with $15 < p_{\text{T,jet}} < 20$ GeV/ c (top) and $20 < p_{\text{T,jet}} < 40$ GeV/ c (bottom). The systematic uncertainties are treated as uncorrelated point-to-point and added to the statistical uncertainties in quadrature. The results are shown in Tabs. 3.5 and 3.6 and are consistent with one. Care should be taken when interpreting the results, as various effects can be in play and medium modifications could give way to a $p_{\text{T,assoc}}$ dependence. This $p_{\text{T,assoc}}$ dependence can be seen as an enhancement of associated yields for $p_{\text{T,assoc}} \geq 2$ GeV/ c on the near-side from the geometric kinematic biases due to the "hard-core" requirement of jet reconstruction.

3.3 Conclusions

The measurement of jet-hadron correlations relative to the event plane is reported for the 20-50% most central events in Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV in STAR. Partonic

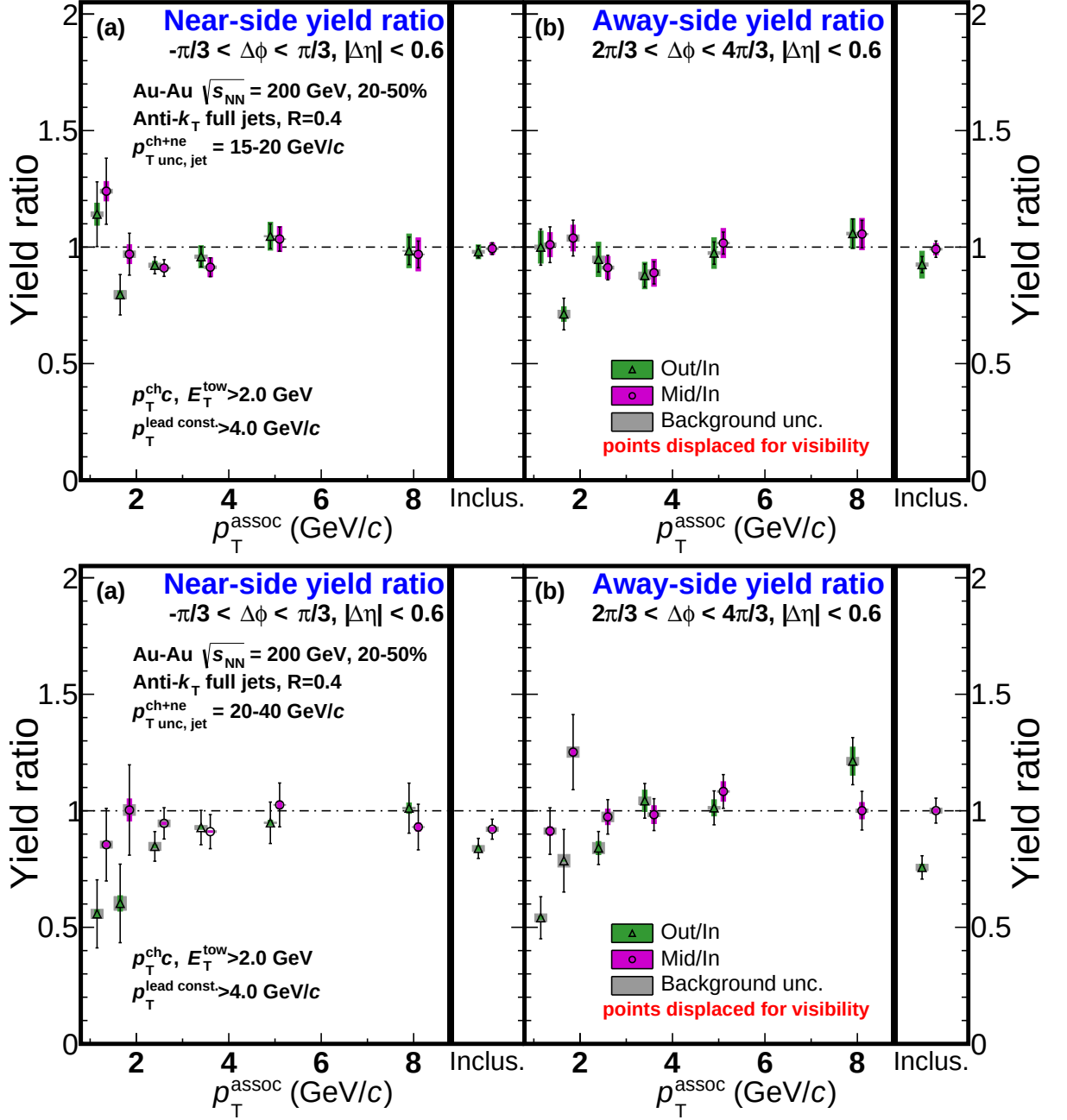


Figure 3.13: Near-side (left) and away-side (right) associated-yield ratios (of out-of-plane and mid-plane to in-plane) vs $p_{T,assoc}$ for 15-20 (top) and 20-40 (bottom) GeV/c full jets in 20-50% centrality collisions. The grey bands describe the systematic uncertainties of the background fits which are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the JES correction. Points are displaced for visibility.

Table 3.6: Results of fits to Fig.3.13 (bottom panel: 20-40 GeV/ c jets) to a constant c , the χ^2 over the number of degrees of freedom (NDF), the number of standard deviations σ of c from one, and the range of c within a 90% confidence limit (CL).

parameter	Near-side		Away-side	
	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$
c	0.874 ± 0.043	0.937 ± 0.043	0.752 ± 0.064	1.02 ± 0.075
χ^2/NDF	1.4	0.19	2.1	0.49
σ	-2.9	-1.5	-3.9	0.2
90% CL	$0.77 - 0.98$	$0.90 - 0.97$	$0.56 - 0.94$	$0.91 - 1.12$

interactions are directly related to the distance traversed in the medium, so it is expected that medium-induced jet modifications should depend on the path length. The angle of the jet, measured with respect to the event plane, is correlated on average with the jet's path length through the medium. In this analysis, the average path length of away-side jets is potentially increased due to the surface bias of the near-side trigger jet. This work utilizes the RPF background-subtraction method to remove the event-plane dependent background while reducing uncertainties and assumptions associated with previous background-subtraction techniques. Associated yields, their ratios, and jet-peak widths are extracted for each event-plane orientation and compared with different average path lengths and JEWEL model calculations. JEWEL performs better in describing the associated yields and widths at higher $p_{\text{T,assoc}}$ when recoil partons are not included. Conversely, including recoil partons leads to JEWEL providing a better description of the lower $p_{\text{T,assoc}}$ region. This study highlights the importance of conducting further tuning of Monte Carlo simulations to accurately describe the results in this analysis for jets that are biased towards hard-fragmented jets due to the HT-trigger and hard-core constituent requirements.

Within the precision of the current measurement at $\sqrt{s_{\text{NN}}} = 200$ GeV, the associated yields and jet-peak widths show no dependence on the event plane. The ratios derived from the associated yields are used to quantify the differences, but they do not deviate significantly from 1.0. For the $20 < p_{\text{T,jet}} < 40$ GeV/ c , there were indications of potential modifications observed in the inclusive bin of $1.0 < p_{\text{T,assoc}} < 10$ GeV/ c . The results presented in this study align with the findings observed in hadron-hadron and jet-hadron

correlations reported in Ref. [113] for RHIC and Ref. [108] for LHC energies. The lack of clear event-plane dependence in our data indicates that any dependence of these modifications on the average path-length is less than our experimental uncertainties.

Chapter 4

Generalized Angularities

Substructure based observables calculated from constituents of clustered jets can potentially have sensitivity to variety of QCD processes, from initial hard (high- Q^2) parton scattering to hadronization. Considering that, a constituent carries two major types of information, it's energy relative to the jet ($z_{\text{const.}} = \frac{p_{\text{T, const.}}}{p_{\text{T, jet}}}$) and its direction with respect to the jet-axis, usually characterized by it's $\eta - \varphi$ coordinate relative to the jet-axis ($\Delta R_{\text{const.,jet}} = \sqrt{(\eta_{\text{const.}} - \eta_{\text{jet}})^2 + (\varphi_{\text{const.}} - \varphi_{\text{jet}})^2}$). An intuitive way to put these two information together is by raising $z_{\text{const.}}$ and $\Delta R_{\text{const.,jet}}$ to exponents κ and β respectively and multiplying them. That would give bunch of per-constituent scalars, which we can add up for a jet to get a family of jet observables called *Generalized angularities*.

$$\lambda_{\beta}^{\kappa} = \sum_{\text{const} \in \text{jet}} z_{\text{const.}}^{\kappa} \left(\frac{\Delta R_{\text{const.,jet}}}{R} \right)^{\beta} \quad (4.1)$$

The parameters κ and β provide tune experimental sensitivity to hard and wide-angle radiation, respectively. λ_{β}^1 's are IRC safe angularities [119], which probe the average angular spread of energy around the jet-axis. They are radial moments of the jet's transverse momentum profile, also known as differential jet-shape ($\rho(r)$) given by Eq. 1.12. Relating with e^+e^- collision measurements, λ_1^1 is the jet *width* or *girth* [120] and λ_2^1 is the jet *thrust* [121]. Further, λ_2^1 is related to the jet-mass (m_{jet}) as $\left(\lambda_2^1 = \frac{m_{\text{jet}}^2}{p_{\text{T, jet}}^2} + \mathcal{O}((\lambda_2^1)^2) \right)$, providing an IRC-safe way of accessing m_{jet} information as it doesn't directly depend on hadron masses. $\lambda_{0.5}^1$ was been shown to have sensitivity to initiating parton-type of the jet, called *LHA angularity* after the Les Houches 2015 where the quark-gluon discrimination study was first

shown [122].

As an experimental observable, it is useful to restrict the summation in Eq. 4.1 to only charged constituents / tracks and mitigate contamination from towers like pileup, especially in the case of heavy-ion collisions. This makes $\lambda_{\beta>0}^1$ IRC unsafe, but as shown in [21, 123] comparison to theoretical predictions [124, 125] is still possible. Comparisons to theoretical predictions can be further simplified by disentangling the pQCD and npQCD contributions to the angularity, through use of grooming algorithms [126].

On the heavy-ion side, angularity measurements can help determine non-trivial biases introduced into nuclear modification factors. As shown in [127], instead of observing expected effects like jet-broadening in central Pb+Pb collisions, an apparent narrowing effect is observed for small-area ($R = 0.2$) jets.

Experimental measurements of IRC unsafe angularities like λ_0^0 , the jet-*multiplicity* and λ_0^2 jet-*momentum dispersion* ($= (p_T^D)^2$) complement the $\lambda_{\beta>0}^1$ with further information about jet fragmentation.

In this chapter, various generalized angularities are measured at RHIC energies. These measurements explore jet fragmentation in a phase space of lower-momentum jets complementing LHC measurements at high energies.

4.1 Angularities of Au+Au jets

A brief exploratory analysis done on jets from the dataset used in chapter 3 is shown. Distributions of Girth (g), p_T^D , and LeSub given by,

$$(g, p_T^D, \text{LeSub}) \equiv \left(\lambda_1^1, \sqrt{\lambda_0^2}, p_{\text{T,const.}}^{\text{lead.}} - p_{\text{T,const.}}^{\text{sublead.}} \right) \quad (4.2)$$

LeSub is the p_T difference between the leading and sub-leading constituents in the jet, which represents the split in momentum at the earliest fragmentation of the jet. The STAR preliminary results shown in this section were presented in the 30th International Conference

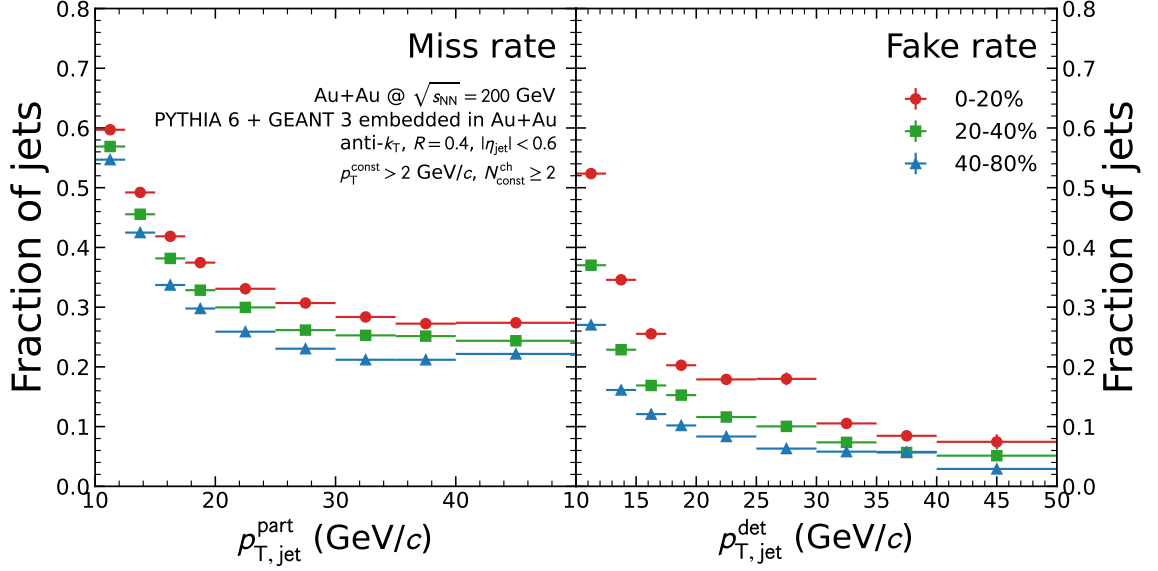


Figure 4.1: Miss rate (left) and fake rate (right) calculated from matched jets from the PYTHIA-6 (STAR tune) + GEANT3 + AuAu 2014 minimum bias pedestal embedding simulation. Each panel shows the miss/fake rates for jets from 0-20%, 20-40%, and 40-80% centrality events as red circles, green squares, and blue triangles respectively.

on Ultra-relativistic Nucleus-Nucleus Collisions (Quark Matter 2023) [128].

4.1.1 Dataset and Selection

Jets are reconstructed with same data (Au + Au collision from 2014), events (HT2-triggered, $|v_z| < 24$ cm; Sect. 3.1.1), track and tower selection (Sect. 3.1.2), constituents (hard-core, $cp_{T, \text{trk}}, E_{T, \text{tow.}} > 2.0$ GeV) and definition ($R = 0.4$, anti- k_T , boost-invariant p_T^2) as in Sect. 3.1.4. The jet selection used in this analysis differs only in that we omit the requirement for a high-tower triggered ($E_{T, \text{tow.}} \geq 5.4$ GeV) inside the jet-cone and add an extra minimum jet-area cut $A_{\text{jet}} > 0.4$ to reduce combinatorial jet contribution. The mathematical requirements to calculate the observables defined in Eq. 4.2 further imposes a $N_{\text{trk.}} \geq 2$ selection on the jets.

4.1.2 Unfolding

Residual background and detector effects are removed by mapping between particle and detector level jets using the embedding simulation and matching procedure described in section 3.1.7. Efficiencies involved in jet reconstruction are shown in Fig. 4.1. A general decrease is seen in miss/fake rates going from smaller to larger $p_{T,\text{jet}}$'s and from central to peripheral collisions. The larger miss/fake rate in central and mid-central collisions (0-20%, 20-40%) can be explained by the presence of significant underlying event / minimum-bias pedestal compared to peripheral collisions (40-80%). MultiFold [129] method was used to simultaneously unfold $p_{T,\text{jet}}$, η_{jet} , ϕ_{jet} , $N_{\text{cons,charged}}$, g , p_T^D and LeSub. MultiFold uses Dense Neural Networks (DNNs) implemented using the EnergyFlow package [130], which are trained on full embedding sample at the detector level and the generator level. See appendix E for more information about unfolding procedure and appendix D for more information on machine learning. Each classifier has three hidden layers of 100 nodes, and is trained with batches of 1024 jets on 80% of the jets, the remaining 20% being used for validation. Training is stopped once the validation loss has not improved for 10 epochs, and the network with the lowest validation loss is kept and used as the starting point in the next iteration. The evolution of the losses during the unfolding is shown in Fig. 4.2 for two runs with the nominal settings, since the training logs of the nominal runs themselves were not kept. In step 1, the validation loss reaches its minimum within the first few epochs of each iteration while the training loss keeps decreasing, i.e. the classifier starts to learn fluctuations of the training sample, which the early stopping prevents from entering the weights. In step 2, the losses change little after the first epoch, as the reweighting between consecutive iterations is small.

Closure of the MultiFold is a measure of confidence in the algorithm determined by applying a pre-trained MultiFold on a detector level jet sample for which the particle level is known and comparing the two through ratios. Here, the matched embedding jets of each centrality class (0-20% and 40-80%) are randomly split into two independent samples: the

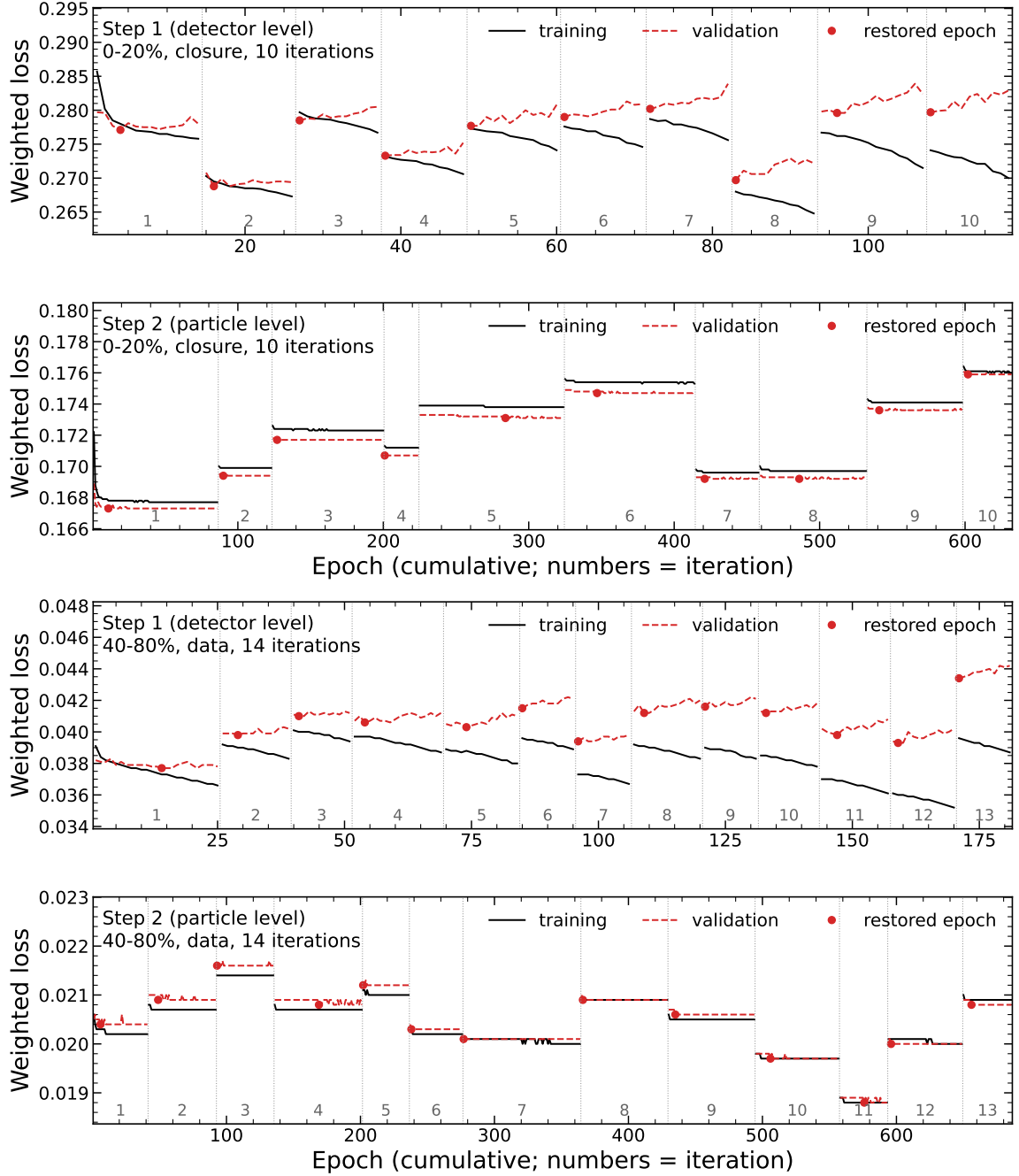


Figure 4.2: Evolution of the training (solid black) and validation (dashed red) losses of the MultiFold classifiers over all epochs of all iterations, for a 0-20% closure run with 10 iterations (left) and a 40-80% data run with 14 iterations (right), both with the settings of the nominal unfolding. The top (bottom) panels show the step-1 (step-2) classifier, which reweights the embedding to the data at detector level (the prior to the reweighted embedding at particle level). Dotted vertical lines separate the iterations, numbered at the bottom, and red dots mark the epoch with the lowest validation loss, whose network weights are kept and used as the starting point of the next iteration. The loss is the categorical cross-entropy weighted by the per-jet weights.

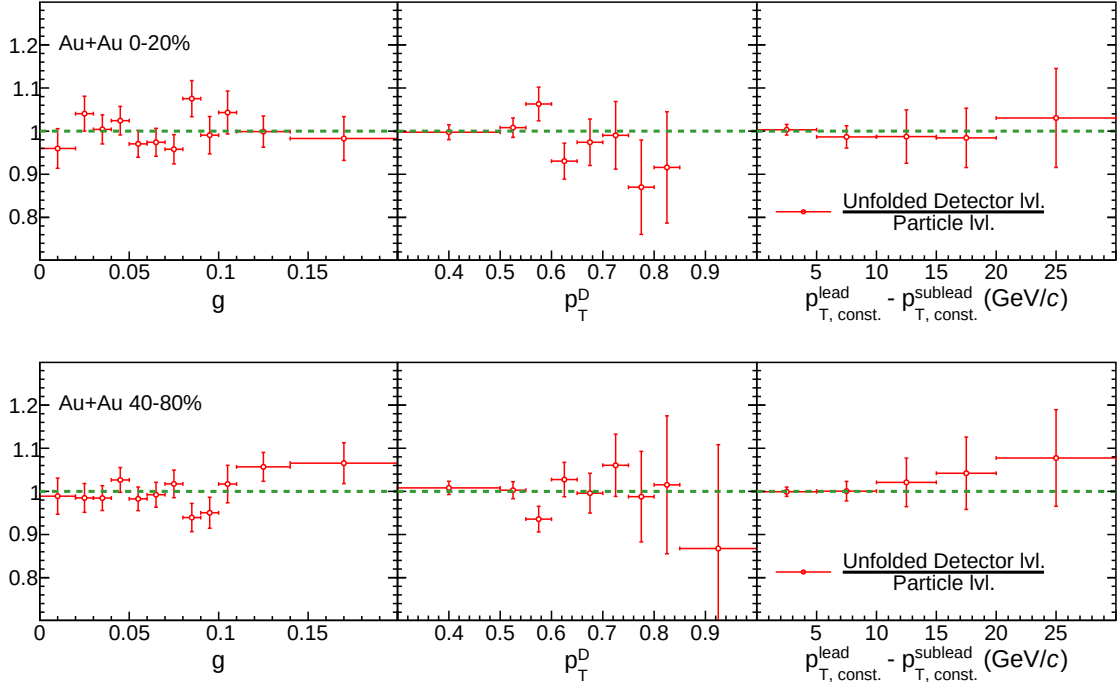


Figure 4.3: Closure test for g , p_T^D and LeSub showing the ratio of distributions calculated using jets from unfolded detector level and particle level events from 0-20% (top) and 40-80% (bottom) central Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV.

detector-level jets of 20% of them serve as pseudo-data, which are unfolded by a MultiFold trained on the remaining 80% with the same settings as for the measured data (10 iterations, batch size of 1024), and the unfolded result is compared to the particle-level jets of the pseudo-data sample (pseudo-truth). Since both samples come from the same PYTHIA-6 simulation, this tests the technical closure of the method rather than its dependence on the prior. Proper closure is demonstrated in Fig.4.3 for the observables studied in these proceedings, with the ratios of distributions being consistent with unity within statistical errors. This study is the first instance of the MultiFold method being used to deconvolute detector effects from heavy-ion collision events.

4.1.3 Systematic Uncertainties

Three sources of systematic uncertainty are considered, all related to the unfolding. The regularization of MultiFold is varied by changing the number of iterations from 10 to 14,

and the relative difference between the two unfolded distributions is taken as an uncertainty. Any residual non-closure from MultiFold, the deviation from unity of the closure ratios in Fig. 4.3 evaluated in each $p_{T,\text{jet}}$ bin, was also included as an uncertainty. A prior variation uncertainty was added from previous calculations using STAR $\sqrt{s} = 200$ GeV p+p collision data. In each bin, the largest of the three relative deviations is taken as the total systematic uncertainty. The breakdown of the systematic uncertainty by source is shown in Fig. 4.4, for both centrality classes, both $p_{T,\text{jet}}$ bins and the ratio of central to peripheral distributions. The total is typically 5-20%, and grows in the sparsely populated tails of the distributions, up to about 90% at large g in central collisions, where the change of the number of iterations dominates. Elsewhere the total is set by either the prior or the iteration variation, while the non-closure is the largest source only in a few bins. The statistical uncertainties are below about 10%, except in the highest LeSub bin at $15 \leq p_{T,\text{jet}} < 20$ GeV/ c , where they reach about 30%. For the ratio of central to peripheral distributions, the total uncertainties of the two centrality classes are added in quadrature, i.e. they are assumed to be uncorrelated. This is a conservative choice, since the unfolding-related variations are expected to shift both centrality classes in the same direction. Detector-level sources such as the tracking efficiency (Sect. 3.1.2) and the tower energy scale were not evaluated in this exploratory analysis.

4.1.4 Result and Discussion

Fully corrected measurements of g , p_T^D and LeSub distributions calculated from jets with $15 \leq p_{T,\text{jet}} < 20$ GeV/ c and $20 \leq p_{T,\text{jet}} < 45$ GeV/ c from events within centrality ranges of 0-20% (central) and 40-80% (peripheral), and R_{CP} ratio comparisons between distributions from central and peripheral events calculated according to Eq. 1.11 are shown in Fig. 4.5. The p_T^D distribution shows a discontinuity at 0.7 due to the strong dependence on number of constituents in jet. Results are consistent between central and peripheral collisions within conservative systematic uncertainties. This can be attributed to the assumption of

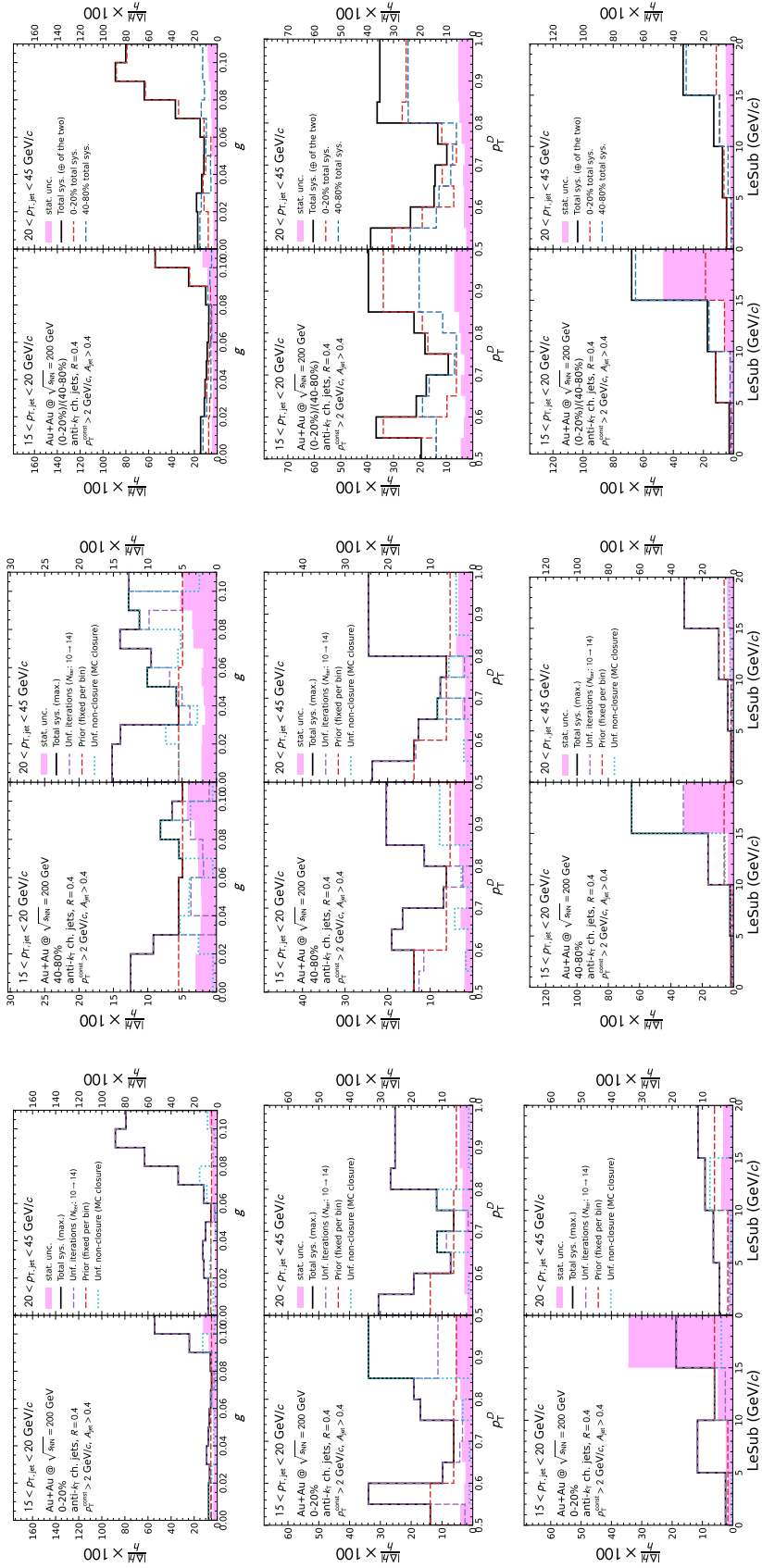


Figure 4.4: Relative systematic uncertainties of the unfolded g (top), p_T^D (middle) and LeSub (bottom) distributions for 0-20% (left) and 40-80% (middle) central Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV, and of their central over peripheral ratio (right). Each plot shows jets with $15 \leq p_{T,jet} < 20$ GeV/c (left panel) and $20 \leq p_{T,jet} < 45$ GeV/c (right panel). The magenta band is the statistical uncertainty. In the left and middle columns, the dashed red and dotted cyan lines are the uncertainties from the number of MultiFold iterations, the prior and the MultiFold non-closure respectively, and the solid black line is the total, their maximum in each bin. In the right column, the dashed red (blue) line is the total uncertainty for 0-20% (40-80%) collisions, and the solid black line is their sum in quadrature.

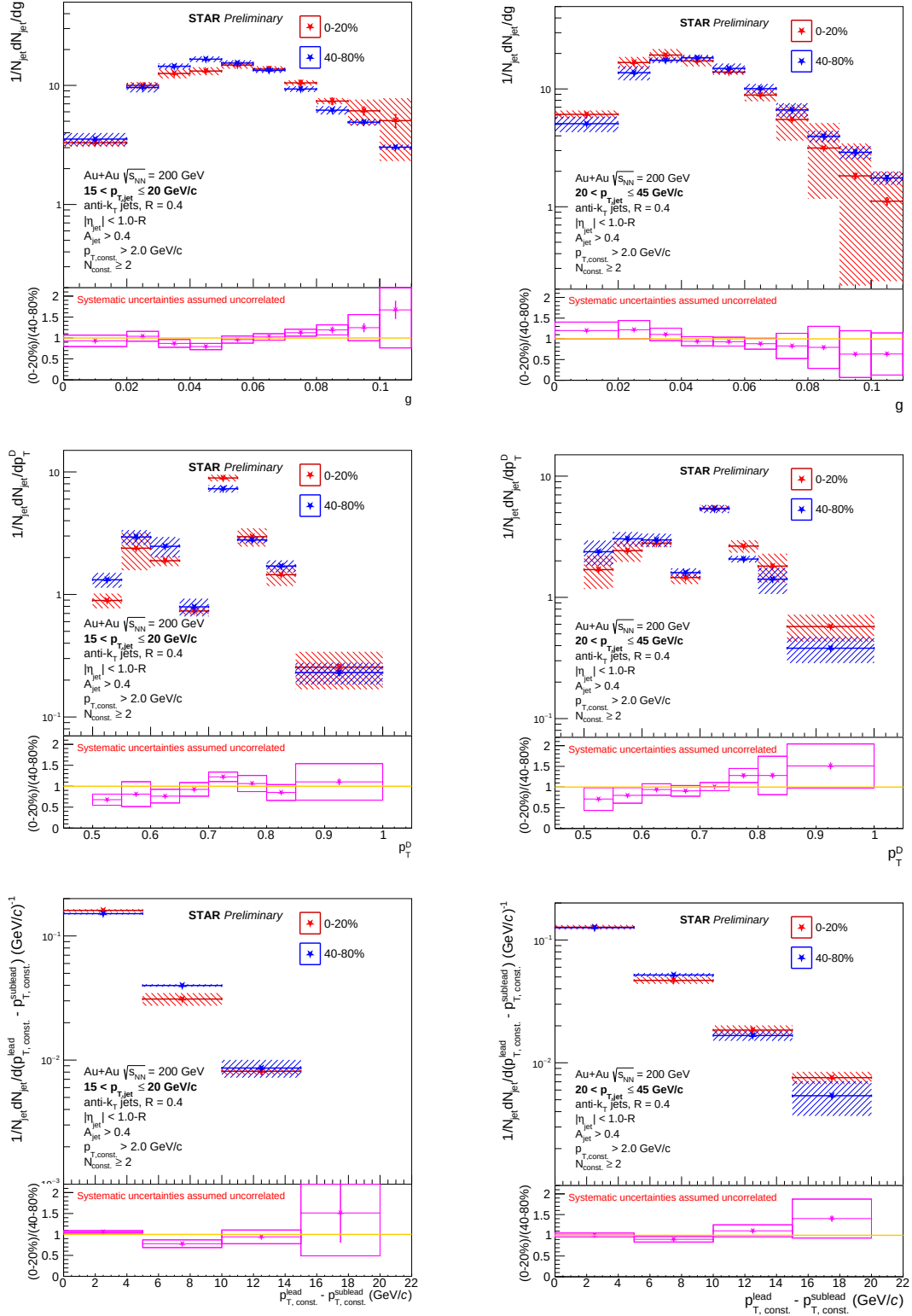


Figure 4.5: Unfolded, normalized distributions of $g = \lambda_1^1$ (top), p_T^D (middle) and LeSub (bottom) for jets with $15 \leq p_{T,\text{jet}} < 20$ GeV/c (left) and $20 \leq p_{T,\text{jet}} < 45$ GeV/c (right) from central (0-20%, red stars) and peripheral (40-80%, blue stars) Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV. Systematic uncertainties are shown as hatched boxes. Central over peripheral ratios (magenta stars) are in the lower panels with their systematic uncertainties represented as magenta boxes. Systematic uncertainties are assumed to be uncorrelated between the centrality ranges.

uncorrelated systematic uncertainties, and a bias in the analysis towards hard-fragmented jets due to the selection of events and constituents for jet clustering. Harder-fragmented jets, on average are narrower and have lower interaction time with QGP, which could make any relative modification in central events compared to the peripheral ones lower than what would be potentially observed in an inclusive jet analysis. This is the first heavy-ion analysis to use unbinned unfolding of detector effects, hence there is room for improvement in application of the technique, which will be explored in the next section.

4.2 Inclusive and groomed angularities of $p + p$ jets

The last section showed that valuable information about jet substructure is lost by over-subtracting background with a naive tool like a hard-core constituent cut. Therefore, a more sophisticated method of isolating perturbative and non-perturbative effects must be used. One such method used in this analysis is the SoftDrop [131] grooming. However, before the last section's analysis is repeated with groomed jets, the unbinned unfolding techniques must be studied, understood and developed for RHIC data in a simpler system. Thus the $p + p$ angularities study presented here serves a three-fold purpose. First, it provides a baseline for future inclusive and groomed angularity studies in heavy-ion collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV with STAR. Second, it develops unbinned unfolding for RHIC energies, by trying out new methods and streamlining the pipeline from unclustered constituents to results. Third, it explores new kinds of measurements beyond plain distributions, which become possible once binning is no longer a hard requirement of the unfolding. Only the IRC-safe angularities are shown here; other variables, such as λ_0^2 , are being worked on for the final publication. Earlier iterations of the $p + p$ angularities have been presented orally at the 11th International Conference on Hard and Electromagnetic Probes of High-Energy Nuclear Collisions (Hard Probes 2023) [132] and as posters at Hard Probes 2024 and Hard Probes 2026. A few of the recommendations of this work for unbinned unfolding are featured in the white paper [23].

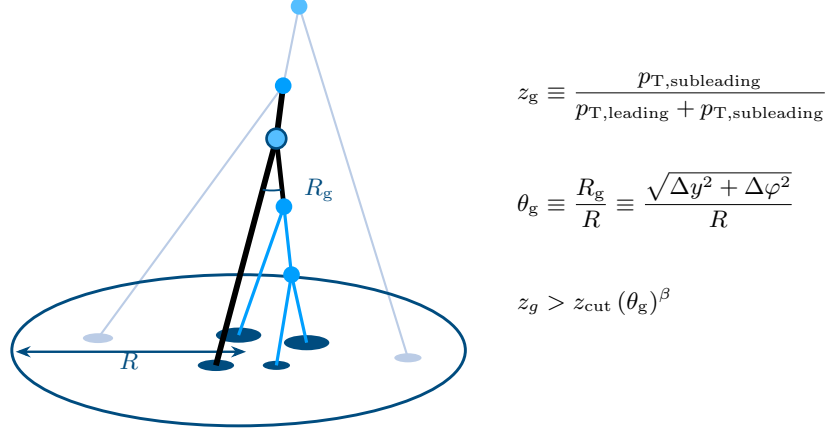


Figure 4.6: Schematic of SoftDrop grooming of a jet of radius R . Going backward through the C/A clustering tree of the jet, soft wide-angle branches (light lines and constituents) fail the SoftDrop criterion $z > z_{\text{cut}} \theta^\beta$ and are removed. The first splitting that satisfies it (open circle) separates the leading (black) and subleading (blue) prongs, and defines the groomed momentum fraction z_g and the groomed radius R_g ($\theta_g = R_g/R$) of the jet.

4.2.1 SoftDrop

SoftDrop is a jet-grooming technique that removes soft, wide-angle radiation from a jet, which is the part most sensitive to hadronization and to the underlying event, while keeping its hard, collinear core. The procedure is illustrated in Fig. 4.6. First, the constituents of the jet are reclustered with the Cambridge/Aachen (C/A) algorithm, which merges the closest pairs in (η, ϕ) first, so that the resulting clustering tree is ordered in angle, with the widest-angle splittings at the top. Second, the tree is traversed backward, starting from the last clustering step: at each step the jet is split into its two branches, and the SoftDrop criterion

$$\frac{\min(p_{T,1}, p_{T,2})}{p_{T,1} + p_{T,2}} > z_{\text{cut}} \left(\frac{\Delta R_{12}}{R} \right)^\beta \quad (4.3)$$

is checked, where $p_{T,\{1,2\}}$ are the transverse momenta of the two branches and ΔR_{12} is their distance in the (η, ϕ) plane. If the criterion fails, the softer branch is removed from the jet and the procedure continues with the harder one. It stops at the first splitting that satisfies the criterion, and the constituents that remain form the groomed jet (the open circle and the black and blue branches in Fig. 4.6). This splitting defines the two groomed observables of the jet: the groomed momentum fraction z_g , the left-hand side of Eq. 4.3, and

the groomed radius $R_g = \Delta R_{12}$, or equivalently $\theta_g = R_g/R$. The soft threshold z_{cut} sets how asymmetric a splitting may be and still be kept, and the angular exponent β sets how much more strongly wide-angle splittings are groomed than collinear ones. For $\beta = 0$ the criterion is independent of the angle and SoftDrop reduces to the modified mass drop tagger (mMDT), while for $\beta \rightarrow \infty$ nothing is groomed. Since the mMDT removes soft radiation at all angles, the groomed jet is largely insensitive to the underlying event and hadronization, and its observables are calculable in perturbative QCD, with z_g directly probing the QCD splitting function [126, 131]. This work uses $z_{\text{cut}} = 0.2$ and $\beta = 0$.

Using SoftDrop, an angularity can be decomposed into a component dominated by perturbative QCD (pQCD) and one dominated by non-perturbative effects (npQCD). The pQCD component is the angularity of the groomed jet, and the npQCD component is the difference between the angularities of the ungroomed and the groomed jet:

$$\begin{aligned}
\mathcal{O}_\beta^\kappa &= \mathcal{O}_{\beta,\text{pQCD}}^\kappa + \mathcal{O}_{\beta,\text{npQCD}}^\kappa \\
\mathcal{O}_{\beta,\text{pQCD}}^\kappa &\equiv \mathcal{O}_{\beta,g}^\kappa \\
\mathcal{O}_{\beta,\text{npQCD}}^\kappa &\equiv \Delta\mathcal{O}_\beta^\kappa = \mathcal{O}_\beta^\kappa - \mathcal{O}_{\beta,g}^\kappa
\end{aligned} \tag{4.4}$$

The momentum balance (z_g) and opening angle (R_g) of the first hard splitting connect the measured jet shapes directly to calculations in perturbative QCD. A larger R_g implies that the most prominent splitting in the jet occurred earlier in its evolution, which, because of the color coherence discussed earlier, opens up more phase space for subsequent perturbative splittings that survive the grooming. Fig. 4.7 illustrates this: the npQCD fraction of the jet width, $\frac{\Delta\lambda_1^1}{\lambda_1^1} \times 100$, decreases with increasing R_g , going from highly collimated, pencil-like jets to jets with a clear two-pronged structure.

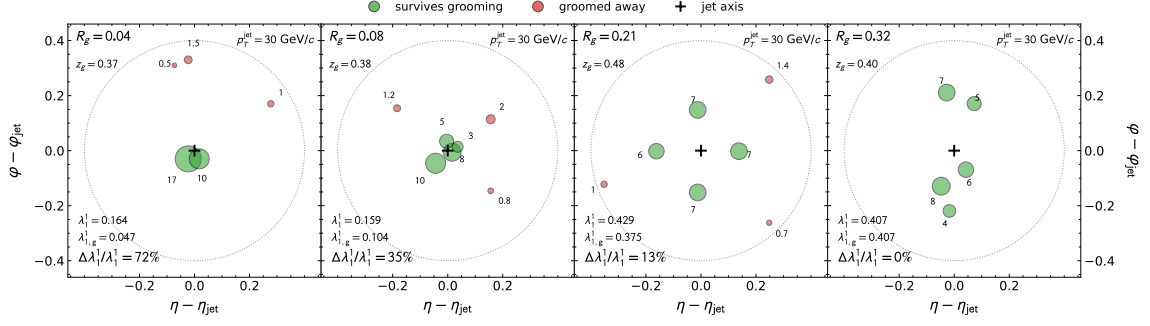


Figure 4.7: Ways of arranging constituents (each represented as a circle whose radius represent $\frac{p_{T,\text{const}}}{p_{T,\text{jet}}}$) in the (η, ϕ) space within a jet with $p_{T,\text{jet}} = 30 \text{ GeV}/c$ such that R_g of the jet increases going from left to right. Green (Red) colored constituents survive (are removed by) the SoftDrop grooming ($z_{\text{cut}} = 0.2$, $\beta = 0$).

4.2.2 Dataset and Methodology

The analysis uses data from $p + p$ collisions at $\sqrt{s} = 200 \text{ GeV}$ collected in 2012 using the STAR detector system. Charged-particle tracks and neutral energy depositions (towers) are measured using STAR's TPC and BEMC detectors respectively.

To enhance jet signal, only High-Tower (HT) triggered events (at least one tower with $E_{T,\text{tower}} \geq 4 \text{ GeV}$) are considered. Events containing a track with $p_{T,\text{track}} \geq 30 \text{ GeV}/c$ are rejected to avoid contamination from cosmic rays, as in the Au+Au analysis (Sect. 3.1.1). Tracks(Towers) with $p_{T,\text{track}} \geq 0.2 \text{ GeV}/c$ ($E_{T,\text{tower}} \geq 0.2 \text{ GeV}$) are considered for jet clustering. The tracks pass the same quality criteria as in the Au+Au analysis (Tab. 3.1), bad and dead towers are removed, and the full momentum of the tracks matched to a tower is subtracted from its energy (Eq. 3.3); the reasons for these choices are discussed in Sect. 3.1.2. Clustered jets completely falling within acceptance ($|\eta_{\text{jet}}| \leq 0.6$), with $p_{T,\text{jet}} \geq 10 \text{ GeV}/c$ and a plurality of charged tracks ($N_{\text{ch},\text{jet}} \geq 2$) are used for further study.

The tracks and towers are clustered into jets using the anti- k_T algorithm with a jet resolution parameter $R = 0.4$, implemented using the FastJet library [133], with the same particle-mass assumptions and recombination scheme as in Sect. 3.1.4. Unlike the hard-core Au+Au jets, all tracks and towers above 0.2 GeV are clustered, so the jets are inclusive. After obtaining the jets, the softer constituents are groomed using Recursive SoftDrop with

$\beta = 0$ and $z_{\text{cut}} = 0.2$. The groomed-jet's constituents lose non-kinematic information like charge, which is regained by matching them to the original jet's constituents.

4.2.3 Detector effects and Unfolding

The embedding simulation starts from PYTHIA6 (STAR Tune) at the particle level. The particles are then reconstructed as detector hits using the STAR detector system implemented in GEANT3 and embedded into zero-bias events from 200 GeV $p + p$ collisions (2012). The jets from particle and detector levels are matched using a $p_{T,\text{jet}}$ ordered nearness criteria, similar to the one used for the Au+Au embedding (Sect. 3.1.7). The matched jets can be used to show detector responses like shown in Fig. 4.8 (top) and $p_{T,\text{jet}}$ -resolution (Eq. 3.12) shown in Fig. 4.8 (bottom).

Unfolding is done using the Multifold method. Every jet is represented as a vector of 23 features. The four classifiers of Multifold are parametrized as DNNs with 3 hidden layers with 256 nodes each. The model weights are trained using the AdamW optimizer with a learning rate of 10^{-3} and weight decay of 10^{-2} . Convergence is achieved in two unfolding iterations. Statistical uncertainties are calculated using bootstrap ensemble with 50 model replicas each sub-sampling 500k each of the data-like and sim-like jets, so each model replica sees 1M jets every epoch. All the required unfolded distributions and profiles are calculated by using the per-jet unfolding weights.

Closure test of the unfolding shown in Fig. 4.10 and 4.11 is done by doing the same unfolding procedure replacing measured data with pseudo-data for which the particle level is known (pseudo-truth). The pseudo-unfolded distributions are then compared to the pseudo-truth via ratios

4.2.4 Statistical Uncertainties

The classifiers of the previous section are trained on finite samples of measured, reconstructed and generated jets, and the training itself is stochastic through the initialization of the model,

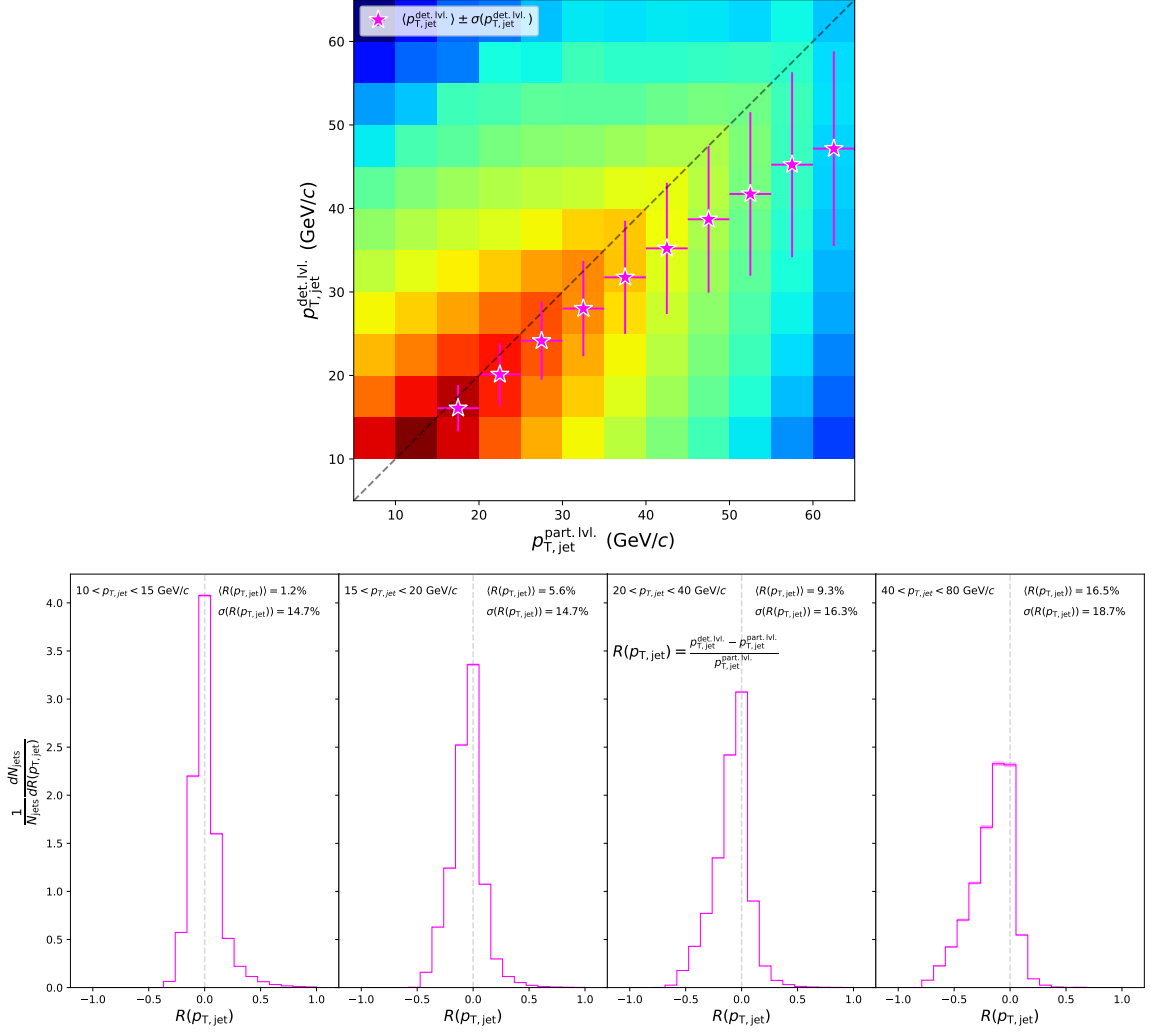


Figure 4.8: Jet- p_T response of the PYTHIA-6 + GEANT3 embedding for matched jets. (Top) Probability of a particle-level jet with $p_{T,jet}^{GEN}$ to be reconstructed with $p_{T,jet}^{RECO}$; the magenta stars are the mean $p_{T,jet}^{RECO}$ in each bin of $p_{T,jet}^{GEN}$, with the standard deviation as error bars. (Bottom) Distributions of the relative difference $R(p_{T,jet}) = (p_{T,jet}^{det.} - p_{T,jet}^{part.}) / p_{T,jet}^{part.}$ in four ranges of $p_{T,jet}^{part.}$; their mean and standard deviation, quoted in each panel, are the jet- p_T scale shift and resolution.

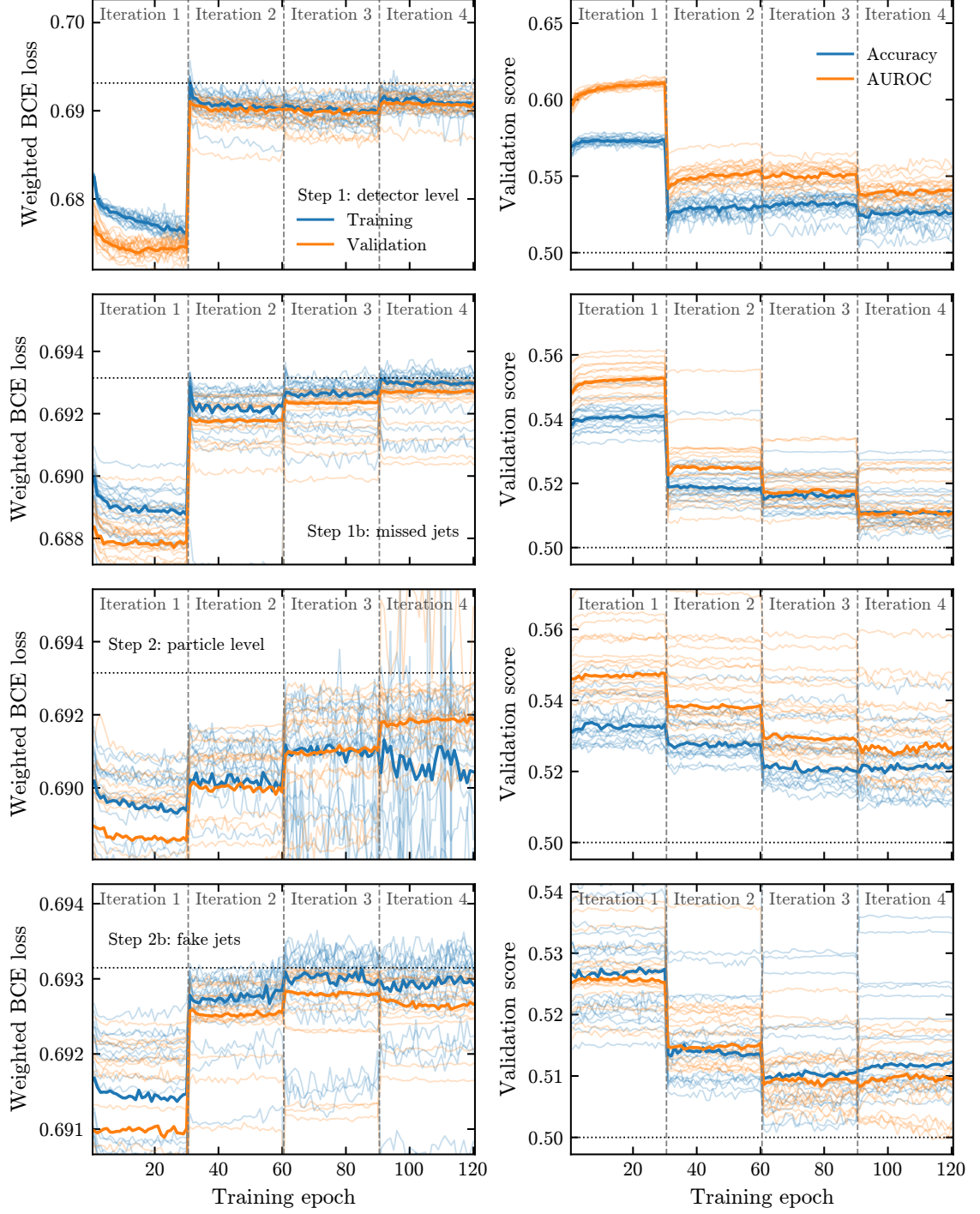


Figure 4.9: Training history of the four MultiFold classifiers over four iterations of 30 epochs each: step 1 (detector level, measured vs. reconstructed jets), step 1b (missed jets), step 2 (particle level) and step 2b (fake jets), from top to bottom. (Left) Weighted binary cross-entropy loss on the training (blue) and validation (orange) samples. (Right) Accuracy (blue) and area under the ROC curve (AUROC, orange) on the validation sample. Thin lines are the individual ensemble replicas and thick lines their median; the dotted lines mark a classifier with no separating power (loss $\ln 2$, score 0.5). The nominal result uses the second iteration.

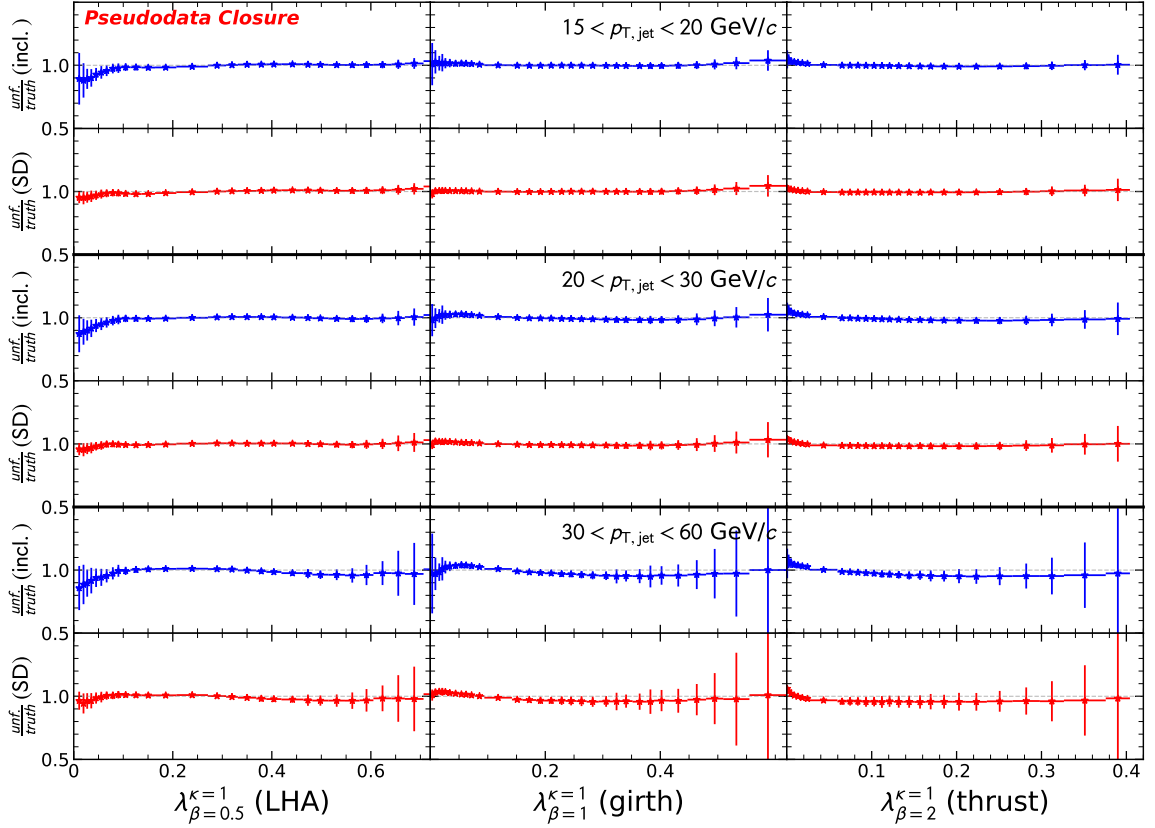


Figure 4.10: Closure test with the data-like prior: ratio of the pseudo-unfolded to the pseudo-truth distributions of $\lambda_{\beta}^{\kappa=1}$ for $\beta = 0.5$ (LHA, left), 1 (girth, middle) and 2 (thrust, right), for inclusive (blue) and SoftDrop-groomed (red) jets, in three $p_{T,\text{jet}}$ ranges (rows). Error bars are the statistical uncertainties.

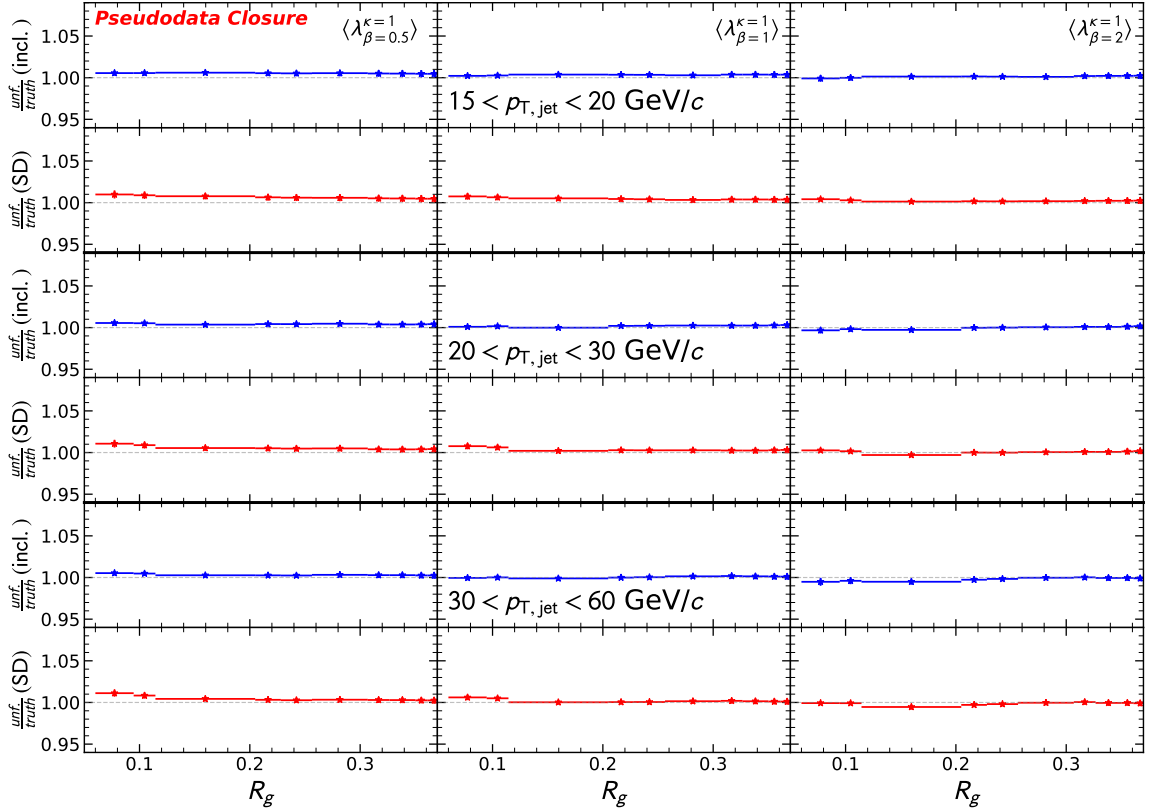


Figure 4.11: Closure test with the data-like prior for the $\langle \lambda_{\beta}^{\kappa=1} \rangle$ vs. R_g profiles: ratio of the pseudo-unfolded to the pseudo-truth profiles for $\beta = 0.5$ (left), 1 (middle) and 2 (right), for inclusive (blue) and SoftDrop-groomed (red) jets, in three $p_{T,\text{jet}}$ ranges (rows). Error bars are the statistical uncertainties.

dropout and the order of the mini-batches. The unfolded distribution inherits all of these, and after several trainings chained across iterations there is no closed form for propagating them. Instead, the whole procedure is repeated on resampled inputs and the spread of the outcomes is taken as the uncertainty.

Every classifier f is replaced by an ensemble of $N_{\text{rep.}}$ replicas. Replica r is trained on a random subsample of m jets drawn without replacement from each of the two classes it separates (e.g. m measured and m reconstructed jets for $f_{\text{meas.,reco}}$), split further into a training and a validation set, with its own initialization and dropout. The same replica index is carried through every classifier of every iteration, so replica r is a complete unfolding on its own, with weights $\omega_{n-1}^{(r)}$ and $\nu_{n-1}^{(r)}$ and, from eq. E.5,

$$p_{\text{gen.},n}^{(r)}(\mathbf{x}) = p_{\text{gen.},0}(\mathbf{x}) \prod_{k=0}^{n-1} \nu_k^{(r)}(\mathbf{x}) \equiv w_n^{(r)}(\mathbf{x}) p_{\text{gen.},0}(\mathbf{x}) \quad (4.5)$$

Where, $w_n^{(r)}(\mathbf{x})$ is the per-jet unfolding weight of replica r after n iterations. The subsampling is only done for training; at inference every replica is evaluated on every jet, so eq. 4.5 gives $N_{\text{rep.}}$ unfolded samples, each with the full statistics of the simulation.

Any binned quantity H , a histogram, a profile or a ratio of two of them, is then calculated once per replica from the weights $w_n^{(r)}$, and its central value and uncertainty are the mean and standard deviation over the ensemble,

$$\bar{H} = \frac{1}{N_{\text{rep.}}} \sum_{r=1}^{N_{\text{rep.}}} H^{(r)}, \quad \sigma_H^2 = \frac{1}{N_{\text{rep.}} - 1} \sum_{r=1}^{N_{\text{rep.}}} \left(H^{(r)} - \bar{H} \right)^2 \quad (4.6)$$

Ratios are formed replica by replica before averaging, so the correlation between numerator and denominator through the shared weights is kept. Where a single diverged replica would pull $\bar{H}$, the median or a trimmed mean over the ensemble is used for the central value instead; σ_H is always the standard deviation.

Since the replicas differ in the measured subsample, the simulated subsample and the training, σ_H contains the statistical uncertainty of the measurement, that of the simulation

and the variance of the training procedure, without separating them. With m smaller than the full samples, the spread is that of samples of size m and is conservative for the purely statistical part. $N_{\text{rep.}}$ only sets how precisely σ_H itself is known, not its size.

4.2.5 Systematic Uncertainties

Jet Reconstruction Related Uncertainties

Analysis was rerun with each of the following changes (compared to the nominal) done to the detector level of the embedding simulation. The differences in the final plots for the source was calculated. The Barlow test was performed on each source of systematic error to remove the statistical fluctuation from the variation. The relative uncertainty from every source, detector and unfolding related, is shown before the Barlow check in Figs. 4.12–4.15 and after it in Figs. 4.18–4.21, for the inclusive and groomed distributions and for the $\langle \lambda_{\beta}^{k=1} \rangle$ vs. R_g profiles.

- Tracking Efficiency - The uncertainty in tracking efficiency (Sect. 3.1.2) is estimated at $\pm 4\%$ for $p + p$ events.
- Variation of Hadronic Correction - For the towers, we remove the full contribution of the track momentum for our central values (Eq. 3.3, $f_{\text{had}} = 1$). As a variation on that, we remove half of the total track momentum contribution.
- Tower Gain Uncertainty - Tower energies increased by 3.8%
- $p_{T,\text{jet}}$ resolution - Use the $p_{T,\text{jet}}$ resolution values from fig. 4.8 to shift the $p_{T,\text{jet}}$ windows in which results are presented.

Unfolding related uncertainties

- Variation of the regularization parameter(number of iterations) - For our central values, we have used 2 iterations as we found the result to converge reliably by that point. The number of iterations is varied to 1 and 3.

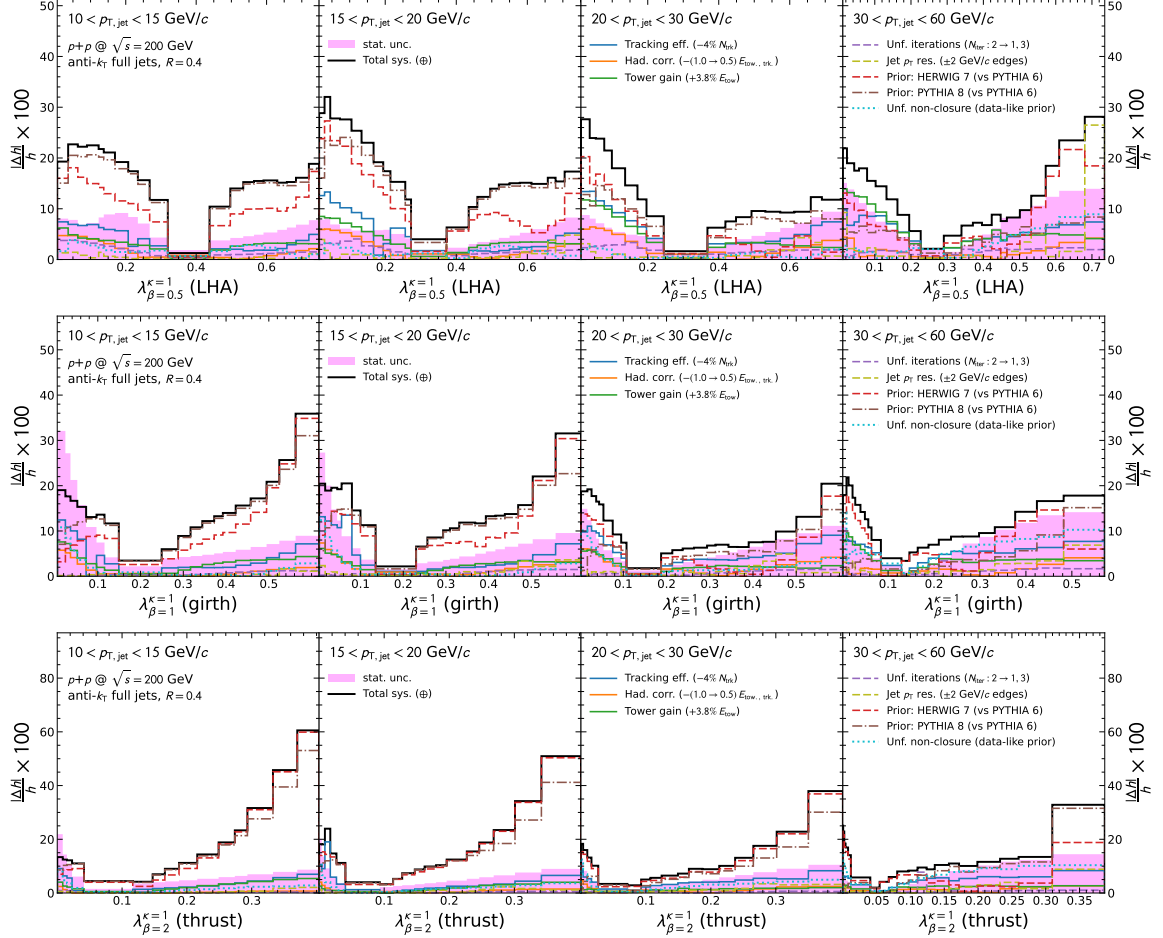


Figure 4.12: Relative systematic uncertainties of the $\lambda_{\beta}^{\kappa=1}$ distributions of inclusive jets, before the Barlow check (Sec. 4.2.5), for $\beta = 0.5, 1$ and 2 (top to bottom). The sources are the tracking efficiency, the hadronic correction and the tower gain (solid), the number of unfolding iterations, the jet- p_T resolution and the HERWIG-7 and PYTHIA-8 priors (dashed), and the non-closure (dotted); the black line is their sum in quadrature and the magenta band is the statistical uncertainty. Jet- p_T ranges are shown as panels.

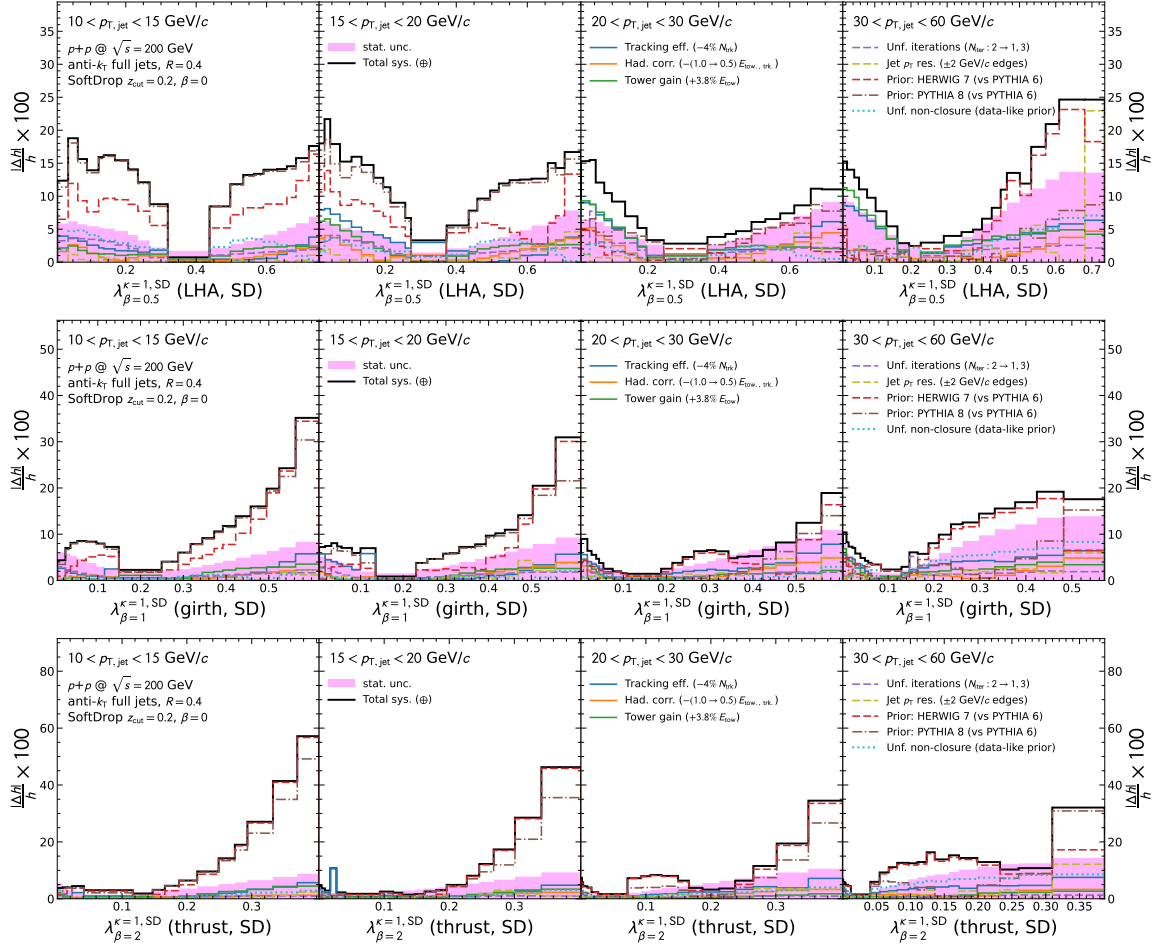


Figure 4.13: Relative systematic uncertainties of the $\lambda_{\beta=1}^{\kappa=1}$ distributions of SoftDrop-groomed jets, before the Barlow check (Sec. 4.2.5), for $\beta = 0.5, 1$ and 2 (top to bottom). The sources are the tracking efficiency, the hadronic correction and the tower gain (solid), the number of unfolding iterations, the jet- p_T resolution and the HERWIG-7 and PYTHIA-8 priors (dashed), and the non-closure (dotted); the black line is their sum in quadrature and the magenta band is the statistical uncertainty. Jet- p_T ranges are shown as panels.

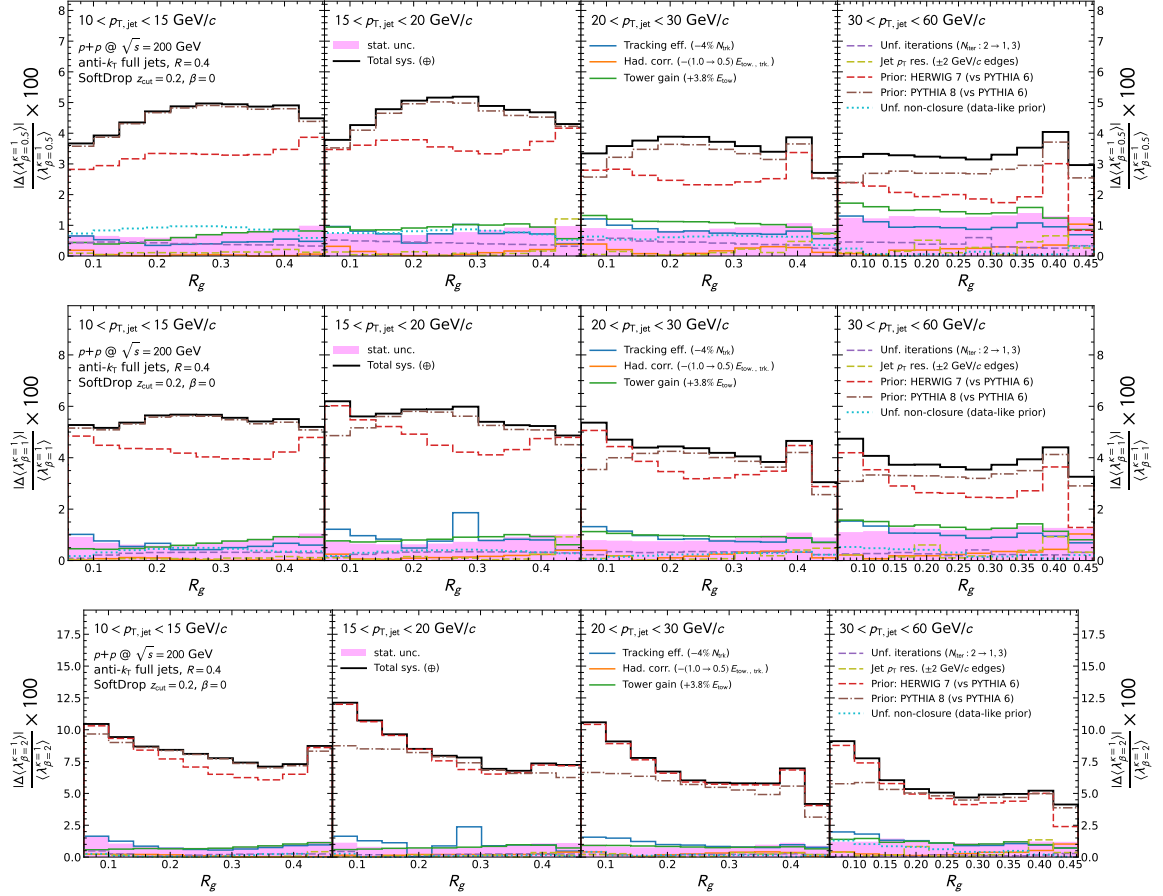


Figure 4.14: Relative systematic uncertainties of the $\langle \chi_{\beta}^{\kappa=1} \rangle$ vs. R_g profiles of inclusive jets, before the Barlow check (Sec. 4.2.5), for $\beta = 0.5, 1$ and 2 (top to bottom). The sources are the tracking efficiency, the hadronic correction and the tower gain (solid), the number of unfolding iterations, the jet- p_T resolution and the HERWIG-7 and PYTHIA-8 priors (dashed), and the non-closure (dotted); the black line is their sum in quadrature and the magenta band is the statistical uncertainty. Jet- p_T ranges are shown as panels.

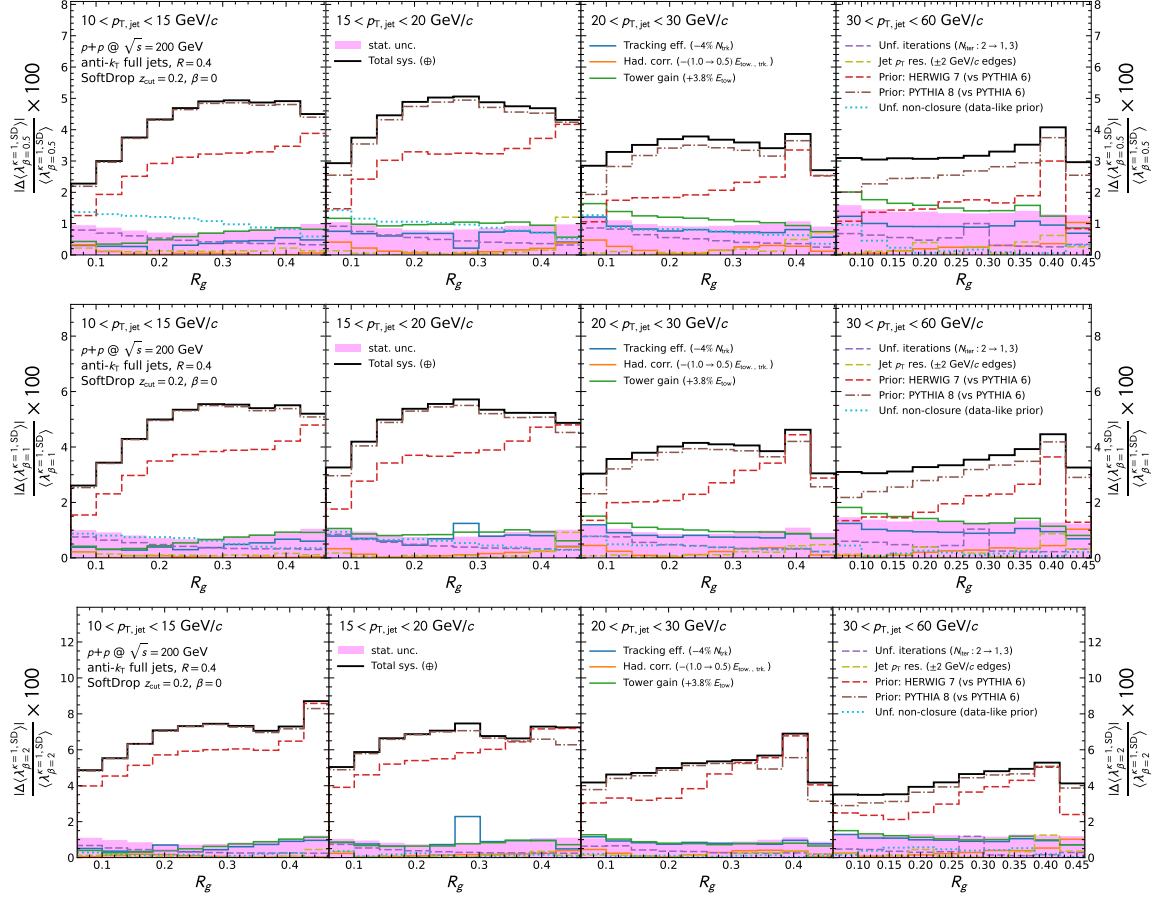


Figure 4.15: Relative systematic uncertainties of the $\langle \lambda_{\beta}^{\kappa=1} \rangle$ vs. R_g profiles of SoftDrop-groomed jets, before the Barlow check (Sec. 4.2.5), for $\beta = 0.5, 1$ and 2 (top to bottom). The sources are the tracking efficiency, the hadronic correction and the tower gain (solid), the number of unfolding iterations, the jet- p_T resolution and the HERWIG-7 and PYTHIA-8 priors (dashed), and the non-closure (dotted); the black line is their sum in quadrature and the magenta band is the statistical uncertainty. Jet- p_T ranges are shown as panels.

- Prior variation - Varied the prior of the embedding sample from PYTHIA6 to HERWIG7 and to PYTHIA8 by reweighting (Sec. 4.2.5)
- Non Closure - Any deviations from 1 for ratios shown in fig. 4.10 and fig. 4.11, with the data-like prior of Sec. 4.2.5

Prior reweighting

The prior of the unfolding is the generated distribution of the embedding simulation, $p_{\text{gen},0}(\mathbf{x})$, together with the reconstructed distribution $p_{\text{reco},0}(\mathbf{y})$ it produces through the response $R(\mathbf{y}, \mathbf{x})$. Two kinds of alternative priors are used: the particle-level distributions of HERWIG7 and PYTHIA8, for the prior variation, and a data-like PYTHIA6 prior, which provides the pseudo-data of the closure test. Neither is obtained by a new detector simulation. Instead, the PYTHIA6 embedding jets are reweighted, which changes the prior while keeping the PYTHIA6+GEANT3 response $R(\mathbf{y}, \mathbf{x})$ fixed, so that each variation probes the dependence on the prior alone.

Each reweighting is a density ratio learned with a classifier, in the same way as ω and ν in the unfolding. A classifier f trained with the binary cross-entropy to separate a target sample (label 1) from the PYTHIA6 sample (label 0) gives the per-jet weight

$$r(\mathbf{x}) = \frac{1}{C} \frac{f(\mathbf{x})}{1 - f(\mathbf{x})} \approx \frac{p_{\text{target}}(\mathbf{x})}{p_{\text{PYTHIA6}}(\mathbf{x})}, \quad (4.7)$$

over the full joint feature vector, so that correlations between the observables are reweighted along with their one-dimensional distributions. The classifier has the architecture and optimizer of the unfolding classifiers, and the jets carry their $\hat{p}_{\text{T}}$ cross-section weights, with the target normalized to the total weight of the PYTHIA6 sample. Since the jet spectrum falls steeply, the loss would be dominated by the lowest- $p_{\text{T,jet}}$ jets. Both classes are therefore given the common training weight $c(p_{\text{T,jet}}) = [p_{\text{PYTHIA6}}(p_{\text{T,jet}})]^{-1/2}$, with $p_{\text{PYTHIA6}}(p_{\text{T,jet}})$ the weighted jet- p_{T} density of the PYTHIA6 sample. Being a function of a feature shared

by both classes, c cancels in eq. 4.7 and only shifts the training towards higher $p_{\text{T,jet}}$. An ensemble of 10 classifiers is trained; their odds are clamped to $[0.1, 10]$ and combined by the geometric mean. The constant C normalizes the weighted mean of r over the PYTHIA6 sample to one, so that only the shape of the prior changes and not its total cross-section.

Generator priors.

The targets are the HERWIG7 and PYTHIA8 particle-level jets, selected with the particle-level cuts of the embedding ($p_{\text{T,jet}} \geq 5 \text{ GeV}/c$, $|\eta_{\text{jet}}| < 0.6$, at least two charged constituents) and described by the same features. The weight $r(\mathbf{x})$ is evaluated on every generated PYTHIA6 jet, matched or missed, and each matched reconstructed jet inherits the weight of its generated partner; fakes, having no generated partner, are unchanged. The reweighted embedding replaces the nominal one in the full unfolding, and the difference of the result to the nominal one is taken as the uncertainty of each generator, entering the envelope of the unfolding uncertainties. The reweighted PYTHIA-6 jets are compared to their HERWIG-7 and PYTHIA-8 targets in Figs. 4.16 and 4.17: the reweighting reproduces the target distributions and $\langle \lambda_{\beta}^{\kappa=1} \rangle$ vs. R_g profiles to within a few percent in most bins, including the differences in their correlation with R_g , which a reweighting of one-dimensional distributions would not capture.

Data-like prior.

The pseudo-data of the closure test is obtained from a one-iteration unfolding of the measured data by the same method. First, a classifier separating measured from reconstructed PYTHIA6 jets (matched and fake) gives $r(\mathbf{y}) \approx p_{\text{meas.}}(\mathbf{y})/p_{\text{reco.,0}}(\mathbf{y})$, which is applied to the reconstructed jets. Matched generated jets inherit the weight of their reconstructed partner. As missed jets have none, a second classifier separates the matched generated jets weighted this way from the unweighted ones, and the resulting smooth $r(\mathbf{x})$ is applied to all generated jets, matched and missed. The reweighted reconstructed jets are the pseudo-data,

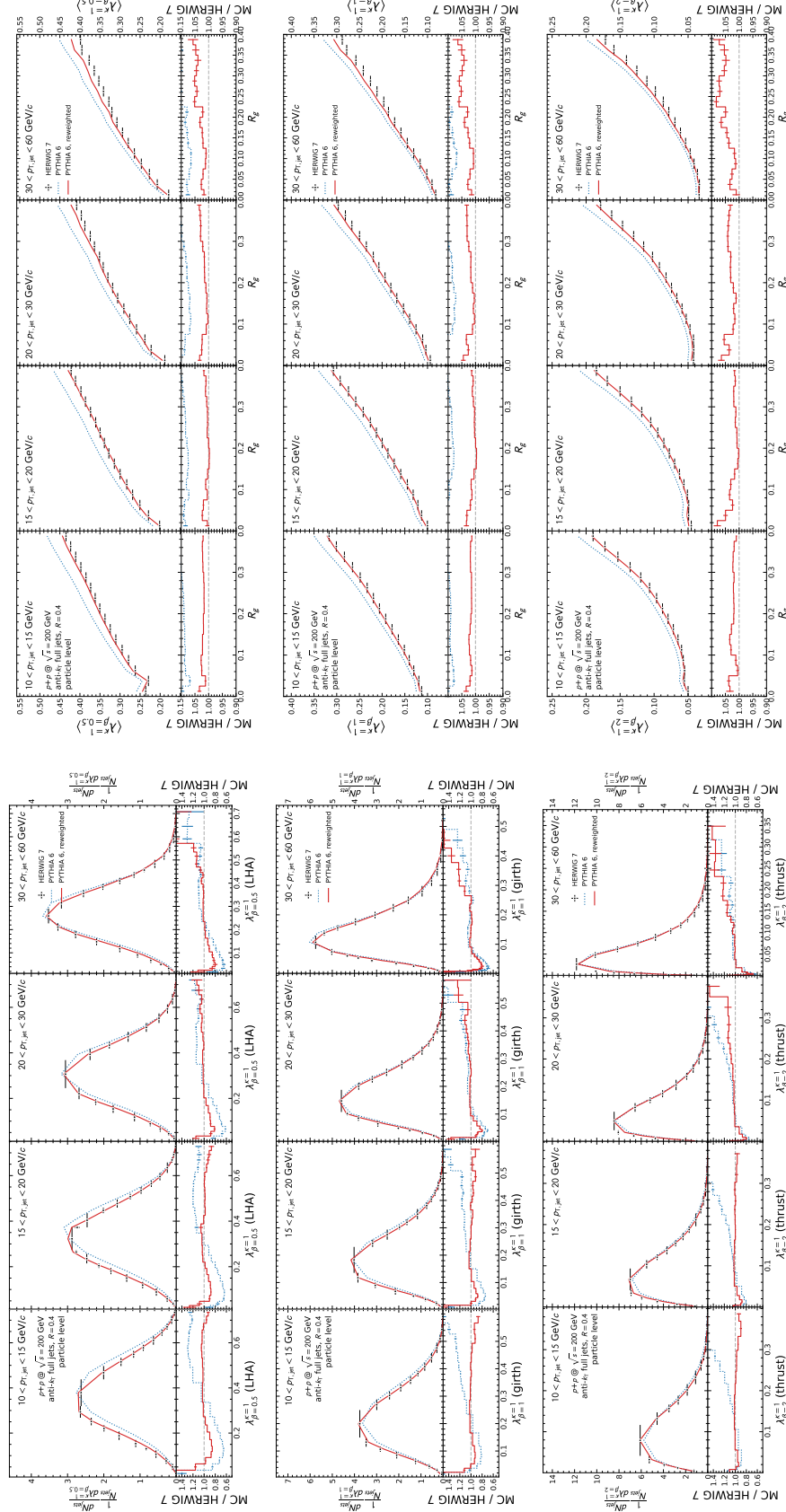


Figure 4.16: Reweighting of the PYTHIA-6 prior to HERWIG-7 at particle level: the distributions of $\lambda_{\beta}^{\kappa=1}$ (left) and the $\langle \lambda_{\beta}^{\kappa=1} \rangle$ vs. R_g profiles (right) for $\beta = 0.5, 1$ and 2 (top to bottom), in four $p_{T,\text{jet}}$ ranges (panels). HERWIG-7 (points) is compared to the nominal (dotted) and the reweighted (solid) PYTHIA-6; the lower panels show their ratios to HERWIG-7.

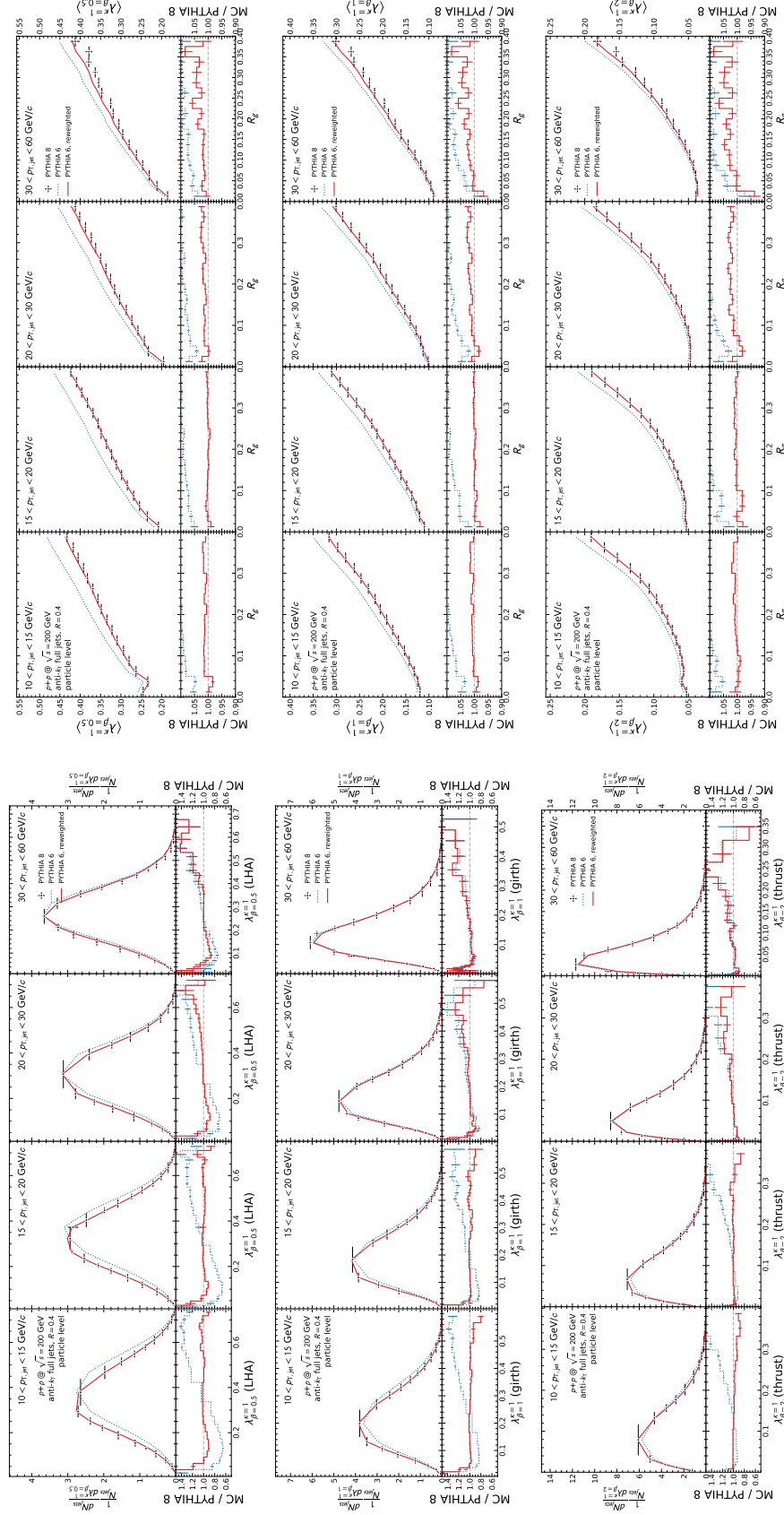


Figure 4.17: Reweighting of the PYTHIA-6 prior to PYTHIA-8 at particle level: the distributions of $\lambda_{\beta}^{\kappa=1}$ (left) and the $\langle \lambda_{\beta}^{\kappa=1} \rangle$ vs. R_g profiles (right) for $\beta = 0.5, 1$ and 2 (top to bottom), in four $p_{T,\text{jet}}$ ranges (panels). PYTHIA-8 (points) is compared to the nominal (dotted) and the reweighted (solid) PYTHIA-6; the lower panels show their ratios to PYTHIA-8.

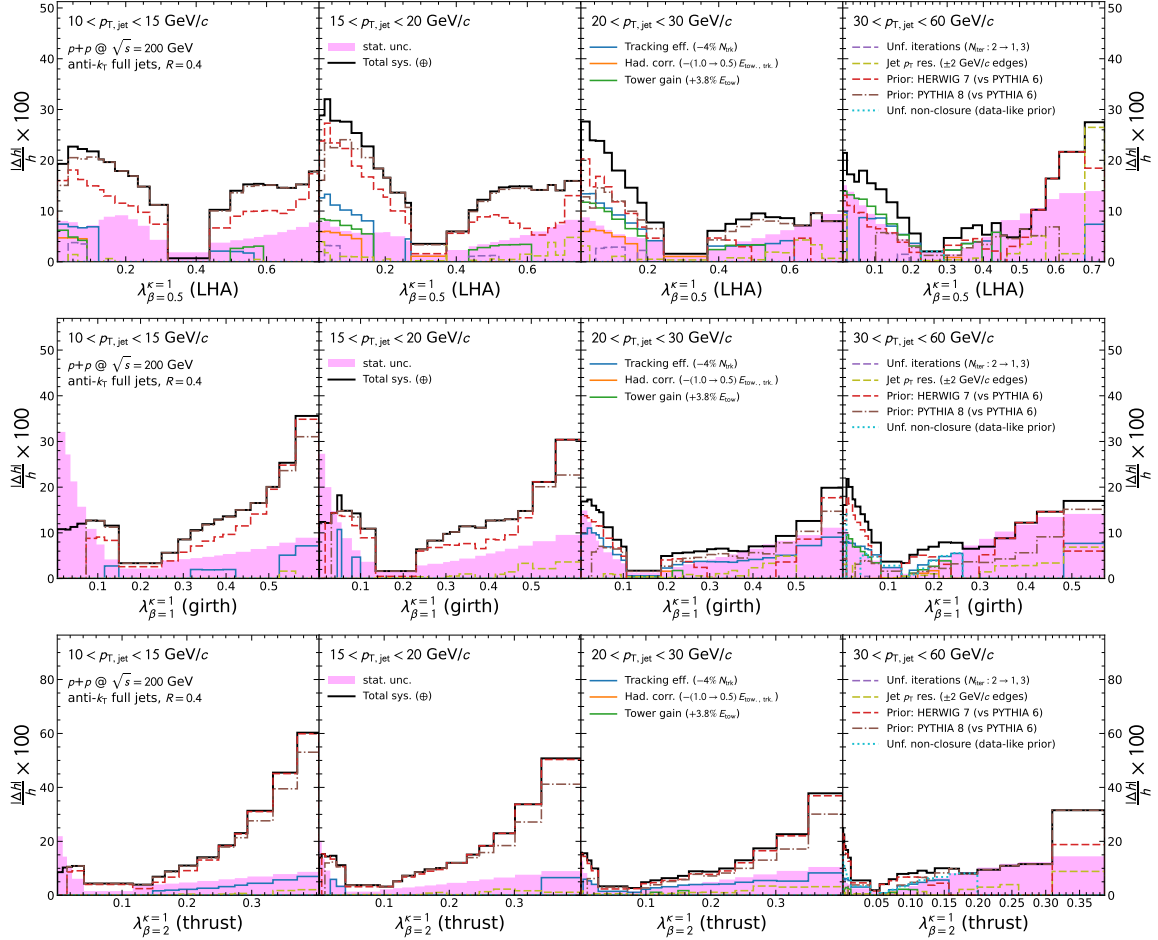


Figure 4.18: Relative systematic uncertainties of the $\lambda_{\beta}^{\kappa=1}$ distributions of inclusive jets, after the Barlow check (Sec. 4.2.5), for $\beta = 0.5, 1$ and 2 (top to bottom). The sources are the tracking efficiency, the hadronic correction and the tower gain (solid), the number of unfolding iterations, the jet- p_T resolution and the HERWIG-7 and PYTHIA-8 priors (dashed), and the non-closure (dotted); the black line is their sum in quadrature and the magenta band is the statistical uncertainty. Jet- p_T ranges are shown as panels.

the reweighted generated jets the pseudo-truth, and the pseudo-data are unfolded with the nominal PYTHIA6 embedding. Since the pseudo-data resemble the measurement, but the pseudo-truth is known, the deviation of the pseudo-unfolded result from the pseudo-truth (Figs. 4.10 and 4.11) is a non-closure uncertainty applied to the measured result as a relative uncertainty.

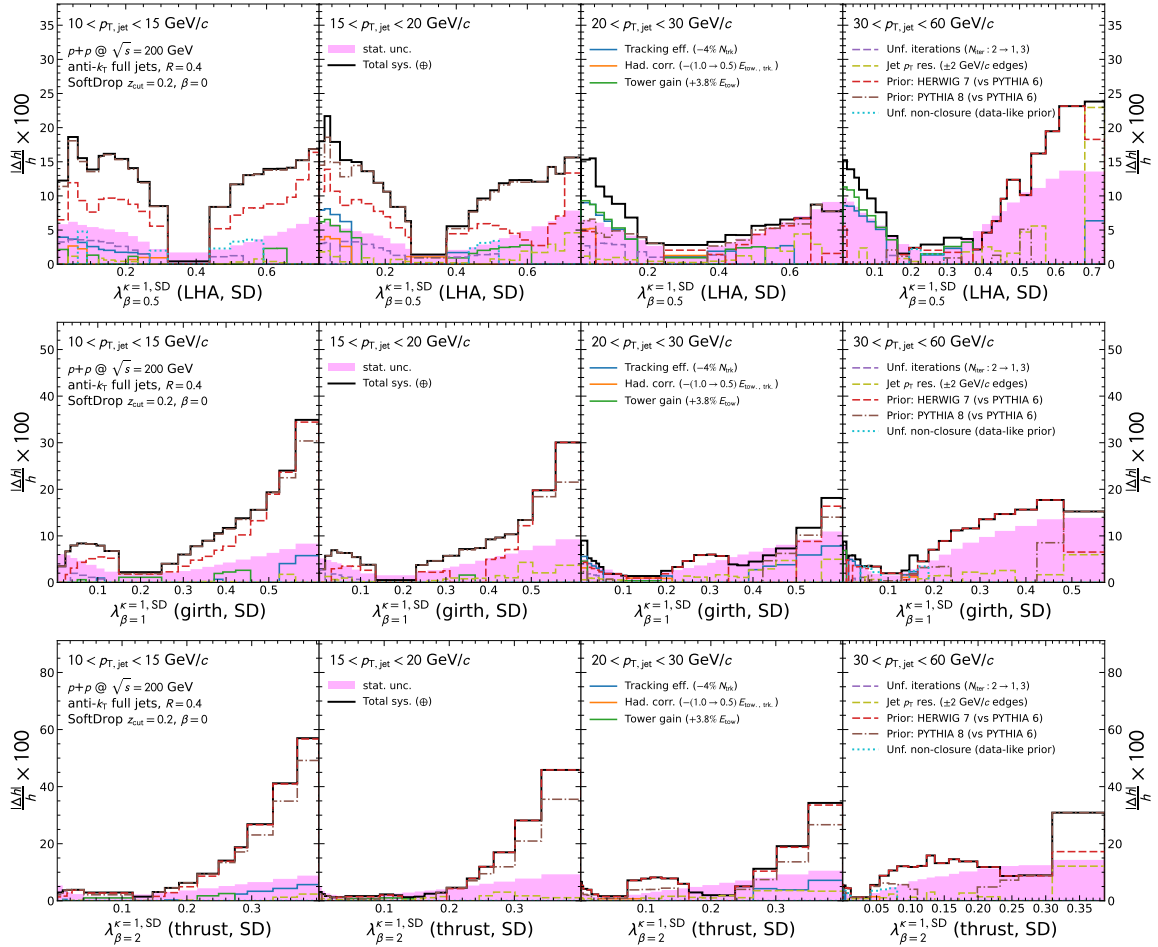


Figure 4.19: Relative systematic uncertainties of the $\lambda_{\beta=1}^{\kappa=1}$ distributions of SoftDrop-groomed jets, after the Barlow check (Sec. 4.2.5), for $\beta = 0.5, 1$ and 2 (top to bottom). The sources are the tracking efficiency, the hadronic correction and the tower gain (solid), the number of unfolding iterations, the jet- p_T resolution and the HERWIG-7 and PYTHIA-8 priors (dashed), and the non-closure (dotted); the black line is their sum in quadrature and the magenta band is the statistical uncertainty. Jet- p_T ranges are shown as panels.

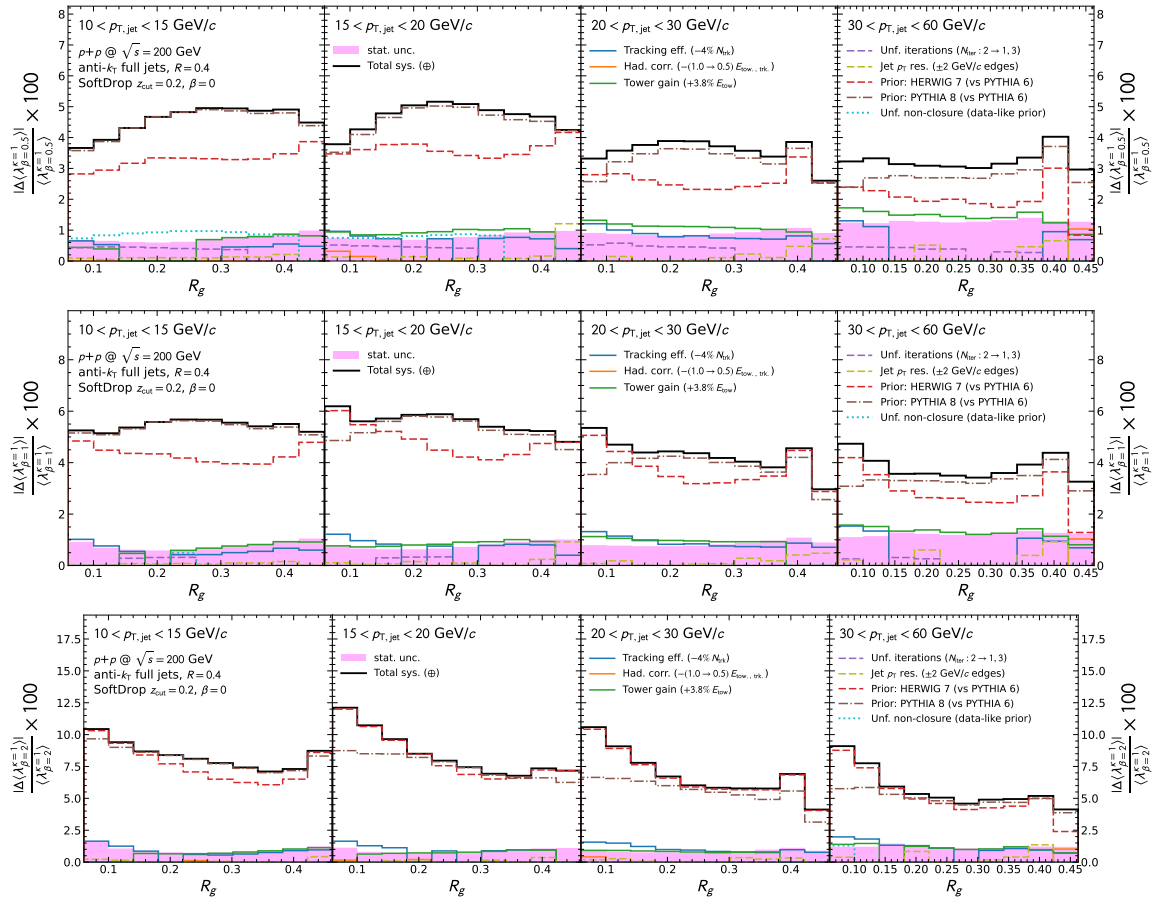


Figure 4.20: Relative systematic uncertainties of the $\langle \lambda_{\beta}^{\kappa=1} \rangle$ vs. R_g profiles of inclusive jets, after the Barlow check (Sec. 4.2.5), for $\beta = 0.5, 1$ and 2 (top to bottom). The sources are the tracking efficiency, the hadronic correction and the tower gain (solid), the number of unfolding iterations, the jet- p_T resolution and the HERWIG-7 and PYTHIA-8 priors (dashed), and the non-closure (dotted); the black line is their sum in quadrature and the magenta band is the statistical uncertainty. Jet- p_T ranges are shown as panels.

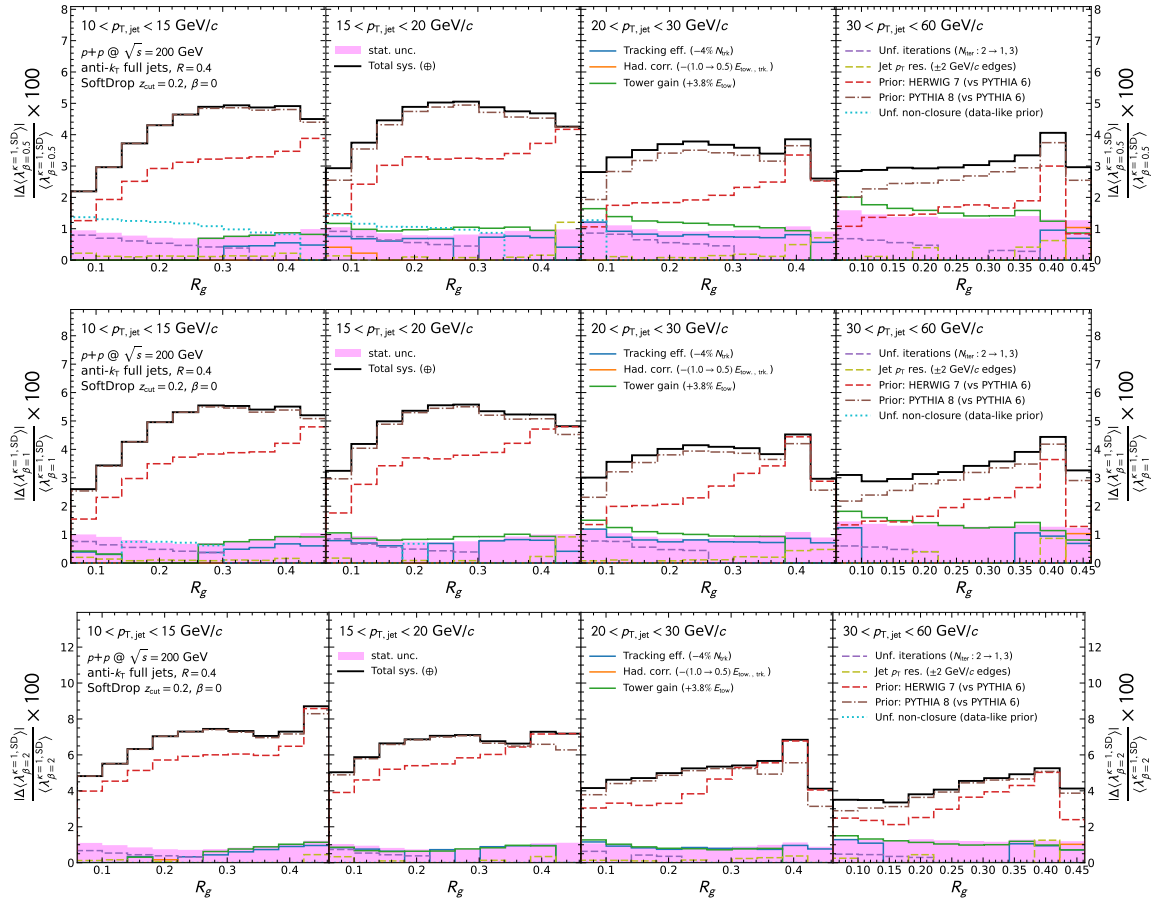


Figure 4.21: Relative systematic uncertainties of the $\langle \lambda_{\beta}^{\kappa=1} \rangle$ vs. R_g profiles of SoftDrop-groomed jets, after the Barlow check (Sec. 4.2.5), for $\beta = 0.5, 1$ and 2 (top to bottom). The sources are the tracking efficiency, the hadronic correction and the tower gain (solid), the number of unfolding iterations, the jet- p_T resolution and the HERWIG-7 and PYTHIA-8 priors (dashed), and the non-closure (dotted); the black line is their sum in quadrature and the magenta band is the statistical uncertainty. Jet- p_T ranges are shown as panels.

Barlow check and pruning systematic uncertainties

The ensembles of Sec. 4.2.4 are also used to decide whether a systematic variation is significant. Each variation is a complete unfolding with its own ensemble, giving $\bar{H}_{\text{var.}}$ and $\sigma_{\text{var.}}$ alongside the nominal $\bar{H}_{\text{nom.}}$ and $\sigma_{\text{nom.}}$. The shift $\bar{H}_{\text{var.}} - \bar{H}_{\text{nom.}}$ is partly a statistical fluctuation between two ensembles, whose size is estimated as $\sigma_{\text{uncorr.}}^2 = |\sigma_{\text{var.}}^2 - \sigma_{\text{nom.}}^2|$, and the shift is only counted as a systematic uncertainty in a bin where

$$\frac{|\bar{H}_{\text{var.}} - \bar{H}_{\text{nom.}}|}{\sigma_{\text{uncorr.}}} \geq 1, \quad (4.8)$$

the *Barlow test*, and is set to zero otherwise.

4.2.6 Comparison with published STAR results

As a cross-check of the unfolding, the jet mass M_{jet} and groomed jet mass $M_{\text{jet,g}}$ [19], and the groomed radius R_g and momentum fraction z_g [20], previously measured by STAR in $p + p$ collisions at $\sqrt{s} = 200$ GeV with the same jet definition, are calculated from the jets unfolded in this work. They are shown in the $p_{\text{T,jet}}$ ranges of the publications, and the groomed observables are obtained by grooming the unfolded jets again with SoftDrop at $z_{\text{cut}} = 0.1$, $\beta = 0$, as in the publications. The systematic uncertainties of this work in these comparisons include the number of iterations, the jet- p_{T} resolution, the HERWIG-7 prior, the non-closure, the tracking efficiency and the hadronic correction. The jet mass, R_g and z_g of this work agree with the published results within uncertainties in most bins (Fig. 4.22), while the groomed mass is higher than the published one in the lowest $M_{\text{jet,g}}$ bins.

4.2.7 Results and Discussions

Distributions.

The unfolded $\lambda_{\beta}^{\kappa=1}$ distributions for $\beta = 0.5$ (LHA), 1 (girth) and 2 (thrust) are shown in Fig. 4.23 for inclusive and SoftDrop-groomed jets in the three $p_{\text{T,jet}}$ ranges. With increasing

$p_{\text{T,jet}}$ all distributions become narrower and move to smaller values; the peak of the inclusive girth distribution, for example, moves from about 0.18 to about 0.11 between the lowest and the highest $p_{\text{T,jet}}$ range. This is the expected behavior of collinear radiation, whose angular spread decreases as the jet energy grows.

Grooming shifts all distributions to smaller angularity values, since the soft, wide-angle constituents it removes contribute the most to λ_β . The change of shape grows with β , because a larger β weights the constituents far from the axis more strongly: for $\beta = 0.5$ the groomed distribution is a moderately shifted version of the inclusive one, while for $\beta = 2$ the groomed thrust falls steeply from the smallest values, most of the groomed jets being reduced to a narrow core.

PYTHIA-6, PYTHIA-8 and HERWIG-7 reproduce the overall shape and the $p_{\text{T,jet}}$ and β dependence of the data, but all three predict jets that are narrower than measured: they overshoot the data at small λ_β , most visibly for the LHA, and undershoot the large- λ_β tails. The deficit in the tails reaches 20–40% for HERWIG-7 and PYTHIA-8, beyond the systematic uncertainties, while PYTHIA-6 stays within them in most bins, for both inclusive and groomed jets. For $\beta = 1$ and 2, grooming reduces the deviations of HERWIG-7 and PYTHIA-8 from the data, most clearly at higher $p_{\text{T,jet}}$, consistent with the removal of the soft wide-angle radiation that is most sensitive to the hadronization modelling; PYTHIA-6 does not show this improvement. Since PYTHIA-6 is also the prior of the unfolding, this agreement alone does not prove a good description, but the prior variations to HERWIG-7 and PYTHIA-8 (Sect. 4.2.5) are part of the systematic uncertainty, so a result pulled towards the prior would show up there.

Angularities as functions of the groomed radius.

The unbinned unfolding provides the full set of jet features for every jet, so that any observable can be projected afterward without being chosen, or binned, before the unfolding. This is used here to measure the mean angularities as a function of the SoftDrop groomed

radius, $\langle\lambda_{\beta}^{\kappa=1}\rangle$ vs R_g , shown in Fig. 4.24, which correlates the radiation pattern of the whole jet with the angle of its first hard splitting.

$\langle\lambda_{\beta}^{\kappa=1}\rangle$ rises monotonically with R_g for all β , for both inclusive and groomed jets. For the groomed jets, all constituents lie roughly within R_g of the axis, so $\langle\lambda_{\beta}\rangle$ grows approximately as R_g^{β} : the rise is concave for $\beta = 0.5$, close to linear for $\beta = 1$ and convex for $\beta = 2$. The inclusive values lie above the groomed ones by the contribution of the soft radiation removed by the grooming, and the two converge at large R_g . This reflects the angular ordering of the clustering sequence: a jet with a large R_g has its widest splitting already passing the SoftDrop condition, so almost nothing is groomed away, while a small R_g means that wide-angle soft branches were removed before the first hard splitting was found. At fixed R_g the mean angularities depend only weakly on $p_{T,\text{jet}}$, decreasing by a few percent between the lowest and the highest $p_{T,\text{jet}}$ range; most of the narrowing of the distributions with $p_{T,\text{jet}}$ seen in Fig. 4.23 therefore comes from the shift of the R_g distribution towards smaller angles.

The profiles separate the two possible origins of the model deviations seen in the distributions. PYTHIA-6 describes the profiles within uncertainties, but HERWIG-7 and PYTHIA-8 lie 10–15% below the data at all R_g , for all β , in all $p_{T,\text{jet}}$ ranges and for inclusive as well as groomed jets. Their narrower jets are thus not caused by a different distribution of the first splitting angle, but by too little radiation at a given splitting angle, which is information that the one-dimensional distributions alone cannot provide.

Comparison with ALICE.

Fig. 4.25 compares the $\lambda_1^{\kappa=1}$ and $\lambda_2^{\kappa=1}$ distributions to the ALICE measurement in $p + p$ collisions at $\sqrt{s} = 5.02$ TeV [21], for charged-particle jets with $20 < p_{T,\text{jet}}^{\text{ch}} < 40$ GeV/ c , which are reconstructed from the unfolded jets of this work for the comparison. The charged-jet and full-jet versions of this work agree closely with each other. At the same charged-jet p_T , the jets at $\sqrt{s} = 200$ GeV are considerably narrower than at $\sqrt{s} = 5.02$ TeV, both with and

without grooming, as expected from the larger fraction of quark-initiated jets at the lower collision energy.

Summary.

In terms of the three goals set out at the beginning of this section: the inclusive and groomed angularities, their $p_{T,\text{jet}}$ dependence and their correlation with R_g provide a $p + p$ baseline for future measurements in heavy-ion collisions at the same energy; the unbinned unfolding passes its closure tests and reproduces earlier STAR measurements of other observables from the same jets (Fig. 4.22); and the R_g profiles, obtained from the same unfolding without a dedicated analysis, trace the deviations of HERWIG-7 and PYTHIA-8 from the data to the radiation at a given splitting angle.

4.3 Conclusion and Outlook

This chapter presented two measurements of jet angularities at $\sqrt{s_{NN}} = 200$ GeV, both corrected for detector effects with unbinned, machine-learning based unfolding.

The distributions of the girth $g = \lambda_1^1$, $p_T^D = \sqrt{\lambda_0^2}$ and LeSub of hard-core jets in Au+Au collisions (Sect. 4.1, Fig. 4.5) are consistent between central (0–20%) and peripheral (40–80%) collisions within the systematic uncertainties. This absence of a visible modification is at least partly a consequence of the analysis choices: the hard-core constituent selection favors hard-fragmented, narrow jets, which interact less with the QGP, and the conservative, uncorrelated treatment of the systematic uncertainties in the central-to-peripheral ratio limits its sensitivity. As the first heavy-ion measurement with unbinned unfolding, it also showed where the method needed to be developed further before being applied to less biased jets.

This development was carried out with the inclusive and SoftDrop-groomed angularities $\lambda_\beta^{\kappa=1}$, $\beta = 0.5, 1$ and 2 , of inclusive jets in $p + p$ collisions at $\sqrt{s} = 200$ GeV (Sect. 4.2.7). The unfolding passes its closure tests and reproduces earlier STAR measurements of the jet

mass, groomed jet mass, R_g and z_g from the same unfolded jets. The measured jets become narrower with increasing $p_{T,\text{jet}}$, and grooming shifts the angularities to smaller values, the more so the larger β . PYTHIA-6, PYTHIA-8 and HERWIG-7 describe the main trends, but predict narrower jets than measured; PYTHIA-8 and HERWIG-7 fall 20–40% below the data in the tails of the distributions, while PYTHIA-6 agrees within uncertainties in most bins. For $\beta = 1$ and 2, grooming reduces the deviations of PYTHIA-8 and HERWIG-7 from the data. Because the unbinned unfolding keeps the full set of jet features, the mean angularities could also be measured as functions of the groomed radius R_g without a dedicated analysis. These profiles rise monotonically with R_g , with the groomed and inclusive values converging at large R_g as expected from the angular ordering of the clustering sequence, and depend only weakly on $p_{T,\text{jet}}$ at fixed R_g . They also show that PYTHIA-8 and HERWIG-7 underestimate the angularities by 10–15% at every R_g , i.e. that their narrower jets come from too little radiation at a given splitting angle rather than from a different distribution of the splitting angle. Compared with ALICE measurements at $\sqrt{s} = 5.02$ TeV at the same charged-jet p_T , the jets at RHIC are considerably narrower, as expected from their larger fraction of quark-initiated jets.

These $p + p$ results provide the baseline for the next step: measuring inclusive and groomed angularities of jets in Au+Au collisions at the same energy, with grooming in place of the hard-core selection to suppress the underlying event, and with the unfolding pipeline developed here. Differential measurements such as the R_g profiles are particularly promising there, since they can separate a medium-induced change of the radiation pattern at a given splitting angle from a change in the distribution of the splitting angles themselves. In $p + p$, the measurement will be extended by the angularities not shown here, such as λ_0^2 , for the final publication.

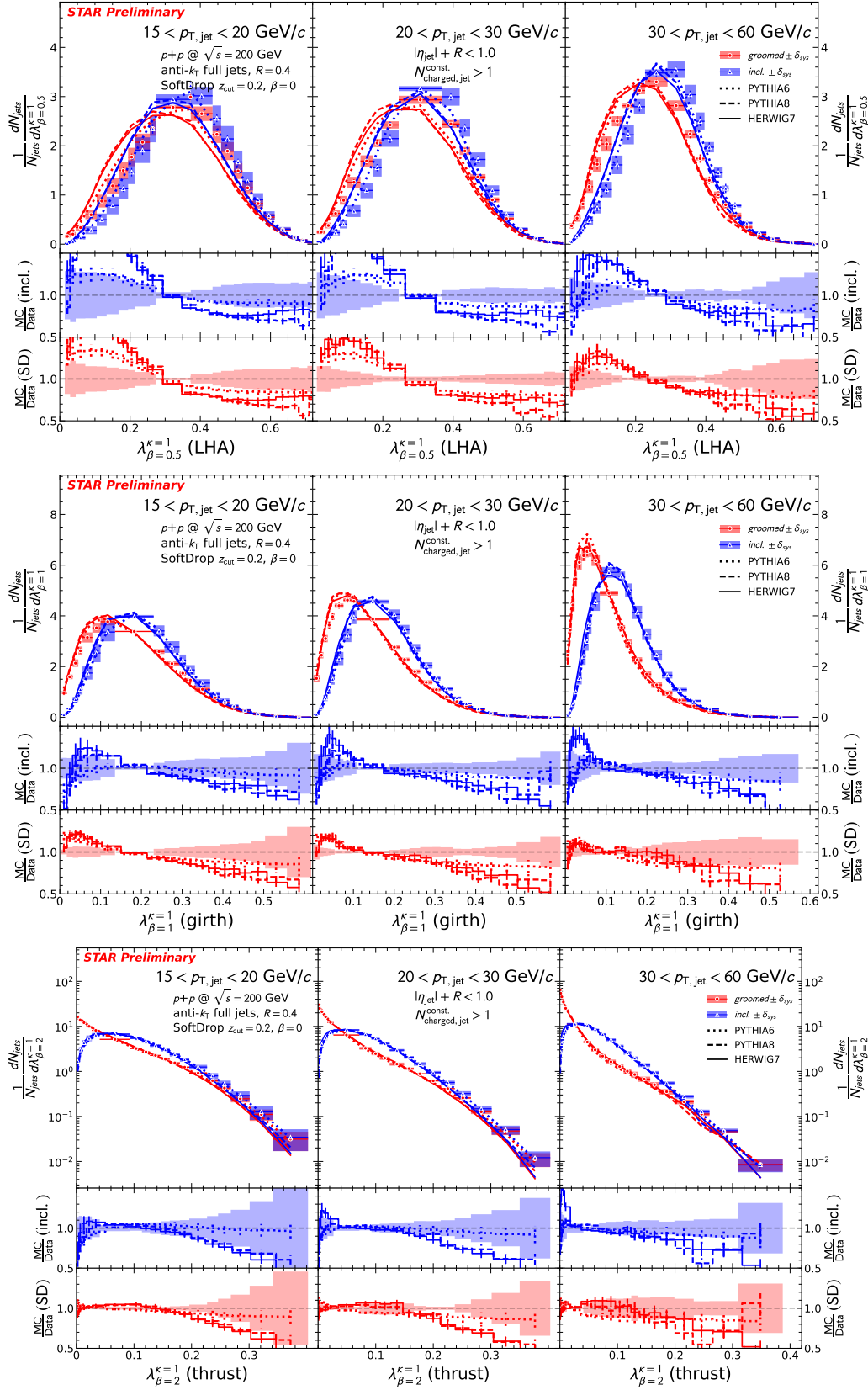


Figure 4.23: Unfolded distributions of $\lambda_{\beta}^{\kappa=1}$ for $\beta = 0.5, 1$ and 2 (top to bottom) for inclusive (blue) and SoftDrop-groomed (red, $z_{\text{cut}} = 0.2$, $\beta = 0$) jets in $p+p$ collisions at $\sqrt{s} = 200$ GeV, in three $p_{T,\text{jet}}$ ranges (columns). Boxes show the systematic and bars the statistical uncertainties. The measurements are compared to PYTHIA-6 (dotted), PYTHIA-8 (dashed) and HERWIG-7 (solid), and the lower panels show the ratios of the event generators to the data for inclusive and groomed jets, with the relative systematic uncertainty of the data as

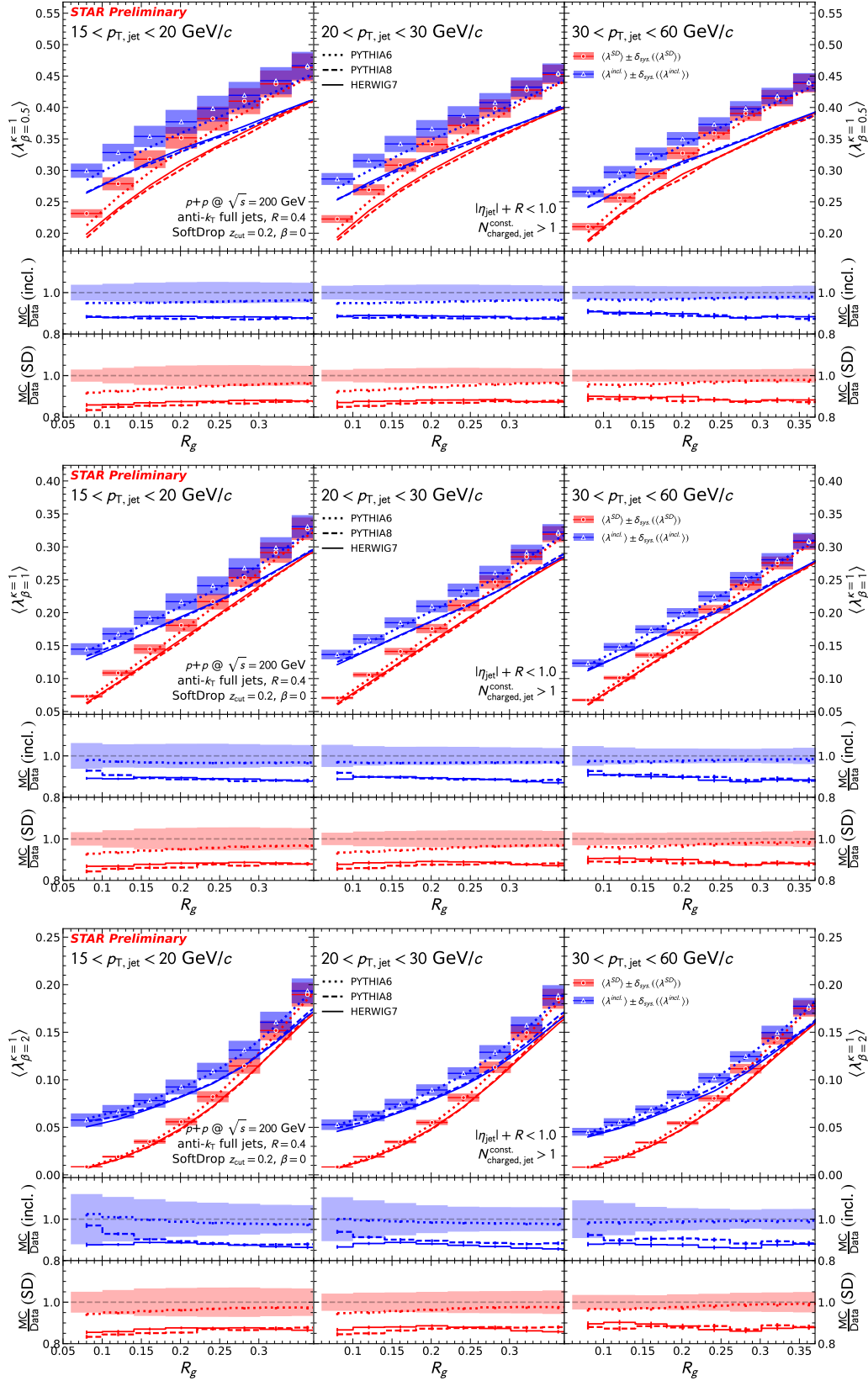


Figure 4.24: Unfolded mean angularities $\langle \lambda_{\beta}^{\kappa=1} \rangle$ as functions of the groomed radius R_g for $\beta = 0.5, 1$ and 2 (top to bottom) for inclusive (blue) and SoftDrop-groomed (red) jets in $p+p$ collisions at $\sqrt{s} = 200$ GeV, in three $p_{T,jet}$ ranges (columns). Boxes show the systematic and bars the statistical uncertainties. The measurements are compared to PYTHIA-6 (dotted), PYTHIA-8 (dashed) and HERWIG-7 (solid), and the lower panels show the ratios of the event generators to the data, with the relative systematic uncertainty of the data as a band.

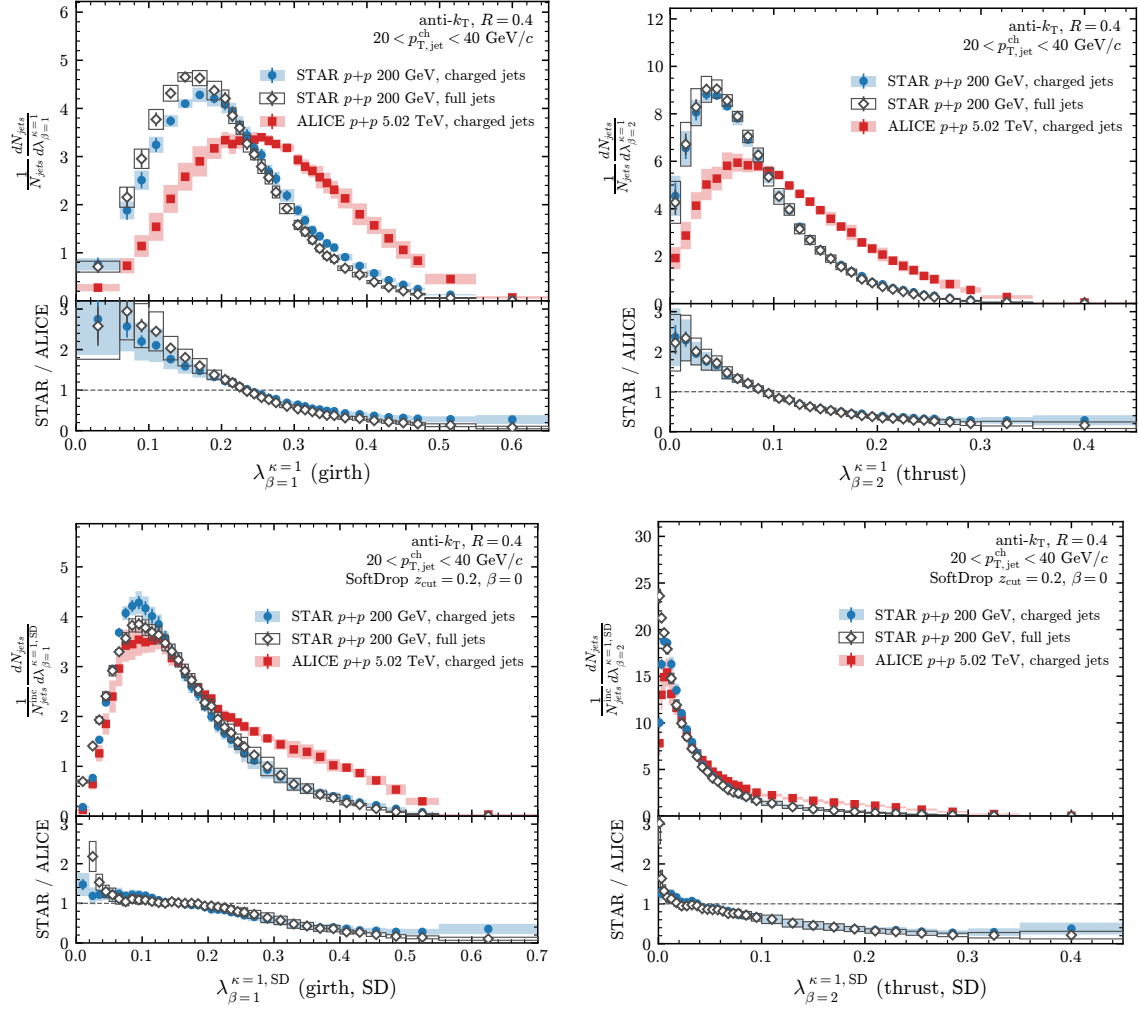


Figure 4.25: Comparison with the ALICE measurement of $p + p$ collisions at $\sqrt{s} = 5.02$ TeV [21] (red) of ungroomed (top) and SoftDrop-groomed (bottom, $z_{cut} = 0.2$, $\beta = 0$) $\lambda_{\beta}^{\kappa=1}$ distributions for $\beta = 1$ (left) and 2 (right), for charged jets with $20 < p_{T,jet}^{ch} < 40$ GeV/c. This work is shown for the charged-particle definition of ALICE (blue) and for the full jets of this analysis (black). The lower panels show the ratios to ALICE.

Chapter 5

Summary

This thesis is about jets, the angular structure of the energy inside them, and what that structure can tell us about QCD in vacuum and in QGP. Ch. 1 set up the theory needed to talk about them. The running of α_s gives asymptotic freedom on one end and color confinement on the other, and the confinement scale Λ separates the perturbative hard process from the non-perturbative hadronization. The evolution of outgoing partons in between is the parton shower, whose collinear enhancements are captured by the DGLAP splitting functions and Sudakov form factors of sec. A.2.1. The soft, wide-angle enhancement is captured by the antenna pattern, which after azimuthal averaging confines soft radiation from a parton to a cone extending up to its color partner, i.e. *angular ordering*. This makes the opening angle (rather than the virtuality) the natural evolution variable of the shower (sec. A.2.4), and is the reason why the angular distribution of energy in a jet is an imprint of the shower's history. The observables studied in ch. 4 are radial moments of exactly this distribution.

Ch. 2 described RHIC, from the ion sources to the storage rings, and the STAR detector. All the analyses presented here use the TPC for charged tracks and the BEMC for neutral energy and triggering, together giving full azimuthal coverage for $|\eta| < 1$. Ch. 1.4 then established the vocabulary of the analyses: jets as defined by IRC-safe sequential recombination algorithms (sec. 1.5), SoftDrop grooming as a way of separating the pQCD and npQCD parts of a jet, the Glauber picture of the initial state, centrality as the experimental proxy for the impact parameter (sec. 1.6.2), anisotropic flow and the event plane (sec. C), and jet quenching along with the R_{AA} and R_{CP} used to quantify it. The di-hadron and

R_{CP} measurements reviewed there are what motivate going from high- p_{T} hadrons to fully reconstructed jets and their substructure.

Ch. ?? collected the data analysis techniques common to the analyses. Two classification studies were shown as illustrations: a CNN acting on jet images separates quark and gluon jets in $p + p$ collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV, and a DNN acting on a vector of substructure observables separates PYTHIA-6 jets from JEWEL jets with an AUC that decreases going from central to peripheral collisions. The main part of the chapter is a general framework for unfolding, starting from Bayes theorem on truth and measured distributions and the assumption that the detector response $R(\mathbf{y}, \mathbf{x})$ is transferable from simulation. Each iteration in eq. E.5 is written in terms of two ratios, ω at the measured level and ν at the particle level, and the convolutions with $R(\mathbf{y}, \mathbf{x})$ which map $\omega(\mathbf{y}) \rightarrow \omega(\mathbf{x})$ and $\nu(\mathbf{x}) \rightarrow \nu(\mathbf{y})$. Binning this gives Iterative Bayesian Unfolding and replacing the ratios with classifier outputs gives MultiFold, with misses and fakes handled by evaluating the same maps at the unmatched jets. Written this way, the binned and unbinned methods are the same algorithm, differing only in how the ratios are estimated.

Ch. 3 presented jet-hadron correlations relative to the event plane in 20-50% central Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV. Since the angle of a jet to the event plane is correlated on average with its path length through the medium, medium-induced modifications were looked for as differences between in-, mid- and out-of-plane trigger jets. The flow modulated combinatorial background was removed with the RPF method, which fits the background dominated region at large $\Delta\eta$ simultaneously for the three orientations. Associated yields, their ratios and jet-peak widths were extracted for $15 < p_{\text{T,jet}} < 20$ GeV/ c and $20 < p_{\text{T,jet}} < 40$ GeV/ c jets. Within the precision of the measurement, neither the yields nor the widths show a dependence on the event plane, and the yield ratios are consistent with 1.0, with indications of potential modification only in the inclusive $1.0 < p_{\text{T,assoc}} < 10$ GeV/ c bin of the higher $p_{\text{T,jet}}$ sample. JEWEL predicts an event-plane dependence well below the systematic uncertainties, consistent with path-length dependence of energy loss

being dominated by jet-by-jet fluctuations, and describes the low- $p_{T,\text{assoc}}$ region better with recoils and the high- $p_{T,\text{assoc}}$ region better without.

Ch. 4 presented the measurements of generalized angularities λ_β^κ . In $p + p$ collisions at $\sqrt{s} = 200$ GeV, inclusive and SoftDrop groomed $\lambda_{\beta \in \{0.5, 1, 2\}}^{\kappa=1}$ distributions and their profiles against R_g were fully corrected using MultiFold with 23 features per jet, converging in two iterations. Grooming shifts the distributions to smaller angularities, more so for higher β where early, soft, wide-angle radiation contributes more. $\langle \lambda_\beta^{\kappa=1} \rangle$ rises monotonically with R_g , steepening with β , and high- R_g jets are unaffected by grooming because their widest split already passes z_{cut} , which is angular ordering seen in data. PYTHIA-6, PYTHIA-8 and HERWIG-7 tuned to RHIC energies describe the distributions and profiles within uncertainties, so these measurements serve as the baselines for the heavy-ion measurements. In Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV, $g = \lambda_1^1$, $p_T^D = \sqrt{\lambda_0^2}$ and LeSub were unfolded with MultiFold for hard-core jets in 0-20% and 40-80% central events, the first use of the method on heavy-ion data. Central to peripheral ratios are consistent with unity within conservative systematic uncertainties. Together with the jet-hadron results, this points to the same conclusion: the HT-trigger and hard-core constituent selections bias both analyses towards hard-fragmented, narrow jets which interact the least with the medium, and a survivor bias is the likely reason why the expected path-length and centrality dependencies are not seen.

5.1 Outlook

The natural next step is to redo the Au+Au angularity measurement with the inclusive constituent selection and grooming used in the $p + p$ analysis, so that the $p + p$ results can serve as the baseline they were meant to be and the R_g profiles can separate the pQCD and npQCD parts of the modification. The unfolding framework of ch. ?? was written with this in mind, as the larger heavy-ion background makes misses and fakes far from negligible. On the jet-hadron side, selecting on the initial state ellipticity through the q_n flow vector

within a centrality class would test whether fluctuations in the initial geometry are what washes out the event-plane dependence. Finally, the quark-gluon and quenched-unquenched classifiers of ch. ?? suggest that the amount of modification can itself be quantified as a discrimination problem, which would extend naturally to the angularities once the groomed heavy-ion measurement is available.

Bibliography

- [1] F. Takahashi et al. Review of Particle Physics. *Int. J. Mod. Phys. A*, 41:2630011, 2026.
- [2] Megan Connors, Christine Nattrass, Rosi Reed, and Sevil Salur. Jet measurements in heavy ion physics. *Reviews of Modern Physics*, 90(2), 2018.
- [3] M. L. Miller et al. Glauber modeling in high energy nuclear collisions. *Ann. Rev. Nucl. Part. Sci.*, 57:205–243, 2007.
- [4] Miklos Gyulassy and Larry McLerran. New forms of qcd matter discovered at rhic. *Nuclear Physics A*, 750(1):30–63, 2005. Quark-Gluon Plasma. New Discoveries at RHIC: Case for the Strongly Interacting Quark-Gluon Plasma. Contributions from the RBRC Workshop held May 14-15, 2004.
- [5] Paul Sorensen. Evidence from identified particles for active quark and gluon degrees of freedom. *Journal of Physics G: Nuclear and Particle Physics*, 32(12):S135, nov 2006.
- [6] J. Adams et al. Evidence from d + Au measurements for final state suppression of high p_T hadrons in Au+Au collisions at RHIC . *Phys. Rev. Lett.*, 91:072304, 2003.
- [7] M J Tannenbaum. How do quarks and gluons lose energy in the qgp? *Journal of Physics: Conference Series*, 589(1):012019, mar 2015.
- [8] Measurement of inclusive charged-particle jet production in au + au collisions at 200 gev. *Phys. Rev. C*, 102:054913, Nov 2020.
- [9] J. Adams et al. Transverse-momentum and collision-energy dependence of high- p_T hadron suppression in Au + Au collisions at ultrarelativistic energies. *Phys. Rev. Lett.*, 91:172302, Oct 2003.
- [10] B. Abelev et al. Measurement of charged jet suppression in pb-pb collisions at $\sqrt{s_{\mathrm{nn}}}=2.76$ tev. *Journal of High Energy Physics*, 2014(3):13, 2014.
- [11] G. Aad et al. Measurement of charged-particle spectra in pb+pb collisions at $\sqrt{s_{\mathrm{nn}}}=2.76$ tev with the atlas detector at the lhc. *Journal of High Energy Physics*, 2015(9):50, 2015.
- [12] Karen McNulty Walsh. 2024.
- [13] Wolfram Fischer. 2026.
- [14] Karen McNulty Walsh. 2026.

- [15] Nuclear science and technology, star detector. 2024.
- [16] M. Anderson et al. The STAR time projection chamber: a unique tool for studying high multiplicity events at RHIC. *Nuc. Inst. and Methods in Physics Research*, 499(2-3):659–678, mar 2003.
- [17] Hans Bichsel. A method to improve tracking and particle identification in tpcs and silicon detectors. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 562(1):154 – 197, 2006.
- [18] H. Agakishiev et al. Event-plane-dependent dihadron correlations with harmonic v_n subtraction in Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV. *Phys. Rev.*, C89(4):041901, 2014.
- [19] Mohamed Abdallah et al. Invariant Jet Mass Measurements in pp Collisions at $\sqrt{s} = 200$ GeV at RHIC. *Phys. Rev. D*, 104(5):052007, 2021.
- [20] Jaroslav Adam et al. Measurement of groomed jet substructure observables in p+p collisions at $\sqrt{s} = 200$ GeV with STAR. *Phys. Lett. B*, 811:135846, 2020.
- [21] Shreyasi Acharya et al. Measurements of the groomed and ungroomed jet angularities in pp collisions at $\sqrt{s} = 5.02$ TeV. *JHEP*, 05:061, 2022.
- [22] Patrick T. Komiske, Eric M. Metodiev, and Matthew D. Schwartz. Deep learning in color: towards automated quark/gluon jet discrimination. *Journal of High Energy Physics*, 2017(1), January 2017.
- [23] Florencia Canelli, Kyle Cormier, Andrew Cudd, Dag Gillberg, Roger G. Huang, Weijie Jin, Sookhyun Lee, Vinicius Mikuni, Laura Miller, Benjamin Nachman, Jingjing Pan, Tanmay Pani, Mariel Pettee, Youqi Song, and Fernando Torales Acosta. A practical guide to unbinned unfolding. *The European Physical Journal C*, 86(2), February 2026.
- [24] Isaac Newton. *Opticks: or, a Treatise of the Reflexions, Refractions, Inflexions and Colours of Light*. Sam. Smith and Benj. Walford, London, 1704.
- [25] Thomas Young. The Bakerian Lecture: Experiments and calculations relative to physical optics. *Phil. Trans. Roy. Soc. Lond.*, 94:1–16, 1804.
- [26] John Dalton. *A New System of Chemical Philosophy*. R. Bickerstaff, Manchester, 1808.
- [27] J. J. Thomson. Cathode rays. *Phil. Mag. Ser. 5*, 44:293–316, 1897.
- [28] Max Planck. Über das Gesetz der Energieverteilung im Normalspectrum. *Annalen Phys.*, 309(3):553–563, 1901.
- [29] A. Einstein. Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt. *Annalen Phys.*, 322(6):132–148, 1905.

- [30] E. Rutherford. The scattering of alpha and beta particles by matter and the structure of the atom. *Phil. Mag. Ser. 6*, 21:669–688, 1911.
- [31] E. Rutherford. LIV. Collision of α particles with light atoms. IV. An anomalous effect in nitrogen. *Phil. Mag. Ser. 6*, 37(222):581–587, 1919.
- [32] J. Chadwick. Possible Existence of a Neutron. *Nature*, 129:312, 1932.
- [33] Paul A. M. Dirac. The quantum theory of the electron. *Proc. Roy. Soc. Lond. A*, 117:610–624, 1928.
- [34] C. D. Anderson. The Positive Electron. *Phys. Rev.*, 43:491–494, 1933.
- [35] S. H. Neddermeyer and C. D. Anderson. Note on the Nature of Cosmic Ray Particles. *Phys. Rev.*, 51:884–886, 1937.
- [36] C. M. G. Lattes, H. Muirhead, G. P. S. Occhialini, and C. F. Powell. PROCESSES INVOLVING CHARGED MESONS. *Nature*, 159:694–697, 1947.
- [37] G. D. Rochester and C. C. Butler. Evidence for the Existence of New Unstable Elementary Particles. *Nature*, 160:855–857, 1947.
- [38] C. L. Cowan, F. Reines, F. B. Harrison, H. W. Kruse, and A. D. McGuire. Detection of the free neutrino: A Confirmation. *Science*, 124:103–104, 1956.
- [39] Murray Gell-Mann. A Schematic Model of Baryons and Mesons. *Phys. Lett.*, 8:214–215, 1964.
- [40] G. Zweig. An SU(3) model for strong interaction symmetry and its breaking. Version 2. Technical Report CERN-TH-412, CERN, 1964.
- [41] O. W. Greenberg. Spin and Unitary Spin Independence in a Paraquark Model of Baryons and Mesons. *Phys. Rev. Lett.*, 13:598–602, 1964.
- [42] M. Y. Han and Yoichiro Nambu. Three Triplet Model with Double SU(3) Symmetry. *Phys. Rev.*, 139:B1006–B1010, 1965.
- [43] Elliott D. Bloom et al. High-Energy Inelastic e p Scattering at 6-Degrees and 10-Degrees. *Phys. Rev. Lett.*, 23:930–934, 1969.
- [44] Martin Breidenbach, Jerome I. Friedman, Henry W. Kendall, Elliott D. Bloom, D. H. Coward, H. C. DeStaebler, J. Drees, Luke W. Mo, and Richard E. Taylor. Observed behavior of highly inelastic electron-proton scattering. *Phys. Rev. Lett.*, 23:935–939, 1969.
- [45] D. J. Gross and F. Wilczek. Ultraviolet behavior of non-abelian gauge theories. *Phys. Rev. Lett.*, 30:1343–1346, Jun 1973.
- [46] H. D. Politzer. Reliable perturbative results for strong interactions? *Phys. Rev. Lett.*, 30:1346–1349, Jun 1973.

- [47] J. J. Aubert et al. Experimental Observation of a Heavy Particle *J. Phys. Rev. Lett.*, 33:1404–1406, 1974.
- [48] J. E. Augustin et al. Discovery of a Narrow Resonance in e^+e^- Annihilation. *Phys. Rev. Lett.*, 33:1406–1408, 1974.
- [49] Martin L. Perl et al. Evidence for Anomalous Lepton Production in $e^+ - e^-$ Annihilation. *Phys. Rev. Lett.*, 35:1489–1492, 1975.
- [50] S. W. Herb et al. Observation of a Dimuon Resonance at 9.5 GeV in 400 GeV Proton-Nucleus Collisions. *Phys. Rev. Lett.*, 39:252–255, 1977.
- [51] G. Hanson et al. Evidence for Jet Structure in Hadron Production by $e^+ e^-$ Annihilation. *Phys. Rev. Lett.*, 35:1609–1612, 1975.
- [52] R. Brandelik et al. Evidence for Planar Events in $e^+ e^-$ Annihilation at High-Energies. *Phys. Lett. B*, 86:243–249, 1979.
- [53] G. Arnison et al. Experimental Observation of Isolated Large Transverse Energy Electrons with Associated Missing Energy at $\sqrt{s} = 540$ GeV. *Phys. Lett. B*, 122:103–116, 1983.
- [54] M. Banner et al. Observation of Single Isolated Electrons of High Transverse Momentum in Events with Missing Transverse Energy at the CERN anti-p p Collider. *Phys. Lett. B*, 122:476–485, 1983.
- [55] G. Arnison et al. Experimental Observation of Lepton Pairs of Invariant Mass Around 95-GeV/ c^2 at the CERN SPS Collider. *Phys. Lett. B*, 126:398–410, 1983.
- [56] F. Abe et al. Observation of top quark production in $\bar{p}p$ collisions. *Phys. Rev. Lett.*, 74:2626–2631, 1995.
- [57] S. Abachi et al. Observation of the top quark. *Phys. Rev. Lett.*, 74:2632–2637, 1995.
- [58] K. Kodama et al. Observation of tau neutrino interactions. *Phys. Lett. B*, 504:218–224, 2001.
- [59] Georges Aad et al. Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC. *Phys. Lett. B*, 716:1–29, 2012.
- [60] Serguei Chatrchyan et al. Observation of a New Boson at a Mass of 125 GeV with the CMS Experiment at the LHC. *Phys. Lett. B*, 716:30–61, 2012.
- [61] R. K. Ellis, W. J. Stirling, and B. R. Webber. Hadroproduction of jets and photons. In *QCD and Collider Physics*, Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology, pages 237–269. Cambridge University Press, Cambridge, 1996.
- [62] Bryan R Webber. A QCD model for jet fragmentation including soft gluon interference. *Nucl. Phys. B*, 238:492–528, 1984.
- [63] B. Andersson, G. Gustafson, G. Ingelman, and T. Sjöstrand. Parton fragmentation

- and string dynamics. *Physics Reports*, 97(2):31–145, 1983.
- [64] T. Sjöstrand, S. Mrenna, and P. Skands. Pythia 6.4 physics and manual. *Journal of High Energy Physics*, 2006(05):026–026, May 2006.
 - [65] Johannes Bellm et al. Herwig 7.0/Herwig++ 3.0 release note. *Eur. Phys. J. C*, 76(4):196, 2016.
 - [66] Peter Z. Skands. Tuning monte carlo generators: The perugia tunes. *Physical Review D*, 82(7), oct 2010.
 - [67] Manny Rosales Aguilar et al. Pythia8 underlying event tune for RHIC energies. *Phys. Rev. D*, 105:016011, Jan 2022.
 - [68] J. Kevin Adkins. PhD thesis, University of Kentucky, 2019.
 - [69] Korinna C. Zapp. JEWEL 2.0.0: directions for use. *Eur. Phys. J.*, C74(2):2762, 2014.
 - [70] R. Brun et al. GEANT3. 9 1987.
 - [71] Edward V. Shuryak. Quantum chromodynamics and the theory of superdense matter. *Physics Reports*, 61(2):71–158, 1980.
 - [72] Kenji Fukushima and Tetsuo Hatsuda. The phase diagram of dense qcd. *Reports on Progress in Physics*, 74(1):014001, December 2010.
 - [73] Eemeli Annala, Tyler Gorda, Aleksi Kurkela, Joonas Nättilä, and Aleksi Vuorinen. Evidence for quark-matter cores in massive neutron stars. *Nature Physics*, 16(9):907–910, 2020.
 - [74] Demircik, Tuna, Ecker, Christian, Järvinen, Matti, Rezzolla, Luciano, Tootle, Samuel, and Topolski, Konrad. Exploring the phase diagram of v-qcd with neutron star merger simulations. *EPJ Web Conf.*, 274:07006, 2022.
 - [75] *Proceedings: 1990 DPF Summer Study on High-energy Physics (Snowmass 1990): Research directions for the decade*, Singapore, 9 1992. World Scientific.
 - [76] Stephen D. Ellis and Davison E. Soper. Successive combination jet algorithm for hadron collisions. *Phys. Rev. D*, 48:3160–3166, Oct 1993.
 - [77] M. Wobisch and T. Wengler. Hadronization corrections to jet cross sections in deep-inelastic scattering, 1999.
 - [78] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez. The anti-kt jet clustering algorithm. *Journal of High Energy Physics*, 2008(04):063, apr 2008.
 - [79] M. Cacciari, G. P. Salam, and G. Soyez. FastJet User Manual. *Eur.Phys.J.*, C72:1896, 2012.
 - [80] A. Adare et al. Transverse energy production and charged-particle multiplicity at midrapidity in various systems from $\sqrt{s_{NN}} = 7.7$ to 200 gev. *Phys. Rev. C*, 93:024901,

Feb 2016.

- [81] J. Adam et al. Measurement of transverse energy at midrapidity in pb-pb collisions at $\sqrt{s_{NN}} = 2.76$ tev. *Phys. Rev. C*, 94:034903, Sep 2016.
- [82] A. Bazavov, Tanmoy Bhattacharya, C. DeTar, H.-T. Ding, Steven Gottlieb, Rajan Gupta, P. Hegde, U. M. Heller, F. Karsch, E. Laermann, L. Levkova, Swagato Mukherjee, P. Petreczky, C. Schmidt, C. Schroeder, R. A. Soltz, W. Soeldner, R. Sugar, M. Wagner, and P. Vranas. Equation of state in $(2 + 1)$ -flavor qcd. *Phys. Rev. D*, 90:094503, Nov 2014.
- [83] Frithjof Karsch. *Lattice QCD at High Temperature and Density*, pages 209–249. Springer Berlin Heidelberg, Berlin, Heidelberg, 2002.
- [84] Edmond Iancu, Andrei Leonidov, and Larry McLerran. Nonlinear gluon evolution in the color glass condensate: I. *Nuclear Physics A*, 692(3):583–645, 2001.
- [85] C. Loizides, J. Nagle, and P. Steinberg. Improved version of the phobos glauher monte carlo. *SoftwareX*, 1-2:13–18, 2015.
- [86] Dmitri Kharzeev and Marzia Nardi. Hadron production in nuclear collisions at rhic and high-density qcd. *Physics Letters B*, 507(1):121–128, 2001.
- [87] BB Back, MD Baker, M Ballintijn, DS Barton, RR Betts, AA Bickley, R Bindel, A Budzanowski, W Busza, A Carroll, et al. Collision geometry scaling of au+ au pseudorapidity density from s nn= 19.6 to 200 gev. *Physical Review C—Nuclear Physics*, 70(2):021902, 2004.
- [88] B. Abelev et al. Centrality determination of pb-pb collisions at $\sqrt{s_{NN}} = 2.76$ tev with alice. *Phys. Rev. C*, 88:044909, Oct 2013.
- [89] Ulrich Heinz and Raimond Snellings. Collective flow and viscosity in relativistic heavy-ion collisions. *Annual Review of Nuclear and Particle Science*, 63(1):123–151, October 2013.
- [90] J. Bielcikova et al. Elliptic flow contribution to two particle correlations at different orientations to the reaction plane. *Phys.Rev.*, C69:021901, 2004.
- [91] S. Voloshin and Y. Zhang. Flow study in relativistic nuclear collisions by Fourier expansion of Azimuthal particle distributions. *Z.Phys.*, C70:665–672, 1996.
- [92] A. Majumder and M. van Leeuwen. The theory and phenomenology of perturbative qcd based jet quenching. *Progress in Particle and Nuclear Physics*, 66(1):41–92, 2011.
- [93] Guang-You Qin and Xin-Nian Wang. Jet quenching in high-energy heavy-ion collisions. *International Journal of Modern Physics E*, 24(11):1530014, 2024/10/14 2015.
- [94] Karen M. Burke et al. Extracting the jet transport coefficient from jet quenching in high-energy heavy-ion collisions. *Phys. Rev. C*, 90:014909, Jul 2014.

- [95] David d’Enterria. 6.4 jet quenching: Datasheet from landolt-börnstein - group i elementary particles, nuclei and atoms · volume 23: “relativistic heavy ion physics” in springer materials (https://doi.org/10.1007/978-3-642-01539-7_16). Copyright 2010 Springer-Verlag Berlin Heidelberg.
- [96] K.J. ESKOLA, R. VOGT, and X.-N. WANG. Nuclear overlap functions. *International Journal of Modern Physics A*, 10(20n21):3087–3090, 1995.
- [97] J. Casalderrey-Solana et al. Simultaneous description of hadron and jet suppression in heavy-ion collisions. *Phys. Rev. C*, 99:051901, May 2019.
- [98] S. Chatrchyan et al. Modification of jet shapes in pbpb collisions at $\sqrt{s_{NN}}=2.76$ tev. *Physics Letters B*, 730:243–263, 2014.
- [99] K.H. Ackermann et al. Star detector overview. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 499(2):624–632, 2003. The Relativistic Heavy Ion Collider Project: RHIC and its Detectors.
- [100] M. Beddo et al. The STAR barrel electromagnetic calorimeter. *Nuc. Inst. and Methods in Physics Research*, 499(2):725–739, 2003. The Relativistic Heavy Ion Collider Project: RHIC and its Detectors.
- [101] K. H. Ackermann et al. STAR detector overview. *Nucl. Instrum. Meth.*, A499:624–632, 2003.
- [102] M. Beddo et al. The STAR Barrel Electromagnetic Calorimeter. *Nucl. Instrum. Meth.*, A499:725–739, 2003.
- [103] L. Adamczyk et al. Measurements of jet quenching with semi-inclusive hadron+jet distributions in Au + Au collisions at $\sqrt{s_{NN}} = 200$ gev. *Phys. Rev. C*, 96:024905, Aug 2017.
- [104] W. M. Castilho, W. Qian, Y. Hama, and T. Kodama. Event-plane dependent di-hadron correlations with harmonic v_n subtraction in a hydrodynamic model. *Physics Letters B*, 777:369–373, 2018.
- [105] J. Adams et al. Azimuthal anisotropy in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. *Physical Review C*, 72(1):014904, Jul 2005.
- [106] M. Cacciari, G. P. Salam, and G. Soyez. The Catchment Area of Jets. *JHEP*, 04:005, 2008.
- [107] L. Adamczyk et al. Dijet imbalance measurements in Au + Au and pp collisions at $\sqrt{s_{NN}} = 200$ GeV at star. *Phys. Rev. Lett.*, 119:062301, Aug 2017.
- [108] S. Acharya et al. Jet-hadron correlations measured relative to the second order event plane in Pb-Pb collisions at $\sqrt{s_{NN}} = 2.76$ TeV. *Phys. Rev. C*, 101:064901, Jun 2020.

- [109] N. Sharma et al. Background subtraction methods for precision measurements of di-hadron and jet-hadron correlations in heavy ion collisions. *Phys. Rev.*, C93(4):044915, 2016.
- [110] G. Aad et al. Measurement of event-plane correlations in $\sqrt{s_{NN}} = 2.76$ TeV lead-lead collisions with the ATLAS detector. *Phys. Rev.*, C90(2):024905, 2014.
- [111] C. Nattrass and T. Todoroki. Event plane dependence of the flow modulated background in dihadron and jet-hadron correlations in heavy ion collisions. *Physical Review C*, 97(5), May 2018.
- [112] C. Nattrass. Re-examining the iconic dihadron correlation measurement demonstrating jet quenching. *Physical Review C*, 97(3), Mar 2018.
- [113] C. Nattrass et al. Disappearance of the Mach Cone in heavy ion collisions. *Phys. Rev.*, C94(1):011901, 2016.
- [114] Korinna Zapp, Gunnar Ingelman, Johan Rathsman, Johanna Stachel, and Urs Achim Wiedemann. A monte carlo model for ‘jet quenching’. *The European Physical Journal C*, 60(4), mar 2009.
- [115] J. G. Milhano and K. C. Zapp. Origins of the di-jet asymmetry in heavy ion collisions. *Eur. Phys. J.*, C76(5):288, 2016.
- [116] K. C. Zapp. Geometrical aspects of jet quenching in JEWEL. *Phys.Lett.*, B735:157–163, 2014.
- [117] J. Noronha-Hostler. Solving the $R_{AA} \otimes v_2$ puzzle. *Journal of Physics: Conference Series*, 736(1), August 2016.
- [118] J. Noronha-Hostler et al. Event-by-event Hydrodynamics+Jet energy loss: A solution to the $R_{AA} \otimes v_2$ puzzle. *Phys. Rev. Lett.*, 116:252301, Jun 2016.
- [119] Carola F Berger and George Sterman. Scaling rule for nonperturbative radiation in a class of event shapes. *Journal of High Energy Physics*, 2003(09):058–058, 2003.
- [120] S. Catani, G. Turnock, and B.R. Webber. Jet broadening measures in e+e- annihilation. *Physics Letters B*, 295(3):269–276, 1992.
- [121] Andrew J. Larkoski, Ian Moult, and Duff Neill. Toward multi-differential cross sections: measuring two angularities on a single jet. *Journal of High Energy Physics*, 2014(9), 2014.
- [122] J. R. Andersen et al. Les Houches 2015: Physics at TeV Colliders Standard Model Working Group Report. In *9th Les Houches Workshop on Physics at TeV Colliders*, 5 2016.
- [123] Hsi-Ming Chang, Massimiliano Procura, Jesse Thaler, and Wouter J. Waalewijn. Calculating track-based observables for the lhc. *Physical Review Letters*, 111(10), 2013.

- [124] Zhong-Bo Kang, Kyle Lee, and Felix Ringer. Jet angularity measurements for single inclusive jet production. *Journal of High Energy Physics*, 2018(4), April 2018.
- [125] Zhong-Bo Kang, Kyle Lee, Xiaohui Liu, and Felix Ringer. Soft drop groomed jet angularities at the LHC. *Phys. Lett. B*, 793:41–47, 2019.
- [126] Mrinal Dasgupta, Alessandro Fregoso, Simone Marzani, and Gavin P. Salam. Towards an understanding of jet substructure. *Journal of High Energy Physics*, 2013(9), 2013.
- [127] S. Acharya et al. Medium modification of the shape of small-radius jets in central Pb-Pb collisions at 2.76 TeV. *Journal of High Energy Physics*, 2018(10), oct 2018.
- [128] Tanmay Pani. Generalized angularities measurements from STAR at $\sqrt{s_{NN}} = 200$ GeV. *EPJ Web Conf.*, 296:11003, 2024.
- [129] Anders Andreassen, Patrick T. Komiske, Eric M. Metodiev, Benjamin Nachman, and Jesse Thaler. Omnifold: A method to simultaneously unfold all observables. *Phys. Rev. Lett.*, 124:182001, May 2020.
- [130] Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler. Energy flow polynomials: a complete linear basis for jet substructure. *Journal of High Energy Physics*, 2018(4), April 2018.
- [131] Andrew J. Larkoski, Simone Marzani, Gregory Soyez, and Jesse Thaler. Soft drop. *Journal of High Energy Physics*, 2014(5), May 2014.
- [132] Tanmay and Pani. Generalized angularities and differential jet shapes measurements from star at $\sqrt{s} = 200$ gev. In *Proceedings of 11th International Conference on Hard and Electromagnetic Probes of High-Energy Nuclear Collisions — PoS(HardProbes2023)*, HardProbes2023, page 172. Sissa Medialab, February 2024.
- [133] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez. FastJet user manual. *The European Physical Journal C*, 72(3), mar 2012.
- [134] Andrew J. Larkoski, Jesse Thaler, and Wouter J. Waalewijn. Gaining (mutual) information about quark/gluon discrimination. *Journal of High Energy Physics*, 2014(11), nov 2014.
- [135] A. M. Poskanzer and S.A. Voloshin. Methods for analyzing anisotropic flow in relativistic nuclear collisions. *Phys.Rev.*, C58:1671–1678, 1998.
- [136] J. Barrette et al. Proton and pion production relative to the reaction plane in Au + Au collisions at AGS energies. *Phys. Rev.*, C56:3254–3264, 1997.
- [137] J. Barrette et al. Energy and charged particle flow in 10.8A GeV/c Au+Au collisions . *Phys. Rev. C*, 55:1420–1430, Mar 1997.
- [138] S. A. Voloshin, A. M. Poskanzer, and R. Snellings. Collective phenomena in non-central nuclear collisions. *Landolt-Bornstein*, 23:293–333, 2010.

- [139] Q. Wang. *Charge Multiplicity Asymmetry Correlation Study Searching for Local Parity Violation at RHIC for STAR collaboration*. PhD thesis, Kansas U., New York, 2012.
- [140] E. Abbas et al. Performance of the ALICE VZERO system. *JINST*, 8:P10016, 2013.
- [141] J. Barrette et al. Energy and charged particle flow in a 10.8A GeV/c Au + Au collisions . *Phys.Rev.*, C55:1420–1430, 1997.
- [142] Yue Shi Lai, James Mulligan, Mateusz Płoskoń, and Felix Ringer. The information content of jet quenching and machine learning assisted observable design, 2022.
- [143] R Brun et al. *GEANT 3: user’s guide Geant 3.10, Geant 3.11; rev. version*. CERN, Geneva, 1987.
- [144] Lydia Brenner et al. Comparison of unfolding methods using RooFitUnfold. *International Journal of Modern Physics A*, 35(24):2050145, 2020.
- [145] Anders Andreassen, Patrick T. Komiske, Eric M. Metodiev, Benjamin Nachman, Adi Suresh, and Jesse Thaler. Scaffolding simulations with deep learning for high-dimensional deconvolution, 2021.

Appendix A

Hard Processes and Parton Showers

$A, \alpha \text{ } \text{---}\text{ } p \text{ } \text{---}\text{ } B, \beta$	$\delta^{AB} \left[-g^{\alpha\beta} + (1-\lambda) \frac{p^\alpha p^\beta}{p^2 + i\epsilon} \right] \frac{i}{p^2 + i\epsilon}$
$A \text{ } \cdots \text{ } p \text{ } \cdots \text{ } B$	$\delta^{AB} \frac{i}{p^2 + i\epsilon}$
$a, i \text{ } \xrightarrow{p} \text{ } b, j$	$\delta^{ab} \frac{i}{(\not{p} - m + i\epsilon)_{ji}}$
	$-g f^{ABC} [(p-q)^\gamma g^{\alpha\beta} + (q-r)^\alpha g^{\beta\gamma} + (r-p)^\beta g^{\gamma\alpha}]$ (all momenta incoming, $p + q + r = 0$)
	$-ig^2 f^{XAC} f^{XBD} [g^{\alpha\beta} g^{\gamma\delta} - g^{\alpha\delta} g^{\beta\gamma}]$ $-ig^2 f^{XAD} f^{XBC} [g^{\alpha\beta} g^{\gamma\delta} - g^{\alpha\gamma} g^{\beta\delta}]$ $-ig^2 f^{XAB} f^{XCD} [g^{\alpha\gamma} g^{\beta\delta} - g^{\alpha\delta} g^{\beta\gamma}]$
	$g f^{ABC} q^\alpha$
	$-ig (t^A)_{cb} (\gamma^\alpha)_{ji}$

Table A.1: Feynman rules for QCD in a covariant gauge with gauge parameter λ : the gluon, ghost and quark propagators, followed by the three- and four-gluon, ghost-gluon and quark-gluon vertices.

A.1 Hard Processes

To consider any nucleus as a singular coherent object, the fluctuations inside it would involve momentum transfers below the confinement scale (Λ) introduced in 1.2.1, and would occur over timescales $\sim 1/\Lambda$. We define *hard-scattering processes* or just *hard processes* as interactions of a hard perturbative probe with high momentum transfer $Q \gg \Lambda$. Since this interaction occurs over a time-scale $\sim 1/Q \ll 1/\Lambda$, the perturbative probe effectively interacts with a near-static “snapshot” of the nucleus with a characteristic resolution $\sim 1/Q$.

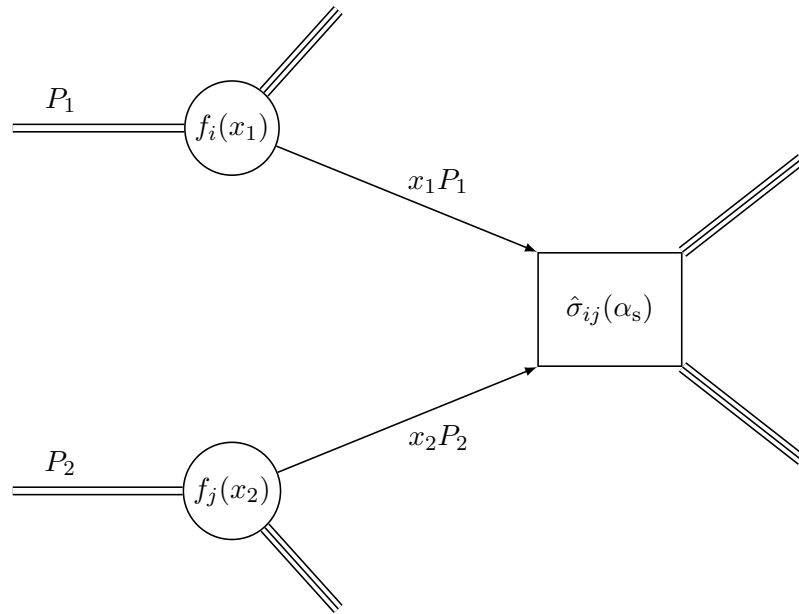


Figure A.1: Collinear factorization of a hadronic hard scattering.

Thus, the cross sections of a collision process of nuclei with momentum P_1 and P_2 depicted by Fig. A.1 can be *factorized* into various independent components as,

$$\frac{d\sigma}{d\mathbf{O}} = \sum_{i,j,f} \int dx_1 dx_2 d\Phi_f f_i(x_1, \mu^2) f_j(x_2, \mu^2) \frac{d\hat{\sigma}_{ij \rightarrow f}}{d\hat{\mathbf{O}}} D_f(\hat{\mathbf{O}} \rightarrow \mathbf{O}, \mu^2) \quad (\text{A.1})$$

Where, the interactions occur between partons i from nucleus 1 and j from nucleus 2 carrying momentum $p_1 = x_1 P_1$ and $p_2 = x_2 P_2$ respectively. μ is an arbitrary *factorization scale*, often just set to Q . The $f_k(x, \mu^2)$ are the *parton densities* giving likelihood of a parton k to carry x fraction of it's parent nucleus's momentum at a particular factorization scale

μ . f labels the partonic final state coming out of the scattering and Φ_f is its phase space. $\hat{\sigma}_{ij \rightarrow f}$ is the *parton cross-section* for scattering of partons i and j into f , kept differential in a partonic observable $\hat{\mathbf{O}}$. We can approximate it to the $(n+k)$ order in α_s as,

$$\frac{d\hat{\sigma}_{ij \rightarrow f}}{d\hat{\mathbf{O}}} = \alpha_s^k \sum_{m=0}^n \alpha_s^m \frac{d\hat{\sigma}_{ij \rightarrow f}^{(m)}}{d\hat{\mathbf{O}}}(p_1, p_2, Q^2/\mu^2) \quad (\text{A.2})$$

Different hard processes will have different leading powers k . W production will have $k=0$ whereas perturbative dijet production with two outgoing quarks will have $k=2$.

The $D_f(\hat{\mathbf{O}} \rightarrow \mathbf{O}, \mu^2)$ are the *fragmentation functions*, giving the likelihood of a partonic final state with $\hat{\mathbf{O}}$ being measured as $\mathbf{O}$ once hadronization is over. Like the parton densities, they cannot be calculated in perturbation theory, but they are *universal*: once measured in one process they can be used in any other, and their dependence on μ is governed by evolution equations of the same form as eq. ???. Eq. A.1 is the skeleton of every measurement presented in this thesis, with the choice of $\mathbf{O}$ being what separates one from the next, and a measurement is comparable to a calculation only if both use the same $\mathbf{O}$. Restricting $\mathbf{O}$ to charged particles, as will be done for the angularities of Ch. 4, puts that restriction into D_f as well, and the non-perturbative input can then no longer be avoided. The nuclear modification factors of Ch. 1.4 are ratios of Eq. A.1 between two collision systems, where the parton densities largely cancel and what is left is the effect of the medium on $\hat{\sigma}_{ij \rightarrow f}$ and D_f .

A.2 From partons to jets

Outgoing partons evolve from higher to lower momentum transfer scales (say, t). Initially their evolution are modeled using the pQCD mechanism called *parton shower*. Eventually, say at an infra-red scale cut-off t_0 , pQCD calculations stop being valid and npQCD processes like hadronization dominate. QCD Monte-Carlo programs combining parton shower at $t < t_0$ and hadronization process at $t > t_0$ are called *QCD event generators*

A.2.1 Parton branching

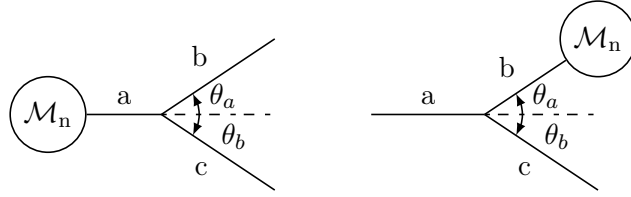


Figure A.2: Kinematics of timelike (left) vs spacelike (right) parton branching

Consider nearly collinear timelike splitting of a parton, $a \rightarrow b + c$ shown in Fig. A.2.1. Let's assume the partons are massless, and define the evolution scale (t) as,

$$t \equiv p_a^2 \approx 2E_b E_c (1 - \cos \theta) \approx z(1 - z)E_a^2 \theta^2 \quad (\text{A.3})$$

Where, $z = E_b/E_a = 1 - E_c/E_a$ and $\theta = \theta_a + \theta_b$ is the small opening angle. The cross-section of a n -parton state branching into $n + 1$ -parton state [61]

Kernel (ij)	Diagram	$\hat{P}_{ij}(z)$
qq		$C_F \left[\frac{1 + z^2}{1 - z} \right]$
gq		$C_F \frac{1 + (1 - z)^2}{z}$
qg		$T_R [z^2 + (1 - z)^2]$
gg		$C_A \left[\frac{z}{1 - z} + \frac{1 - z}{z} + z(1 - z) \right]$

Table A.2: The four leading-order collinear splittings entering the DGLAP evolution of Eq. A.4. The daughter parton i carrying momentum fraction z is drawn on the upper leg, the recoiling parton $(1 - z)$ on the lower [61].

$$d\sigma_{n+1} = d\sigma_n \frac{dt}{t} dz \frac{\alpha_s}{2\pi} \hat{P}_{ba} \quad (\text{A.4})$$

The $\hat{P}_{ba}(z)$ are the *unregularized Altarsi-Parisi splitting functions*, giving the probability of the branching $a \rightarrow b + c$ with b taking a fraction z of a 's energy. Their leading order forms are shown in Tab. A.2. $\hat{P}_{qq}$ diverges as $z \rightarrow 1$, $\hat{P}_{gq}$ as $z \rightarrow 0$, and $\hat{P}_{gg}$ at both ends, each of these limits being the emission of a soft gluon. Only $\hat{P}_{qg}$, the splitting of a gluon into a $q\bar{q}$ pair, stays finite over the whole range of z .

Same equation would hold for the space-like case. To obtain the evolution equations, we can frame the evolution in terms of the change in parton distribution function $f(x, t)$ when the evolution scale t goes from t to $t + \delta t$,

$$\begin{aligned} \delta f_i(x, t) &= \delta f_{\text{in}} - \delta f_{\text{out}} \\ &= \frac{\delta t}{t} \sum_j \int_0^1 dz \frac{\alpha_s}{2\pi} \left[\frac{\hat{P}_{ij}(z)}{z} f_j(x/z, t) - \hat{P}_{ji} f_i(x, t) \right] \end{aligned} \quad (\text{A.5})$$

The second term of the RHS gives *Sudakov form factor*, which gives the probability of a parton i evolving from scale t to a softer scale t_0 without a resolvable emission,

$$\Delta_i(t) \equiv \exp \left(- \sum_j \int_{t_0}^t \frac{dt'}{t'} \int dz \frac{\alpha_s}{2\pi} \hat{P}_{ji}(z) \right) \quad (\text{A.6})$$

We get the familiar DGLAP-like evolution equations,

$$t \frac{\partial}{\partial t} \left(\frac{f_i}{\Delta_i} \right) = \frac{1}{\Delta_i} \sum_j \int \frac{dz}{z} \frac{\alpha_s}{2\pi} \hat{P}_{ij}(z) f_j(x/z, t) \quad (\text{A.7})$$

To deal with singularities corresponding to soft gluons at $z \rightarrow 1$, we introduce an explicit infrared cut-off $z < 1 - \epsilon(t)$. Requiring the transverse momentum in timelike branching

$a \rightarrow bc$ to be positive,

$$\begin{aligned} p_T^2 &= z(1-z)p_a^2 - (1-z)p_b^2 - zp_c^2 > 0 \\ \implies z(1-z) &> t_0/t \implies t_0/t < z < 1 - t_0/t \end{aligned} \quad (\text{A.8})$$

and also, $t > 2t_0$. Thus the Sudakov factors become,

$$\Delta_i(t) \equiv \exp \left(- \sum_j \int_{2t_0}^t \frac{dt'}{t'} \int_{t_0/t'}^{1-t_0/t'} dz \frac{\alpha_s}{2\pi} \hat{P}_{ji}(z) \right) \quad (\text{A.9})$$

A.2.2 Angular Ordering

Sec. A.2.1 took collinear enhancements into account to all orders in α_s . However it does not take the enhancement in soft gluon emission at wide emission angle. This is demonstrated in Eq. A.10 by considering an on-mass-shell external line of momentum p , mass m emitting a gluon with momentum q at an angle θ

$$\frac{1}{(p \pm q)^2 - m^2} = \frac{\pm 1}{2p \cdot q} = \frac{\pm 1}{2\omega E(1 - v \cos \theta)}, \quad (\text{A.10})$$

where ω is the gluon energy, E and v the energy and velocity of the emitter and θ the emission angle. For light partons $v \rightarrow 1$ and the collinear enhancement appears as $\theta \rightarrow 0$; the enhancement as $\omega \rightarrow 0$ holds for any velocity and angle. Internal lines are off-mass-shell and carry no such enhancement, since $(p + q)^2 - m^2 \rightarrow p^2 - m^2 \neq 0$. The cross section carries a sum over all pairs of external lines $\{i, j\}$,

$$d\sigma_{n+1} = d\sigma_n \frac{d\omega}{\omega} \frac{d\Omega}{2\pi} \frac{\alpha_s}{2\pi} \sum_{i,j} C_{ij} W_{ij} \quad (\text{A.11})$$

The weighted sum $\sum_{i,j} C_{ij} W_{ij}$ is called the *antenna pattern* of the process. The *color factor* C_{ij} and *radiation function* W_{ij} are defined as,

$$\begin{aligned} C_{ij} &= -\mathbf{Q}_i \cdot \mathbf{Q}_j \\ W_{ij} &= \frac{\omega^2 p_i \cdot p_j}{p_i \cdot q \, p_j \cdot q} = \frac{1 - v_i v_j \cos \theta_{ij}}{(1 - v_i \cos \theta_{iq})(1 - v_j \cos \theta_{jq})} \end{aligned} \quad (\text{A.12})$$

The vectors $\mathbf{Q}_i$ represents color charge of parton i .

$$\mathbf{Q}_i^2 = \begin{cases} C_F = 4/3 & i \in \text{quarks} \\ C_A = 3 & i \in \text{gluons} \\ 0 & i \in \text{color singlet} \end{cases} \quad (\text{A.13})$$

W_{ij} comes from interference between lines i and j . W_{ij} can be separated into two parts $W_{ij}^{[i/j]}$ each with an *incoherent* and a *coherent* contribution from line i/j . Assuming massless emitters ($v_i = v_j = 1$) for simplicity,

$$\begin{aligned} W_{ij} &= W_{ij}^{[i]} + W_{ij}^{[j]} \\ W_{ij}^{[i]} &= \frac{1}{2} \left(\overbrace{\frac{1}{1 - \cos \theta_{iq}}}^{\text{incoherent}} + \overbrace{\frac{\cos \theta_{iq} - \cos \theta_{ij}}{(1 - \cos \theta_{jq})(1 - \cos \theta_{iq})}}^{\text{coherent}} \right) \\ &= \frac{1}{2(1 - \cos \theta_{iq})} \left(1 + \frac{\cos \theta_{iq} - \cos \theta_{ij}}{1 - \cos \theta_{ij} \cos \theta_{iq} - \sin \theta_{ij} \sin \theta_{iq} \cos \phi_{iq}} \right) \end{aligned} \quad (\text{A.14})$$

Using the direction of line i as the z-axis, $d\Omega$ from Eq. A.11 becomes $d \cos \theta_{iq} d\phi_{iq}$, We can carry out the ϕ_{iq} integration,

$$\begin{aligned} \int_0^{2\pi} \frac{d\phi_{iq}}{2\pi} W_{ij}^{[i]} &= \frac{1}{1 - \cos \theta_{iq}} \left(\frac{1 + \frac{\cos \theta_{iq} - \cos \theta_{ij}}{|\cos \theta_{iq} - \cos \theta_{ij}|}}{2} \right) \\ &= \begin{cases} \frac{1}{1 - \cos \theta_{iq}} & \cos \theta_{iq} > \cos \theta_{ij} \implies \theta_{iq} < \theta_{ij} \\ 0 & \text{otherwise,} \end{cases} \end{aligned} \quad (\text{A.15})$$

Eq. A.15 shows that azimuth-averaged soft gluon emission from a line i is confined to a

cone, centered on i 's direction, extending up to the direction of line j , i 's sibling from the preceding split. Since the emission probability of Eq. A.11 is weighted by the color charges, a gluon radiates $C_A/C_F = 9/4$ times as much as a quark of the same energy into the same cone. Gluon jets are therefore wider and carry more particles than quark jets of the same momentum, a difference the classifiers of Ch. ?? are trained to pick up, and one that has to be kept in mind whenever angularities are compared between samples with different quark and gluon fractions.

A.2.3 Color Coherence

Lets take the case of three partons $\{i, j, k\}$ forming a color singlet ($\mathbf{Q}_i + \mathbf{Q}_j + \mathbf{Q}_k = 0$) e.g., $e^+e^- \rightarrow q\bar{q}g$. First, partons l and k come from a color singlet state ($\mathbf{Q}_l + \mathbf{Q}_k = 0$). Then l splits into i and j ($\mathbf{Q}_l = \mathbf{Q}_i + \mathbf{Q}_j = -\mathbf{Q}_k$). The antennae function for this state is:

$$\begin{aligned}
\sum_{a,b} C_{ab} W_{ab} &= -\mathbf{Q}_i \cdot \mathbf{Q}_j W_{ij} - \mathbf{Q}_j \cdot \mathbf{Q}_k W_{jk} - \mathbf{Q}_i \cdot \mathbf{Q}_k W_{ik} \\
&= \frac{1}{2} [\mathbf{Q}_i^2 (W_{ij} + W_{ik} - W_{jk}) + \mathbf{Q}_j^2 (W_{ij} + W_{jk} - W_{ik}) + \mathbf{Q}_k^2 (W_{ik} + W_{jk} - W_{ij})] \\
&= \mathbf{Q}_i^2 \left(W_{ij}^{[i]} + \tilde{W}_{jk}^{[i]} - \tilde{W}_{ik}^{[j]} \right) + \mathbf{Q}_j^2 \left(W_{ij}^{[j]} + \tilde{W}_{ik}^{[j]} - \tilde{W}_{jk}^{[i]} \right) \\
&\quad + \mathbf{Q}_k^2 \left(\tilde{W}_{jk}^{[i]} + \tilde{W}_{ik}^{[j]} + \frac{W_{ik}^{[k]} + W_{jk}^{[k]}}{2} \right)
\end{aligned} \tag{A.16}$$

Where,

$$\tilde{W}_{jk}^{[i]} = \frac{W_{ik}^{[i]} - W_{ij}^{[i]}}{2}, \quad \tilde{W}_{ik}^{[j]} = \frac{W_{jk}^{[j]} - W_{ij}^{[j]}}{2} \tag{A.17}$$

are shifted radiation functions for lines i and j , with the collinear singularity canceled out. This allows us to make certain useful approximations. Taking $\theta_{ij} \rightarrow 0$, we can approximate the directions of i and j by l .

$$W_{ik}^{[k]} \simeq W_{jk}^{[k]} \simeq W_{lk}^{[k]}, \text{ and } \tilde{W}_{jk}^{[i]} \simeq \tilde{W}_{ik}^{[j]} \simeq W_{lk}^{[ij]}/2 \tag{A.18}$$

Where,

$$\tilde{W}_{lk}^{[ij]} = \begin{cases} W_{lk}^{[l]} & \theta_{lq} > \theta_{ij} \\ 0 & \theta_{lq} < \theta_{ij}, \end{cases} \quad (\text{A.19})$$

$$W \simeq \mathbf{Q}_i^2 W_{ij}^{[i]} + \mathbf{Q}_j^2 W_{ij}^{[j]} + \mathbf{Q}_k^2 W_{lk}^{[k]} + \mathbf{Q}_l^2 \tilde{W}_{lk}^{[ij]}, \quad (\text{A.20})$$

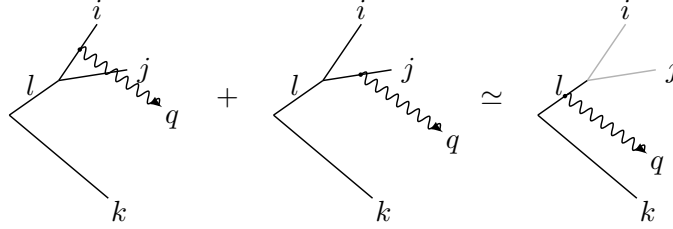


Figure A.3: Wide-angle emission of soft gluon q , off i and j , acts as if the emission came off the parent parton l , imagined to be on shell.

Each parton radiates in proportion to its color charge squared. When i and j close in angle their incoherent contributions are limited to cones of half-angle θ_{ij} , while at larger angles, out to the direction of k , they radiate coherently in proportion to their combined charge $\mathbf{Q}_l^2$, as if from an internal line of momentum $p_l = p_i + p_j$ as shown in Fig. A.3.

A.2.4 Coherent Parton Branching

Extended to higher orders this gives a *coherent parton branching* formalism. In place of the virtual mass-squared t the evolution variable is the opening angle, or more precisely

$$\zeta = \frac{p_b \cdot p_c}{E_b E_c} \simeq 1 - \cos \theta \quad (\text{A.21})$$

for $a \rightarrow bc$, with angular ordering $\zeta' < \zeta$ imposed on successive branchings and the propagator factor dt/t of Sec. A.2.1 replaced by $d\zeta/\zeta$. Using the splitting function $\hat{P}_{ba}(z)dz$ rather than the soft approximation $\mathbf{Q}_a^2 d\omega/\omega$ treats both soft and collinear enhancements correctly, so that

$$d\sigma_{n+1} = d\sigma_n \frac{d\zeta}{\zeta} dz \frac{\alpha_s}{2\pi} \hat{P}_{ba}(z). \quad (\text{A.22})$$

The virtual mass-squared cut-off is replaced by an angular one, $\zeta_0 = t_0/E^2$ for a parton of energy E , and the convenient evolution variable $\tilde{t} = E^2\zeta \geq t_0$ in terms of which the ordering condition $\zeta_b, \zeta_c < \zeta_a$ and the resulting cut-off on z read

$$\tilde{t}_b < z^2\tilde{t}, \quad \tilde{t}_c < (1-z)^2\tilde{t}, \quad \sqrt{t_0/\tilde{t}} < z < 1 - \sqrt{t_0/\tilde{t}}, \quad (\text{A.23})$$

with $\tilde{t} = \tilde{t}_a$ and $z = E_b/E_a$. Neglecting the masses of b and c ,

$$t = z(1-z)\tilde{t}, \quad p_t^2 = z^2(1-z)^2\tilde{t}, \quad (\text{A.24})$$

and the Sudakov form factor of Sec. A.2.1 becomes

$$\tilde{\Delta}_i(\tilde{t}) \equiv \exp\left(-\sum_j \int_{4t_0}^{\tilde{t}} \frac{d\tilde{t}'}{\tilde{t}'} \int_{\sqrt{t_0/\tilde{t}'}}^{1-\sqrt{t_0/\tilde{t}'}} dz \frac{\alpha_s(z^2(1-z)^2\tilde{t}')}{2\pi} \hat{P}_{ji}(z)\right) \quad (\text{A.25})$$

At large $\tilde{t}$ this falls more slowly than its collinear counterpart, though still faster than any negative power of $\tilde{t}$. The slower fall implies less branching, which is the suppression of soft gluon emission by angular ordering.

A.2.5 Angular structure of a jet

Sec. A.2.4 is enough to estimate what the radiation around a hard parton looks like. At a hadron collider, the angle θ between two particles is measured as their distance ΔR in the (η, ϕ) plane and the energy E as the transverse momentum p_T (see Sec. 2.2.1), and a *jet* is, for now, all the radiation within $\Delta R < R$ of the initiating parton. Taking z to be the momentum fraction of the emitted gluon, as in $\hat{P}_{gq}$ and $\hat{P}_{gg}$ of Tab. A.2, the soft limit is $\hat{P}_{gi}(z) \rightarrow 2C_i/z$ and $d\zeta/\zeta \rightarrow 2d\theta/\theta$, so Eq. A.22 becomes,

$$dP = \frac{2\alpha_s C_i}{\pi} \frac{d\hat{\theta}}{\hat{\theta}} \frac{dz}{z}, \quad \hat{\theta} \equiv \frac{\Delta R}{R} < 1 \quad (\text{A.26})$$

Where, $C_i = \mathbf{Q}_i^2$ is the color charge of the initiating parton from Sec. A.2.2. Eq. A.26 is flat in $\ln(1/z)$ and $\ln(1/\hat{\theta})$, so a jet is filled uniformly in this plane, the *Lund plane*, down to the cut-off. The *generalized angularity* of a jet is the sum over its emissions,

$$\lambda_\beta = \sum_{i \in \text{jet}} z_i \hat{\theta}_i^\beta \quad (\text{A.27})$$

Eq. 4.1 is the same quantity written in terms of the measured constituents of a jet, with $z_i \rightarrow p_{\text{T, const}}/p_{\text{T, jet}}$ and $\hat{\theta}_i \rightarrow \Delta R_{\text{const, jet}}/R$, and with a second exponent κ on z_i which is 1 here. For $\beta > 0$, an emission contributes nothing to λ_β in either the soft or the collinear limit, which is what makes it an IRC safe observable. Since λ_β is additive, the probability of a jet having an angularity below $\hat{\lambda}_\beta$ is the probability of no emission with $z\hat{\theta}^\beta > \hat{\lambda}_\beta$, which is the Sudakov form factor of Eq. A.25 with the no-emission condition replaced accordingly. Calling this cumulative distribution $\Sigma_i(\hat{\lambda}_\beta)$, with the hat marking it as a parton level quantity like in Eq. A.1, the hadron level cross section follows from it through the fragmentation function of Eq. A.1 for $\mathbf{O} = \lambda_\beta$, summed over the flavor of the parton initiating the jet,

$$\frac{d\sigma}{d\lambda_\beta} = \sum_{i \in \{q, g\}} \int d\hat{\lambda}_\beta \frac{d\sigma_i}{d\hat{\lambda}_\beta} D_i(\hat{\lambda}_\beta \rightarrow \lambda_\beta) = \sum_{i \in \{q, g\}} \sigma_i \int d\hat{\lambda}_\beta \frac{d\Sigma_i}{d\hat{\lambda}_\beta} D_i(\hat{\lambda}_\beta \rightarrow \lambda_\beta) \quad (\text{A.28})$$

Where, σ_i is the cross section for producing a jet initiated by parton i , i.e. Eq. A.1 with everything but the flavor of the outgoing parton integrated over, and $d\sigma_i/d\hat{\lambda}_\beta = \sigma_i d\Sigma_i/d\hat{\lambda}_\beta$ is its parton level distribution. Normalized to the total cross section, the left hand side is the distribution measured in Ch. 4. The rest of this section is about Σ_i , and comes back to D_i at the end. At fixed coupling,

$$\begin{aligned} \Sigma_i(\hat{\lambda}_\beta) &= \exp \left(-\frac{2\alpha_s C_i}{\pi} \int_0^1 \frac{d\hat{\theta}}{\hat{\theta}} \int_0^1 \frac{dz}{z} \Theta(z\hat{\theta}^\beta - \hat{\lambda}_\beta) \right) \\ &= \exp \left(-\frac{\alpha_s C_i}{\pi\beta} \ln^2 \hat{\lambda}_\beta \right) \end{aligned} \quad (\text{A.29})$$

Quark and gluon jets differ only through C_i , so $\Sigma_g = \Sigma_q^{C_A/C_F}$ [134]. A convenient choice for scale of α_s is $p_{\text{T,jet}}R$, the largest transverse momentum an emission inside the jet can have.

Tab. A.3 evaluates Eq. A.29 for the three β values measured in Ch. 4.

β	median $\hat{\lambda}_\beta$		fraction with $\hat{\lambda}_\beta > 0.1$	
	quark	gluon	quark	gluon
0.5	0.13	0.26	59%	87%
1	0.06	0.15	36%	64%
2	0.02	0.07	20%	40%

Table A.3: Parton level median angularity and fraction of jets with $\hat{\lambda}_\beta > 0.1$ from Eq. A.29 at $\alpha_s = 0.2$, corresponding to $p_{\text{T,jet}} \approx 20$ GeV/ c and $R = 0.4$, for quark ($C_F = 4/3$) and gluon ($C_A = 3$) initiated jets.

Median angularities are larger by a factor of 2–3 for gluon jets compared to quark jets. Increasing β pushes both to smaller values and widens the gap between them, as the observable weighs the wide angle radiation more heavily. Going up in $p_{\text{T,jet}}$ only changes α_s logarithmically, so the distributions are expected to narrow slowly.

Appendix B

Relativistic Kinematics

B.1 Laboratory vs Center-of-mass frames

A typical collision is a two-to-many reaction like:

$$\underbrace{A}_{\text{projectile}} + \underbrace{B}_{\text{target}} \rightarrow C_1 + C_2 + C_3 + \dots, \quad (\text{B.1})$$

In *laboratory frame*, the target is at rest, so,

$$\mathbf{p}_B^{\text{lab}} = \vec{0} \implies p_B^{\text{lab}} = (m_B, 0, 0, 0), \quad (\text{B.2})$$

In *center-of-mass* frame, the sum of three-momentum vectors vanish in the initial and final states

$$\mathbf{p}_A^{\text{cm}} + \mathbf{p}_B^{\text{cm}} = \sum_i \mathbf{p}_{C,i}^{\text{cm}} = \vec{0}, \quad (\text{B.3})$$

Using Lorentz invariance of the Mandelstam variable $s \equiv (p_A + p_B)_\mu (p_A + p_B)^\mu$:

$$\begin{aligned} s &= (E_A^{\text{lab}} + m_B)^2 - |\mathbf{p}_A^{\text{lab}}|^2 = (E_A^{\text{cm}} + E_B^{\text{cm}})^2 \\ \implies s &= m_A^2 + m_B^2 + 2m_B E_A^{\text{lab}} = (E^{\text{cm}})^2 \end{aligned} \quad (\text{B.4})$$

As relativistic energies ($\mathcal{O}(10^{2-3}\text{GeV})$) are much higher than reactant masses $\mathcal{O}(1 - 10\text{GeV})$,

$$\sqrt{s} = E^{\text{cm}} \approx \sqrt{2m_B E_A^{\text{lab}}} \quad (\text{B.5})$$

As the center-of-mass energy that produces new particles increases only as $\sqrt{E^{\text{lab}}}$, making any gain in particle production more expensive at relativistic energies. Thus modern colliders usually operate in center-of-mass frames.

B.2 Coordinate system

Considering the non-linear addition of velocities in relativistic mechanics,

$$\beta = \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2}, \quad (\text{B.6})$$

Considering the transformation $y = \tanh^{-1} \beta$,

$$\begin{aligned} \tanh^{-1} \beta_1 + \tanh^{-1} \beta_2 &= \tanh^{-1} \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2} = \tanh^{-1} \beta \\ \implies y_1 + y_2 &= y \end{aligned} \quad (\text{B.7})$$

The addition is linear again. We call this transformed velocity *rapidity*, given by:

$$y = \tanh^{-1} \beta = \frac{1}{2} \ln \frac{1 + \beta}{1 - \beta} = \frac{1}{2} \ln \frac{E + p_z}{E - p_z} \quad (\text{B.8})$$

Rapidity can grow without bound, and at small velocities, $y \approx \beta$, making it a relativistic analogue of velocity. Since it transforms linearly under Lorentz boosts, rapidity distribution of particles retain their shapes. For relativistic energies, $E \approx p$, making B.8,

$$y \approx \frac{1}{2} \ln \frac{p + p_z}{p - p_z} = -\ln \left(\tan \frac{\theta}{2} \right) = \eta \quad (\text{B.9})$$

η is called the *pseudo-rapidity*, the massless limit of rapidity. As shown in B.2, pseudo-rapidity can be directly determined from particle production angle θ measured with respect

to the beam axis.

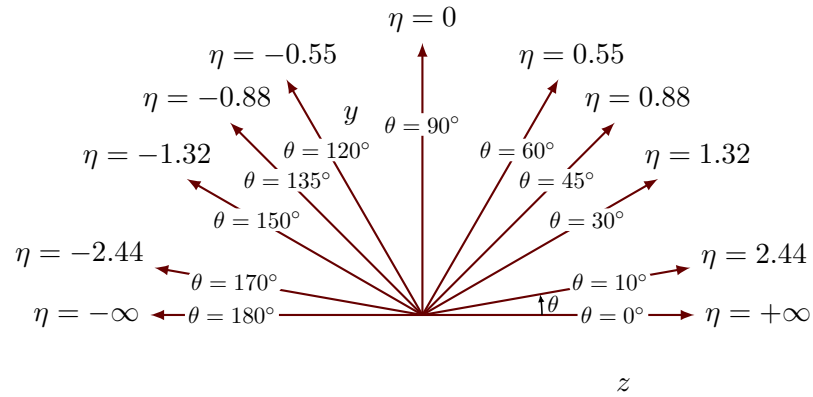


Figure B.1: η vs θ

Also considering *azimuthal angle* (φ) and *transverse-momentum* ($p_T = \sqrt{p_x^2 + p_y^2}$) of the particles, which are invariant under boosts along the beam-line by design, we can use (p_T, η, φ) as the coordinate system for the particles produced in the collision. Figure 2.6 illustrates this coordinate system for a cylindrical collider detector.

Appendix C

Event plane reconstruction

The reaction plane, Ψ_{RP} cannot be directly measured and is approximated by the second-order symmetry plane, $\Psi_{2,EP}$, referred to as the *event plane* in this text, for simplicity. $\Psi_{\text{RP}} \rightarrow \Psi_{2,EP}$ if nucleon distributions were in their average positions and devoid of fluctuations from interactions among nucleons [108]. By measuring the charged particle azimuthal distribution, the n-th order event plane can be extracted by [135],

$$\Psi_{n,EP} = \frac{1}{n} \arctan \left(\frac{Q_{y,n}}{Q_{x,n}} \right), \quad (\text{C.1})$$

the weighted vector $\mathbf{Q}_n = Q_{x,n}\hat{\mathbf{x}} + Q_{y,n}\hat{\mathbf{y}}$ are given by,

$$\mathbf{Q}_n = \hat{\mathbf{x}} \sum_{\text{tracks}} p_{\text{T,track}} \cos(n\phi_{\text{track}}) + \hat{\mathbf{y}} \sum_{\text{tracks}} p_{\text{T,track}} \sin(n\phi_{\text{track}}). \quad (\text{C.2})$$

where the sum is calculated for all reconstructed charged particles (tracks) in the event, ϕ_{track} is the track's azimuthal angle, and $p_{\text{T,track}}$ the track's transverse momentum.

Due to finite acceptance and multiplicities, the calculated event plane has an underlying anisotropy that is corrected by applying two separate correction methods. First, a calibration and recentering correction procedures are applied to remove bias introduced by non-uniform acceptance of the TPC tracking system and further account for potential beam-condition effects [135–137]. This procedure involves recentering the weighted $\mathbf{Q}_2$ -vectors such that, $\langle \mathbf{Q}_2 \rangle = \mathbf{0}$. It is done by calculating a modified $\mathbf{Q}_2$, obtained by subtracting an event averaged $\langle \mathbf{Q}_2 \rangle$ from each event's nominal $\mathbf{Q}_2$.

The higher harmonics of $\Psi_{n,EP}$ still persist after recentering [135], and are removed through a second correction step, referred to as the shifting method, applied event-by-event to make the event-plane angle isotropic in the lab frame [138]. This method defines a new angle:

$$\Psi'_{2,EP} = \Psi_{2,EP} + \sum_n \frac{2}{n} (-\langle \sin(n\Psi_{2,EP}) \rangle \cos(n\Psi_{2,EP}) + \langle \cos(n\Psi_{2,EP}) \sin(n\Psi_{2,EP}) \rangle), \quad (C.3)$$

Usually 20 orders of Fourier moment are removed. Additional details of the recentering and shifting corrections can be found in Refs. [135, 136, 139]

The difference between Ψ_{RP} and $\Psi_{2,EP}$ is quantified by event-plane resolution, R_n or $R_n\{\Psi_{2,EP}\}$ given by Eqn. C.4. The observed v_n , v_n^{obs} is corrected for this limited resolution by doing, $v_n = v_n^{\text{obs}}/R_n$ [135, 140].

$$R_n = R_n\{\Psi_{2,EP}\} = \langle \cos(n[\Psi_{RP} - \Psi_{2,EP}]) \rangle \quad n \text{ is even} \quad (C.4)$$

Furthermore, individual events are divided into two random sub-events by assigning charged tracks to sub-events, "a" and "b". The sub-events are unique, with approximately equal multiplicities. We can write the correlation of two event planes by taking the product of two sub-events, [135, 141]

$$\langle \cos(n[\Psi_{2,EP}^a - \Psi_{2,EP}^b]) \rangle = \langle \cos(n[\Psi_{2,EP}^a - \Psi_{RP}]) \rangle \langle \cos(n[\Psi_{2,EP}^b - \Psi_{RP}]) \rangle, \quad (C.5)$$

This allows calculation of the event-plane resolution directly from data. Since a and b have equal multiplicities, the resolution of each sub-event can be calculated from the correlation between the two sub-events as [135]:

$$\langle \cos(n[\Psi_{2,EP}^a - \Psi_{RP}]) \rangle = \sqrt{\langle \cos(n[\Psi_{2,EP}^a - \Psi_{2,EP}^b]) \rangle}. \quad (C.6)$$

The event-plane resolution is multiplicity dependent and calculated for separate ranges

of collision centrality using the two sub-events method. Narrower bins are calculated and then combined accordingly to match the ranges required, by averaging the results from the narrow bins weighted by the multiplicity of each bin [138].

Appendix D

Jets and Machine Learning

D.1 Machine Learning Basics

Fortunately, many of the tasks encountered in HEP can be naturally reformulated as machine learning problems. Typically, the problems are formulated in terms of a search for some function $f : X \rightarrow Y$, from the space of the observed data X to a low-dimensional space of a desired target label Y , which optimizes some metric of our choosing. This metric is called a loss function and is written as $L[\mathbf{y}, f(\mathbf{x})]$.

Ideally, a learning algorithm would find the function that optimizes L over all possible values of $(\mathbf{x}, \mathbf{y})$, but this is intractable owing to the curse of dimensionality and an infinite number of functions to choose from. Instead, in supervised learning, one has labeled training data $\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^N$ sampled from $p(\mathbf{x}, \mathbf{y})$. Furthermore, the function space is restricted to a model—a highly flexible family of functions $f_\phi(\mathbf{x})$ parameterized by ϕ . In this case, the algorithms minimize the loss function directly with respect to the model parameters ϕ . Neural networks, support vector machines, and decision trees are examples of types of models commonly used in machine learning. These models often have a large number of parameters, and in the case of neural networks, finding the optimal f_ϕ can be a difficult problem.

An essential goal in machine learning is generalization—the ability of the model to perform well on data that were not used in training. Failure in this task is called overtraining or overfitting. A vast array of techniques exist to avoid overtraining, all of which can be considered forms of regularization. Regularization techniques such as dropout were key to advances in image recognition with deep learning.

D.2 Jet Classification

For jets, classifiers can be used as a tagging mechanism. For almost every way of representing a jet, there is a class of machine learning models. We discuss two cases here

D.2.1 Quark-Gluon jet discrimination

Quark and gluon jet $p + p$ events are generated as $qq \rightarrow qq$ and $gg \rightarrow gg$ dijet events at $\sqrt{s_{\text{NN}}} = 200$ GeV using PYTHIA-8 (RHIC Tune). Each jet is represented as an 24×24 image following [22]. The RGB color channels of the image are made as,

- R: $\sum p_T$ of charged constituents,
- G: $\sum p_T$ of neutral constituents,
- B: number of charged constituents.

giving jets as shown in D.2.1. Preprocessing involves centering the images around each jet's p_T -weighted centroid $(\bar{y}, \bar{\phi}) = \sum_i p_{T,i}(y_i, \phi_i) / \sum_i p_{T,i}$, followed by normalization. Data is augmented using procedures like random reflections ($\Delta y \rightarrow -\Delta y$ or $\Delta \phi \rightarrow -\Delta \phi$) and random jitter.

The images were input for a Convolutional Neural Network (CNN) implemented using PyTorch as shown in fig. D.2. Each convolutional layer consisted of 64 filters, with filter sizes of 5×5 , 3×3 , and 3×3 , respectively (8×8 , 4×4 , and 4×4 for the 72×72 images). The max-pooling layers performed a 2×2 down-sampling with a stride length of 2. The dense layer consisted of 128 units, with a dropout rate of 0.2 before and after it.

The network was trained with the AdamW optimizer at a learning rate of 10^{-3} and a weight decay of 0.1, using the cross-entropy loss. Training used a batch size of 128 for 20 epochs, and the model with the lowest validation loss was retained. Training images were augmented with random reflections in $\Delta \eta$ and $\Delta \phi$ and random translations of up to 0.5 pixels. The Receiver Operator Characteristic (ROC) curve is shown differential in $p_{T,\text{jet}}$ by evaluating the model on a holdout test set.

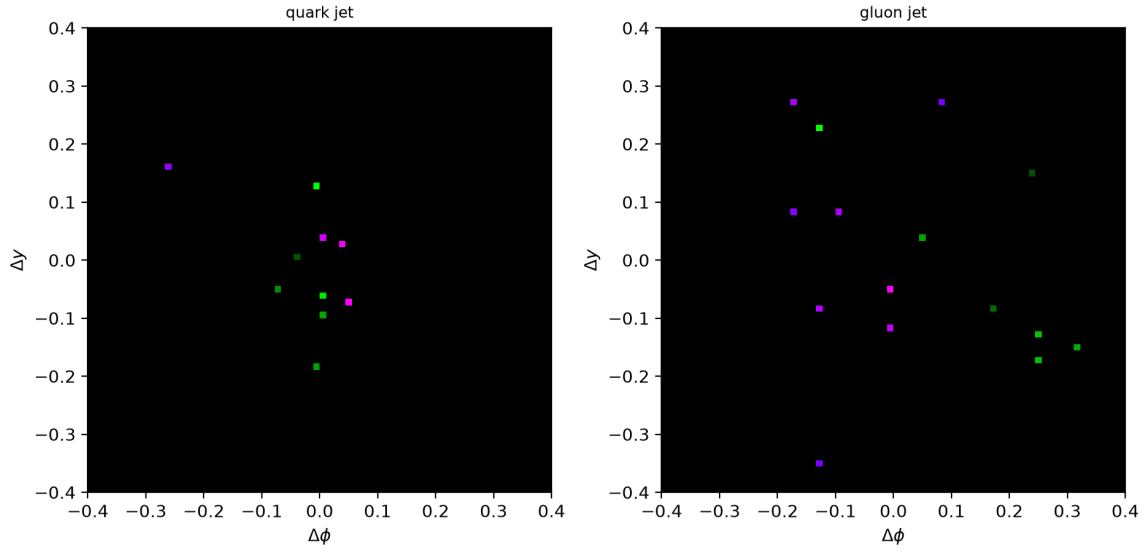


Figure D.1: Typical quark (left) and gluon (right) jet. Gluon jets exhibit a broader spatial radiation pattern and higher particle multiplicity compared to quark jets

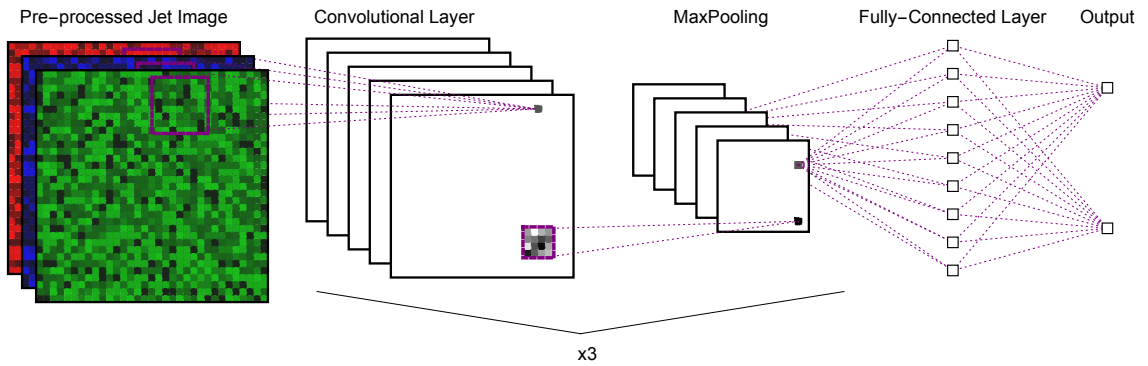


Figure D.2: An illustration of the deep convolutional neural network architecture. The first layer is the input jet image, followed by three convolutional layers, a dense layer and an output layer. [22]

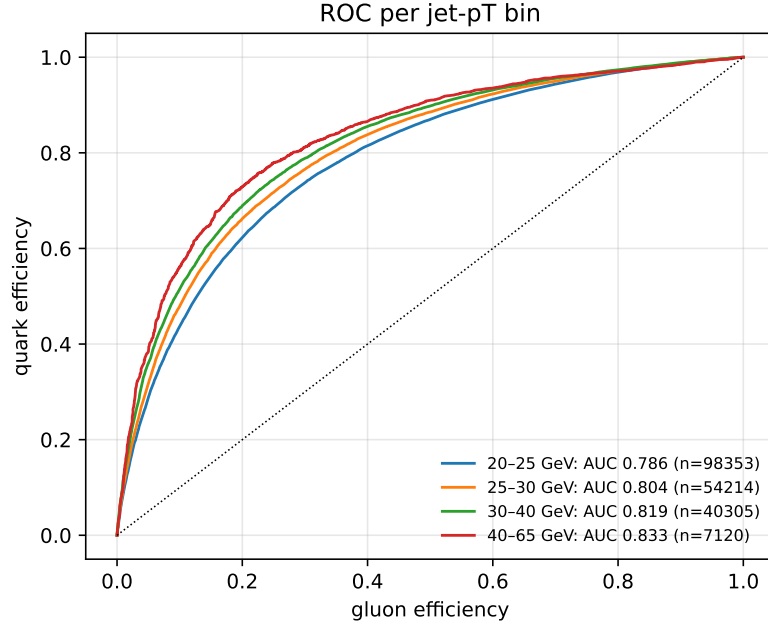


Figure D.3:

A way to gain some understanding of how the model looks at jet substructure is through *interpretability* methods. We use a simple method of just noting the average change in classifier output upon masking a pixel. As seen in fig. D.4,

D.2.2 Characterizing jet quenching

Another way to think about jet quenching/modification, is to ask how easily can a classifier algorithm tell apart the quenched and unquenched jet. Here we take PYTHIA-6 [64] as the source of unquenched jets and JEWEL [69] generates quenched jets in different centrality bins with/without recoil centers. Jets are represented as a 9-D vectors of features: $p_{T,\text{jet}}$, $N_{\text{constit.},\text{jet}}^{\text{charged}}$, $N_{\text{constit.},\text{jet}}^{\text{neutral}}$, LeSub , λ_0^2 , $\lambda_1^{0.5}$, λ_1^1 , λ_1^2 . The jets are input into Dense Neural Networks (DNNs) with three hidden layers with 100 nodes each. After training, we calculate ROC curves shown in D.2.2 using a holdout test set. The DNN performs better than chance in all the centralities and for both with/without recoils case. The with-recoils case gives exaggerated ROC curves due to the background of recoil centers. AUC score decreases from

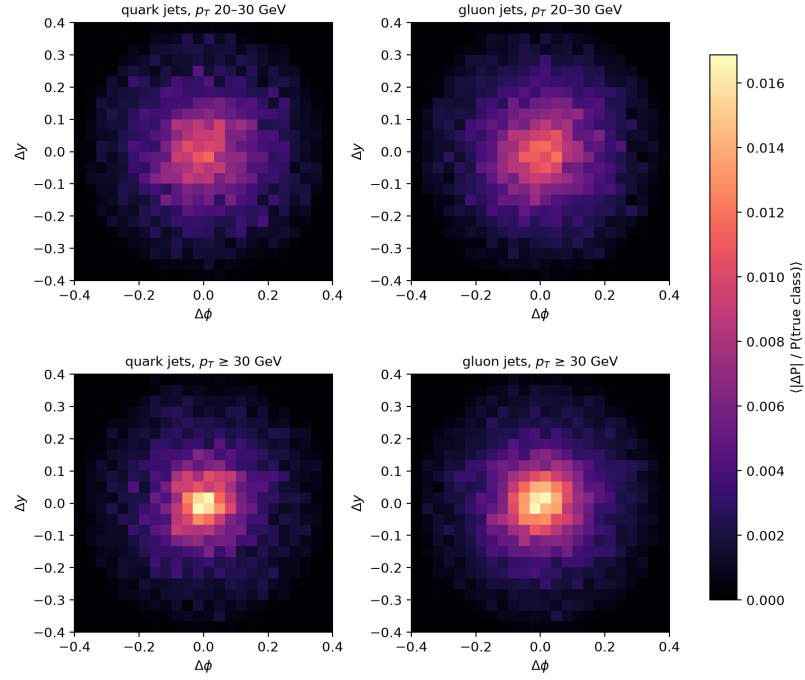


Figure D.4: Per-pixel feature importance

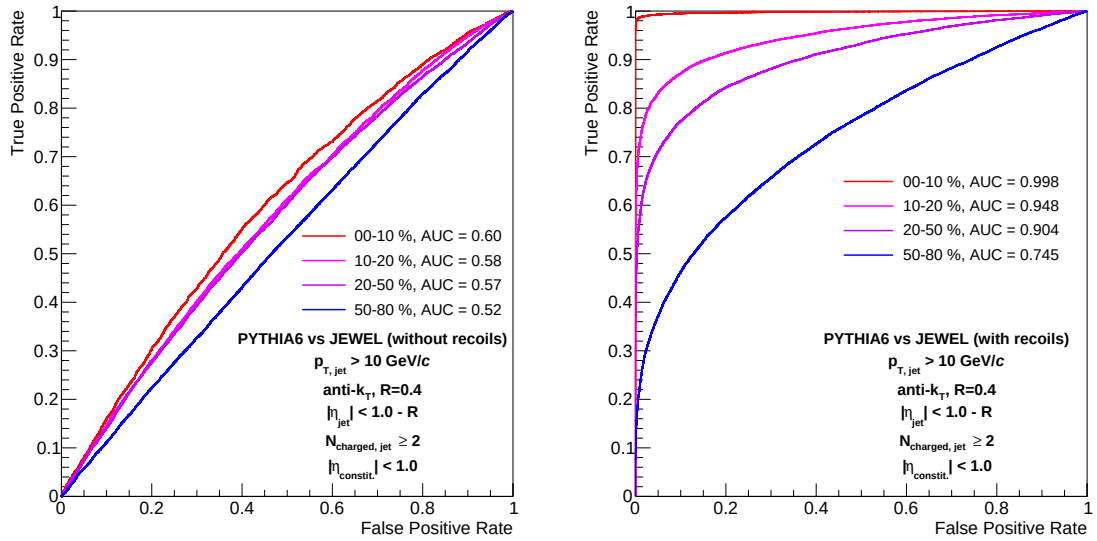


Figure D.5: ROC curves for PYTHIA6 vs. JEWEL without recoils (left) and with recoils (right).

central (0-10%) to peripheral (50-80%) events possibly implying more modification in central compared to peripheral. More information about jet quenching and machine learning can be found in [142]

Appendix E

Unfolding

For any experiment, there is a disconnect between the desired phenomena (*truth* level) and the measurement (*measured* level) due to limitations of the detector setup. This is portrayed in fig. E through a dijet event at the truth level, whose final state charged and neutral particles are recorded as TPC tracks and BEMC towers. Each of the detectors smear the energy and momenta and for almost all experiments, the information about particle species is lost. For example, at STAR often a π^0 mass assumption is taken for measured constituents. Thus jets, which are composite objects of these tracks and towers are non-trivially modified by a complicated convolution of these effects. Even with the best background removal procedure, jets will contain some residual background fluctuations from the underlying event.

Thus distributions of any jet observable $\mathbf{O}$ will have these *detector effects* which will need to be removed in order for the measurement to be comparable to theory calculations. In other terms, a truth value of $\mathbf{x}$ can manifest as a range of measured values $(\mathbf{x}_0, \mathbf{x}_1)$ similarly, a measured value of $\mathbf{y}$ can come from a range of truth values $(\mathbf{y}_0, \mathbf{y}_1)$. Thus, using Bayes theorem, the probability either measured value being $\mathbf{y}$ or truth value being $\mathbf{x}$ would be,

$$\begin{aligned} p_{\text{meas.}}(\mathbf{y}) &= \int d\mathbf{x} \, p_{\text{meas.}|\text{tr.}}(\mathbf{y}|\mathbf{x}) p_{\text{tr.}}(\mathbf{x}) \\ p_{\text{tr.}}(\mathbf{x}) &= \int d\mathbf{y} \, p_{\text{tr.}|\text{meas.}}(\mathbf{x}|\mathbf{y}) p_{\text{meas.}}(\mathbf{y}) \end{aligned} \tag{E.1}$$

From eq. E.1, we measure $p_{\text{meas.}}$ through histograms, density estimates, etc. and $p_{\text{tr.}}$ is what we want to measure. Now, $p_{\text{meas.}|\text{tr.}}$ represents the detector's *response*, which can

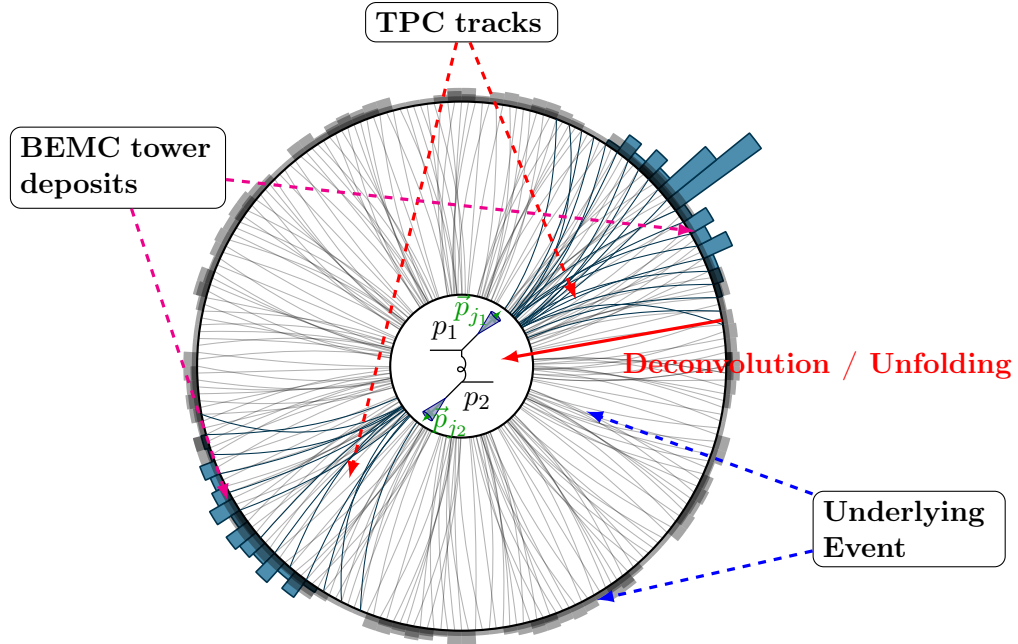


Figure E.1: Schematic figure showing the unfolding procedure. The desired physics / truth / particle level is represented by the $2 \rightarrow 2$ feynman diagram in the middle. The blue (gray) tracks and tower deposits are the dijet (underlying) event as recorded by the TPC and BEMC.

be obtained to a high degree of accuracy from simulation programs like GEANT [143]. We can start from a *generated* truth distribution $p_{\text{gen},0}$ and obtain the *reconstructed* measured distribution $p_{\text{reco},0}$. Thus the assumption is,

$$p_{\text{meas.}|tr.}(\mathbf{y}|\mathbf{x}) = p_{\text{reco.}|gen.,0}(\mathbf{y}|\mathbf{x}) \equiv R(\mathbf{y}, \mathbf{x}), \quad (\text{E.2})$$

Substituting $R(\mathbf{y}, \mathbf{x})$ in eq. E.1 and using $\omega_0(\mathbf{y}) = p_{\text{meas.}}(\mathbf{y})/p_{\text{reco.,0}}(\mathbf{y})$

$$\begin{aligned} p_{\text{gen.,1}}(\mathbf{x}) &\equiv \int d\mathbf{y} \, p_{\text{gen.}|reco.,0}(\mathbf{x}|\mathbf{y}) p_{\text{meas.}}(\mathbf{y}) \\ &= \int d\mathbf{y} \, \frac{R(\mathbf{y}, \mathbf{x}) p_{\text{gen.,0}}(\mathbf{x})}{p_{\text{reco.,0}}(\mathbf{y})} p_{\text{meas.}}(\mathbf{y}) \\ &= p_{\text{gen.,0}}(\mathbf{x}) \int d\mathbf{y} \, R(\mathbf{y}, \mathbf{x}) \omega_0(\mathbf{y}) \end{aligned} \quad (\text{E.3})$$

Using $\nu_0(\mathbf{x}) = p_{\text{gen.},1}(\mathbf{x})/p_{\text{gen.},0}(\mathbf{x})$ measured distribution correspondingly becomes,

$$\begin{aligned} p_{\text{reco.},1}(\mathbf{y}) &= \int d\mathbf{x} \ (\mathbf{x}) \ R(\mathbf{y}, \mathbf{x}) \ p_{\text{gen.},1}(\mathbf{x}) \\ &= p_{\text{reco.},0}(\mathbf{y}) \int d\mathbf{x} \ \frac{R(\mathbf{y}, \mathbf{x}) \ p_{\text{gen.},0}(\mathbf{x})}{p_{\text{reco.},0}(\mathbf{y})} \nu_0(\mathbf{x}) \end{aligned} \quad (\text{E.4})$$

The convolutions $\int d\mathbf{y} \ R(\mathbf{y}, \mathbf{x}) \ \omega_0(\mathbf{y})$ and $\int d\mathbf{x} \ R(\mathbf{y}, \mathbf{x}) \ \frac{p_{\text{gen.},0}(\mathbf{x})}{p_{\text{reco.},0}(\mathbf{y})} \nu_0(\mathbf{x})$ are mapping $\omega_0(\mathbf{y}) \rightarrow \omega_0(\mathbf{x})$ and $\nu_0(\mathbf{x}) \rightarrow \nu_0(\mathbf{y})$ respectively. For the n^{th} iteration,

$$\begin{aligned} \omega_{n-1}(\mathbf{y}) &= \frac{p_{\text{meas.}}(\mathbf{y})}{p_{\text{reco.},n-1}(\mathbf{y})} \\ p_{\text{gen.},n}(\mathbf{x}) &= p_{\text{gen.},n-1}(\mathbf{x}) \int d\mathbf{y} \ R(\mathbf{y}, \mathbf{x}) \ \omega_{n-1}(\mathbf{y}) \\ \nu_{n-1}(\mathbf{x}) &= \frac{p_{\text{gen.},n}(\mathbf{x})}{p_{\text{gen.},n-1}(\mathbf{x})} \\ p_{\text{reco.},n}(\mathbf{y}) &= p_{\text{reco.},n-1}(\mathbf{y}) \int d\mathbf{x} \ R(\mathbf{y}, \mathbf{x}) \ \frac{p_{\text{gen.},n-1}(\mathbf{x})}{p_{\text{reco.},n-1}(\mathbf{y})} \nu_{n-1}(\mathbf{x}) \end{aligned} \quad (\text{E.5})$$

Given convergence, $p_{\text{reco.},n}(\mathbf{y}) \rightarrow p_{\text{gen.}}(\mathbf{y})$ and $p_{\text{tr.},n}(\mathbf{x}) \rightarrow p_{\text{unf.}}(\mathbf{x})$ is the unfolded distribution.

This algorithm can be adapted for binned data by expressing the distributions as vectors representing bin counts of histograms and $R(\mathbf{y}, \mathbf{x})$ as a matrix, giving the traditional method of *Iterative Bayesian Unfolding* [144]. Although it has been the standard for a long time, the required binning of data limits its usefulness beyond 3-dimensions. Also, not all measurements can be straightforwardly represented in the response matrix.

One of the first unbinned approach uses classifiers to calculate ratios ω and ν . For i^{th} iteration, $\omega_{i-1}(\mathbf{y})$ can be calculated using classifier $f_{\text{meas.},\text{reco}}$ which classifies a jet represented by $\mathbf{y}$ as measured or reconstructed. Thus,

$$\omega_{i-1}(\mathbf{y}) = \frac{f_{\text{meas.},\text{reco}}(\mathbf{y})}{1 - f_{\text{meas.},\text{reco}}(\mathbf{y})} \quad (\text{E.6})$$

and $\nu_{i-1}(\mathbf{y})$ can be calculated using classifier $f_{i,i-1}$ which classifies a jet represented by $\mathbf{x}$

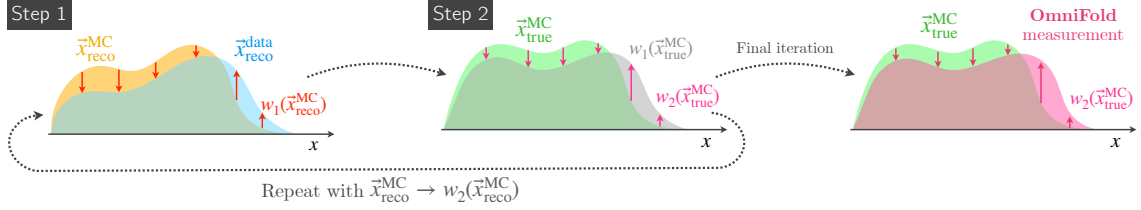


Figure E.2: Illustration of the OmniFold method. In Step 1, MC is corrected to match data at the detector level, producing a weighting function $w_1(\vec{x}_{\text{reco}}^{\text{MC}})$. In Step 2, a new function $w_2(\vec{x}_{\text{true}}^{\text{MC}})$ is learned based only on particle-level quantities. The method proceeds iteratively, refining the function $w_2(\vec{x}_{\text{true}}^{\text{MC}})$ such that the particle-level MC can be reweighted to give event yields and kinematics that match those observed in the data. [23]

as being generated from i^{th} or $(i-1)^{\text{th}}$ iteration.

$$\nu_{i-1}(\mathbf{x}) = \frac{f_{i,i-1}(\mathbf{x})}{1 - f_{i,i-1}(\mathbf{x})} \quad (\text{E.7})$$

For misses, a classifier $f_{i,i-1}^{\text{gen.}}$ separates matched generated jets weighted by $\omega_{i-1}(\mathbf{y})$ from those of the $(i-1)^{\text{th}}$ iteration; for fakes, a classifier $f_{i,i-1}^{\text{reco.}}$ separates matched reconstructed jets weighted by $\nu_{i-1}(\mathbf{x})$ from those of the $(i-1)^{\text{th}}$ iteration,

$$\omega_{i-1}(\mathbf{x}) = \frac{f_{i,i-1}^{\text{gen.}}(\mathbf{x})}{1 - f_{i,i-1}^{\text{gen.}}(\mathbf{x})}, \quad \nu_{i-1}(\mathbf{y}) = \frac{f_{i,i-1}^{\text{reco.}}(\mathbf{y})}{1 - f_{i,i-1}^{\text{reco.}}(\mathbf{y})}, \quad (\text{E.8})$$

evaluated at the miss's $\mathbf{x}$ and the fake's $\mathbf{y}$ respectively. Thus, it takes 2-4 classifier training per unfolding iteration [145].